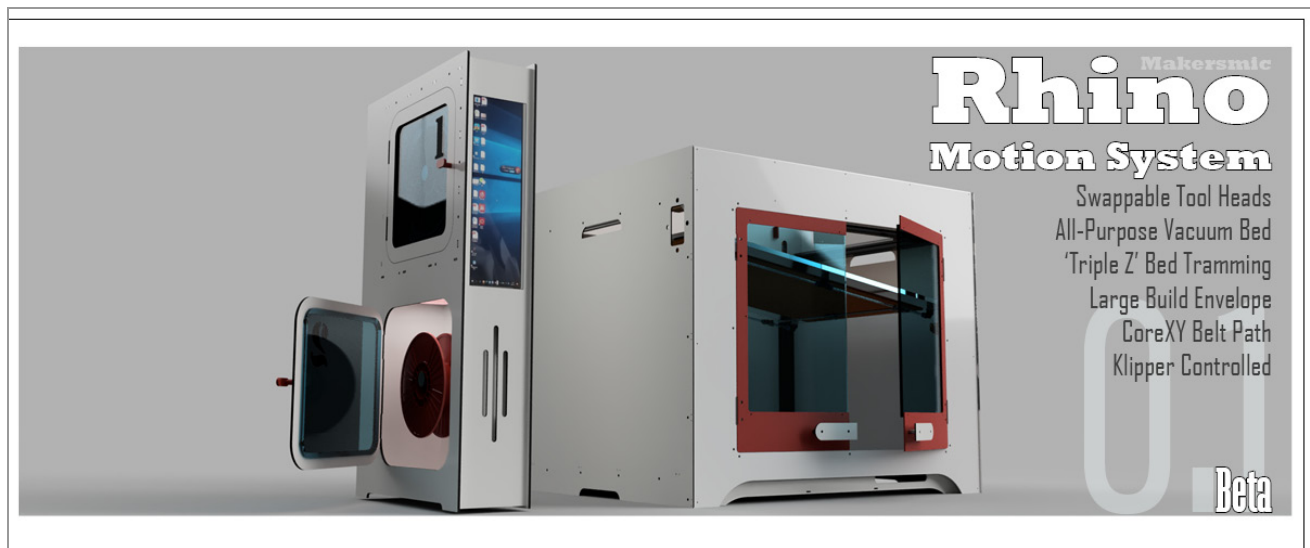


# Rhino Multi-Tool Motion System

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## User Manual



The Rhino is a multi-tool motion system equipped with a variety of tools to help you build your dreams. It uses Klipper firmware. Information can be found online at the Makersmic GitHub page.



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## PART 1

# The Rhino

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The machine, what it needs, and how it is wired.

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About the Rhino

---

System necessities

---

The machine

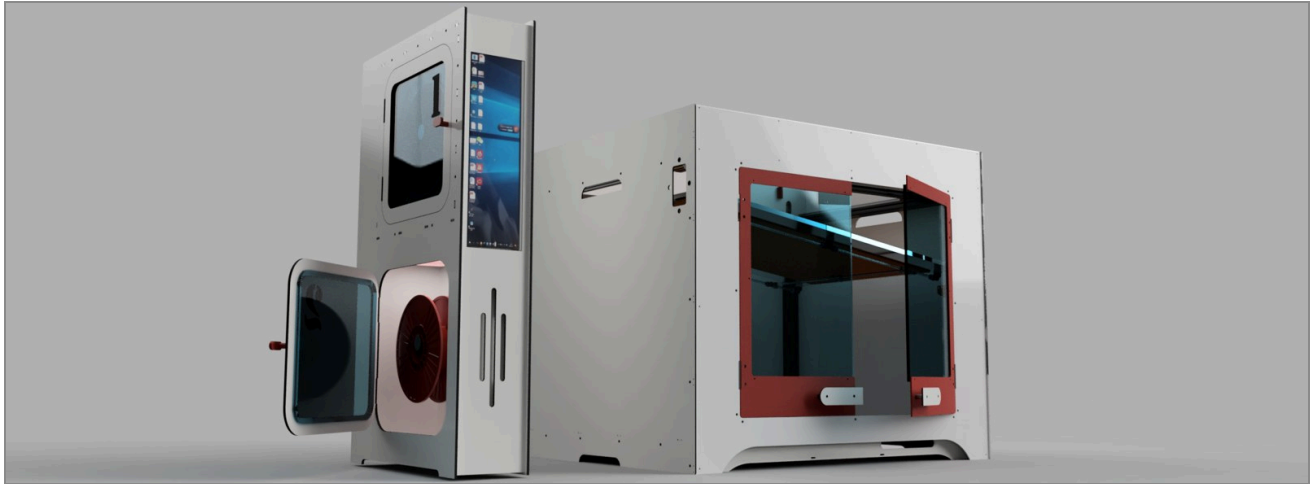
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Wiring and pin mapping

---

## About the Rhino

One motion system, five built-in tools, and room for more: the Rhino changes discipline by changing the tool on its X-carriage, and Klipper changes behaviour to match.



*Full assembly render: the enclosed machine and its side tower with display and filament bay*

| Feature                                      | Rhino                                                                                                                                             |
|----------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------|
| Motion                                       | CoreXY, 15 mm GT2 belts on X and Y (3 m per belt, 6 m total)                                                                                      |
| Axis travel (from <code>printer.cfg</code> ) | X 576 mm, Y 406 mm, Z 400 mm                                                                                                                      |
| X axis                                       | MGW7 linear rail with MGW7H carriage on an aluminium-tube gantry                                                                                  |
| Z axis                                       | Three leadscrews (rear T12×2, front T8×2) on kinematic couplings; automatic tramming with <code>Z_TILT_ADJUST</code>                              |
| Build plate                                  | Aluminium heat spreader with silicone heater mats, air-gap insulated; designed as an all-purpose vacuum bed                                       |
| Enclosure                                    | 3 mm ACP panels with twin front doors                                                                                                             |
| Tool change                                  | Tool slides onto the carriage's V-groove, held by one M3×50 bolt and knurled nut; one 21-pin umbilical plug                                       |
| Tools                                        | Print heads (BlockOne, SwitchFly), LightSaber laser, HotJoe CNC spindle, DragKnife, plus tools you add                                            |
| Air                                          | "Fresh Air" lines: print part cooling, laser blow-air, spindle chip evacuation                                                                    |
| Controller                                   | BTT Octopus (STM32F446)                                                                                                                           |
| Software                                     | Klipper, Moonraker, Mainsail, the Rhino Tool Manager; OrcaSlicer for printing, LaserWeb 4 for laser and drag knife (Kiri:Moto is being looked at) |

The design files, drawings and build notes are in the Makersmic `Rhino-3d-Printer` repository on GitHub (see *Build reference*). Still in development there: the “Cheeky Monkey” filament box (filament monitoring, air handling, chip exhaust) and a needle-cutter tool.

## System necessities

Klipper runs on a wide range of controller boards, so the Rhino is not tied to one. This manual and the configuration are written for the Big Tree Tech Octopus v1.1, but any controller will do if its microcontroller and mainboard provide what the two tables below list. The third table is the additional hardware the build needs: power, switching, protection and wiring for the electronics cabinet, plus the mechanical parts.

| Microcontroller |                                              |
|-----------------|----------------------------------------------|
| 1               | Capable of running Klipper firmware          |
| 2               | 1 × 24 V hotend PWM channel                  |
| 3               | 1 channel capable of controlling a 5 V servo |
| 4               | 7 × 24 V PWM channels                        |
| 5               | 3 × 24 V on/off channels                     |
| 6               | 3–5 thermistor input channels                |
| 7               | 6–8 individual stepper outputs               |
| 8               | 1 RGB LED channel                            |
| 9               | 3 endstop inputs                             |

| Mainboard |                                     |
|-----------|-------------------------------------|
|           | Capable of running Klipper firmware |
|           | 2 USB channels                      |

| Qty | Recommended additional hardware          |
|-----|------------------------------------------|
| 3   | DIN-mount SSR rated 3–30 V / 220 V       |
| 3   | DIN-mount 24 V / 24 V SSR                |
| 1   | 21W4 female socket connector             |
| 1   | 40 mm double-stack Delta fan             |
| 1   | 40 mm fan                                |
| 2   | DIN-mount 20 A circuit breaker, 12–120 V |
| 1   | DIN-mount CP40-1P 275 V surge protection |
| 1   | DIN-mount DZ47-63 3P breaker             |
| 24  | 211B terminal block                      |
| 50  | Fusable-style terminal block             |
| 4   | DIN-mount UK10-DREHSI connector          |
| 6   | 300 mm DIN rail                          |
| 3   | Solid-state relay heatsinks              |
| 1   | WJ175 JQX-38F high-current relay         |
| 3   | 12 V white LED strips                    |
| 1   | 12 V RGB LED strip                       |

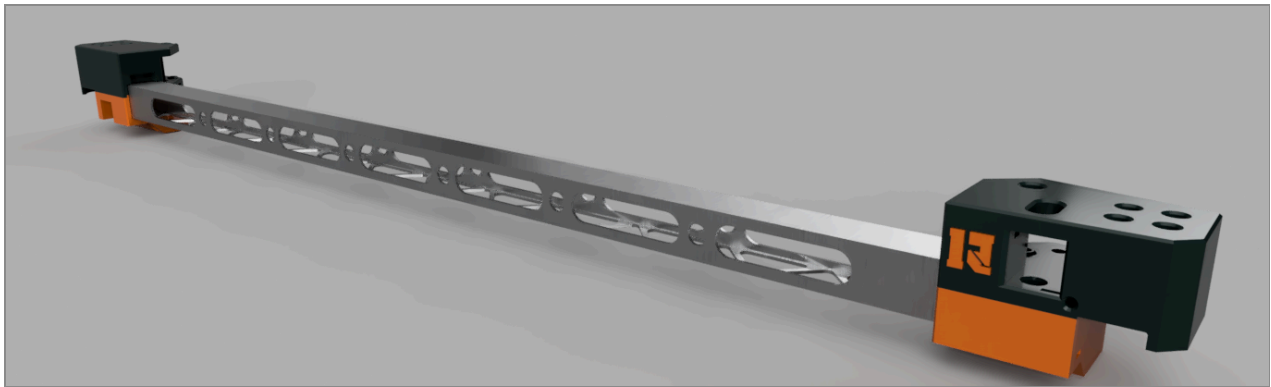
| Qty | Recommended additional hardware            |
|-----|--------------------------------------------|
| 1   | 12 V power supply                          |
| 1   | 24 V power supply                          |
| 1   | T12 leadscrew                              |
| 2   | T8 leadscrew with the same lead as the T12 |
| 4   | 20T 15 mm belt 2GT pulleys (5 mm bore)     |
| 12  | 608zz ABEC-7 bearing                       |
| 4   | 12 mm bearings                             |
| 16  | 15 mm idler pulleys                        |
| 6   | 5 mm × 120 mm stainless steel toy shafts   |
| 2   | 8 mm × 200 mm stainless steel toy shafts   |
| 4   | M/F 8 mm × 20 mm camera rosette            |
| 1   | 110 V fused power plug                     |
| 1   | 220 V power plug                           |
| 6–8 | NEMA 17 stepper motors                     |
| 3   | Optical endstops                           |
| 1   | 8 ft CPAP hose                             |
| 1   | PC817 12 V optocoupler                     |
|     |                                            |
|     |                                            |

## The machine

Every tool rides the same gantry and plugs into the same 21-pin umbilical; only the wiring of each individual tool travels with it.

### Gantry

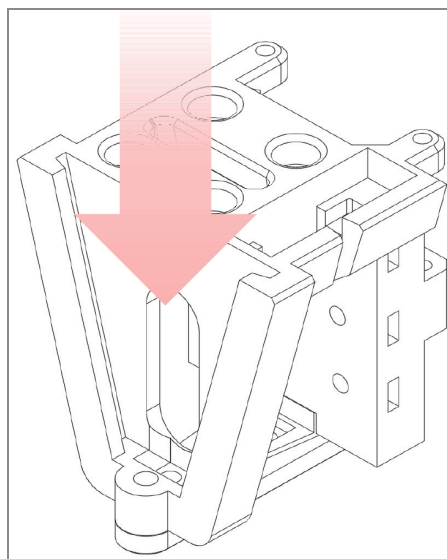
The X beam is 1" aluminium square tube (1/16" wall), lightened with cut-outs made from 3D-printed router templates. The "Bulkup" XY joiners (April 2023) carry extra material at the pulleys and stopped the beam sagging where it meets the joiners. The A/B stepper motors must be no more than 48 mm long with a shaft under 30 mm; they stand on M3×30 mm aluminium standoffs.



*X gantry beam with the Bulkup XY joiners*

### X-carriage and tool slot

The carriage runs on an MGW7 rail and carries the tool lock, the belt clamps and the optical X endstop. A tool slides down into the V-groove on the carriage face, which aligns it, and one M3×50 bolt with a knurled nut (or wing nut) clamps it.



*X-carriage detail: the tool drops into the wedge-type slot*

## Z axis

The bed rides three leadscrews — one T12×2 at the rear, two T8×2 at the front — joined to the plate by kinematic couplings. Each screw is belt-driven through a 2:1 reduction on closed-loop 15 mm GT2 belts of 180–190 mm, and `printer.cfg` matches it with `gear_ratio: 2:1` on all three Z steppers. Because each corner has its own motor, Klipper trams the bed with `Z_TILT_ADJUST`.

## Electrical cabinet

The BTT Octopus controller switches every heavy load through a DIN-rail relay, which keeps high currents off the board and makes a failed relay a quick swap.

| Circuit                                           | Switched by         |
|---------------------------------------------------|---------------------|
| 12 V low-current: LED and NeoPixel lighting, fans | 700-TBR24 6 A relay |
| 12 V high-current: laser module, vacuum pump      | JQX-13FL 10 A relay |
| 24 V spindle (ESC power)                          | JQX-38F 40 A relay  |
| Heated bed mats                                   | Solid-state relays  |

Power comes from separate 24 V, 12 V and 5 V supplies. DIN-rail fuse holders protect the stepper motor lines against shorts. Bridgeable distribution blocks spread power; Wago-style blocks (the “Hub”) terminate the umbilical inside the cabinet.

## The umbilical

An essential assembly of the Rhino, the umbilical provides all the connections each toolhead needs. It uses a 21W4 socket connector for every tool connection, with automatic reconfiguration between tool changes, and the umbilical LED strip signals user-specified alarms.

Each tool ends in a male 21W4 D-sub (four high-current contacts plus seventeen signal pins); the female half is fixed at the top rear of the frame with the Hub below it. The cable is braided Cat5 plus two 14 AWG conductors. Pins 1, 2, 11 and 12 are the high-current contacts.

The pin-by-pin assignment of the umbilical, with the board pin behind each contact, is in *Wiring and pin mapping*. Each tool's own page in *The tools* shows which contacts that tool uses.

**Why every tool has a thermistor.** Klipper's extruder expects a temperature reading at all times. Rather than editing `printer.cfg` for each tool, every tool carries the same thermistor type on pins 3–4 (the laser's is on its heatsink). The swap wizard reads it to confirm the umbilical is seated.

**Constant connections** do not pass through the umbilical; they run to the carriage through the *Back Bone*: the X endstop (3 wires), the umbilical LED strip, the chamber temperature sensor and two 20 mm air tubes.

## Wiring and pin mapping

These pages record how the Rhino is wired to its Big Tree Tech Octopus v1.1 controller: the 21 contacts of the umbilical, the board pins that stay on the machine, every stepper driver, and the wiring diagrams for the special cases. Blank rows and cards are for your own additions.

Where these sheets and `printer.cfg` disagree, Klipper follows `printer.cfg`. The differences found while preparing this manual are listed in *Still to check on the machine*.

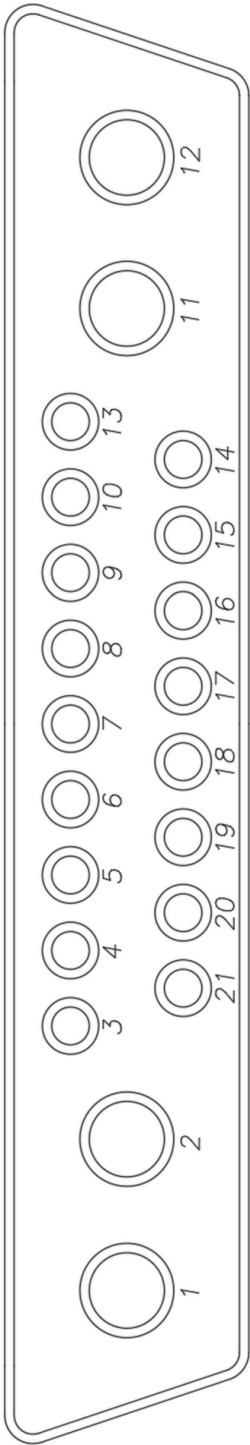
## Rhino Wiring: Pin Mapping

Board: Big Tree Tech Octopus v1.1

| Umbilical pin | Board pin     | Function                                 | Board label |
|---------------|---------------|------------------------------------------|-------------|
| 1             | <b>PA2 +</b>  | Hotend heater                            | HE0         |
| 2             | –             | Hotend heater                            |             |
| 3             | <b>PF4 –</b>  | Hotend thermistor                        | T0          |
| 4             | <b>+</b>      | Hotend thermistor                        |             |
| 5             | <b>PE5</b>    | Hotend fan                               | Fan0        |
| 6             | –             | Hotend fan                               |             |
| 7             | <b>A1</b>     | Stepper A1                               | Motor2      |
| 8             | <b>A2</b>     | Stepper A2                               |             |
| 9             | <b>B1</b>     | Stepper B1                               |             |
| 10            | <b>B2</b>     | Stepper B2                               |             |
| 11            | <b>PA1 +</b>  | 24 V power activate                      | Bed Out     |
| 12            | –             | 24 V power activate                      |             |
| 13            | –             | Probe                                    | Probe       |
| 14            | <b>+</b>      | Probe                                    |             |
| 15            | <b>PG10</b>   | Probe signal                             |             |
| 16            | <b>PB0</b>    | Laser PWM (5 V required)                 | RGB         |
| 17            |               |                                          |             |
| 18            | <b>PB11</b>   | PWM signal 24 V –, ESC signal to GND pin |             |
| 19            |               |                                          |             |
| 20            | <b>PD12 +</b> | 12 V power activate +                    | Fan2        |
| 21            | –             | 12 V power activate –                    |             |

| Umbilical constant connections | Board pin   |
|--------------------------------|-------------|
| X-limit switch                 | <b>PG6</b>  |
| Chamber temperature            | <b>PF5</b>  |
| Umbilical LED                  | <b>PD14</b> |

# Umbilical colour coding



| Pin | Wire colour |
|-----|-------------|
| 1   |             |
| 2   |             |
| 3   |             |
| 4   |             |
| 5   |             |
| 6   |             |
| 7   |             |
| 8   |             |
| 9   |             |
| 10  |             |
| 11  |             |
| 12  |             |
| 13  |             |
| 14  |             |
| 15  |             |
| 16  |             |
| 17  |             |
| 18  |             |
| 19  |             |
| 20  |             |
| 21  |             |

Record the wire colour on each contact of your umbilical as you build it.

Other board pins · Big Tree Tech Octopus v1.1

[illegible]

Model no: \_\_\_\_\_

[illegible]

## Rhino Wiring: Stepper Assignments

Big Tree Tech Octopus v1.1

|   |                              |      |             |
|---|------------------------------|------|-------------|
| X | Driver type and manufacturer |      | BTT TMC2240 |
|   | SPI / UART                   |      |             |
|   | Run current                  |      | 1.1 A       |
|   | Board socket                 | 0    |             |
|   | Step                         | PF13 |             |
|   | Dir                          | PF12 |             |
|   | Enable                       | PF14 |             |
|   | CS / UART                    | PC4  |             |

|   |                              |      |             |
|---|------------------------------|------|-------------|
| Y | Driver type and manufacturer |      | BTT TMC2240 |
|   | SPI / UART                   |      |             |
|   | Run current                  |      | 1.1 A       |
|   | Board socket                 | 1    |             |
|   | Step                         | PG0  |             |
|   | Dir                          | PG1  |             |
|   | Enable                       | PF15 |             |
|   | CS / UART                    | PD11 |             |

## Rhino Wiring: Stepper Assignments

Big Tree Tech Octopus v1.1

|                      |                              |      |             |
|----------------------|------------------------------|------|-------------|
| <b>E<sub>0</sub></b> | Driver type and manufacturer |      | BTT TMC2240 |
|                      | SPI / UART                   |      |             |
|                      | Run current                  |      | 0.8 A       |
|                      | Board socket                 | 4    |             |
|                      | Step                         | PF9  |             |
|                      | Dir                          | PF10 |             |
|                      | Enable                       | PG2  |             |
|                      | CS / UART                    | PF2  |             |

Pins and current from printer.cfg [extruder]

|                      |                              |      |                  |
|----------------------|------------------------------|------|------------------|
| <b>Z<sub>0</sub></b> | Driver type and manufacturer |      | BTT TMC2208 v3.0 |
|                      | SPI / UART                   |      |                  |
|                      | Run current                  |      | 0.65 A           |
|                      | Board socket                 | 5    |                  |
|                      | Step                         | PE6  |                  |
|                      | Dir                          | PA14 |                  |
|                      | Enable                       | PE0  |                  |
|                      | CS / UART                    | PD3  |                  |

## Rhino Wiring: Stepper Assignments

Big Tree Tech Octopus v1.1

|                      |                              |      |                  |
|----------------------|------------------------------|------|------------------|
| <b>Z<sub>1</sub></b> | Driver type and manufacturer |      | BTT TMC2208 v3.0 |
|                      | SPI / UART                   |      |                  |
|                      | Run current                  |      | 0.8 A            |
|                      | Board socket                 | 6    |                  |
|                      | Step                         | PC13 |                  |
|                      | Dir                          | PF0  |                  |
|                      | Enable                       | PF1  |                  |
|                      | CS / UART                    | PE4  |                  |

|                      |                              |     |                  |
|----------------------|------------------------------|-----|------------------|
| <b>Z<sub>2</sub></b> | Driver type and manufacturer |     | BTT TMC2208 v3.0 |
|                      | SPI / UART                   |     |                  |
|                      | Run current                  |     | 0.8 A            |
|                      | Board socket                 | 7   |                  |
|                      | Step                         | PE2 |                  |
|                      | Dir                          | PE3 |                  |
|                      | Enable                       | PD4 |                  |
|                      | CS / UART                    | PE1 |                  |

## Rhino Wiring: Stepper Assignments

Big Tree Tech Octopus v1.1

|                      |                              |  |             |
|----------------------|------------------------------|--|-------------|
| <b>X<sub>1</sub></b> | Driver type and manufacturer |  | BTT TMC2240 |
|                      | SPI / UART                   |  |             |
|                      | Run current                  |  | 0.8 A       |
|                      | Board socket                 |  |             |
|                      | Step                         |  |             |
|                      | Dir                          |  |             |
|                      | Enable                       |  |             |
|                      | CS / UART                    |  |             |

|                      |                              |  |             |
|----------------------|------------------------------|--|-------------|
| <b>Y<sub>1</sub></b> | Driver type and manufacturer |  | BTT TMC2240 |
|                      | SPI / UART                   |  |             |
|                      | Run current                  |  | 0.8 A       |
|                      | Board socket                 |  |             |
|                      | Step                         |  |             |
|                      | Dir                          |  |             |
|                      | Enable                       |  |             |
|                      | CS / UART                    |  |             |

## Rhino Wiring: Stepper Assignments

Big Tree Tech Octopus v1.1

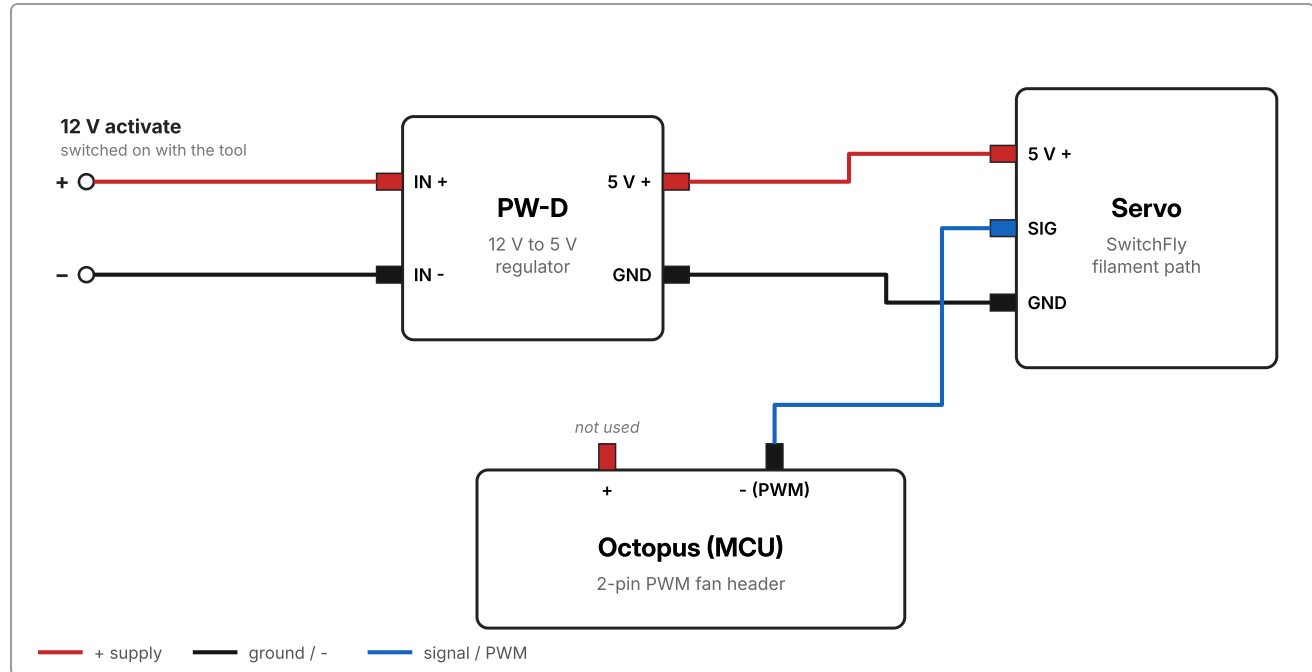
|  |                              |  |
|--|------------------------------|--|
|  | Driver type and manufacturer |  |
|  | SPI / UART                   |  |
|  | Run current                  |  |
|  | Board socket                 |  |
|  | Step                         |  |
|  | Dir                          |  |
|  | Enable                       |  |
|  | CS / UART                    |  |

|  |                              |  |
|--|------------------------------|--|
|  | Driver type and manufacturer |  |
|  | SPI / UART                   |  |
|  | Run current                  |  |
|  | Board socket                 |  |
|  | Step                         |  |
|  | Dir                          |  |
|  | Enable                       |  |
|  | CS / UART                    |  |

## Rhino Wiring: Diagrams

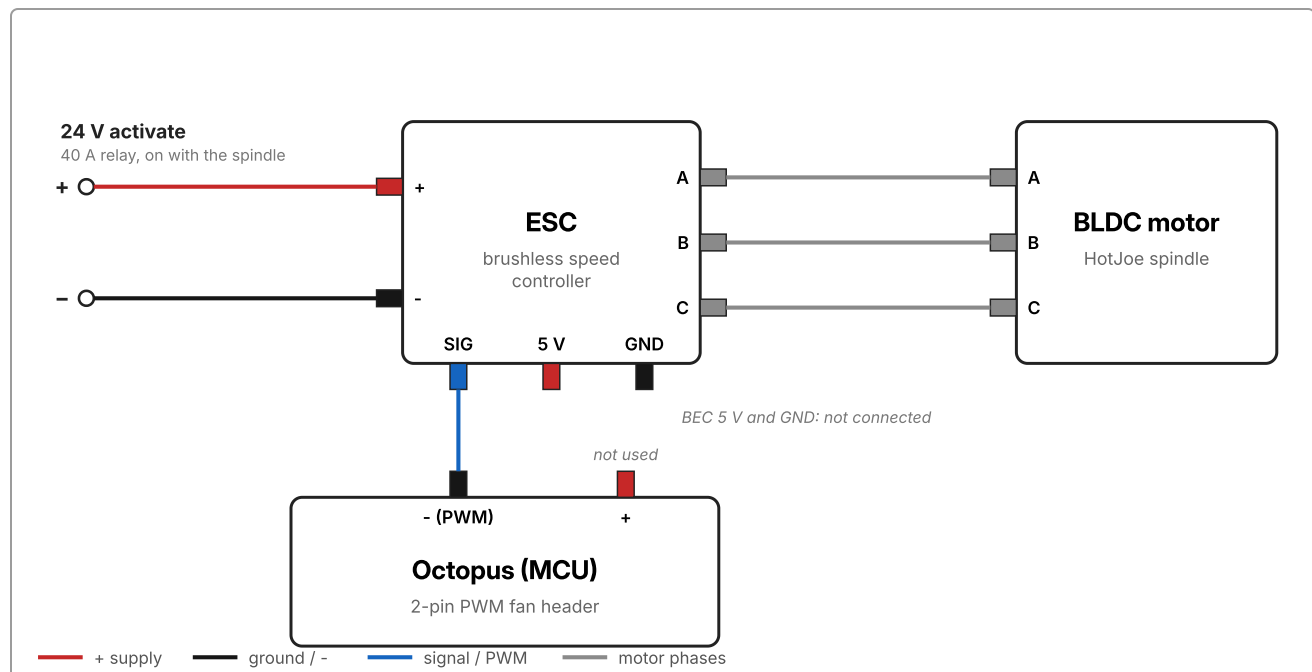
Red: + supply · black: ground / - · blue: signal or PWM · grey: motor phases

### SwitchFly servo



A PW-D regulator turns the switched 12 V into 5 V for the servo; the board's PWM output drives its signal line.

### HotJoe BLDC spindle

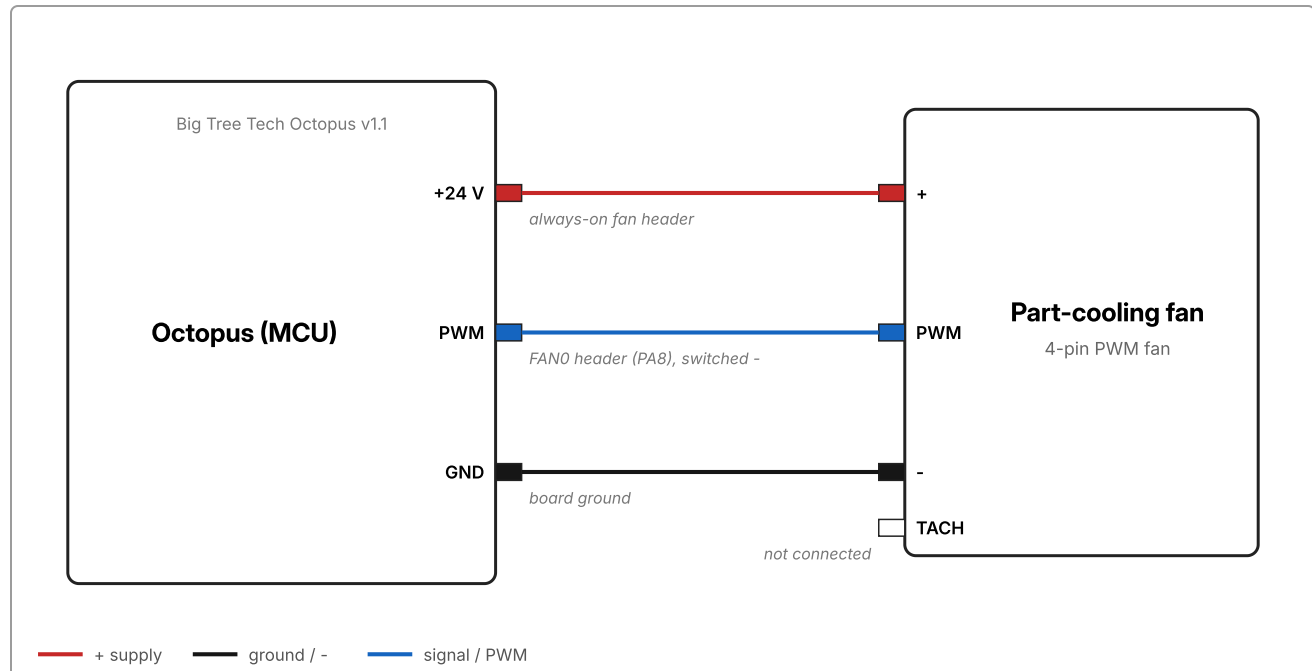


The relay-switched 24 V powers the ESC; the board's PWM output (SPINDLE\_SPEED) drives its signal lead.

## Rhino Wiring: Diagrams

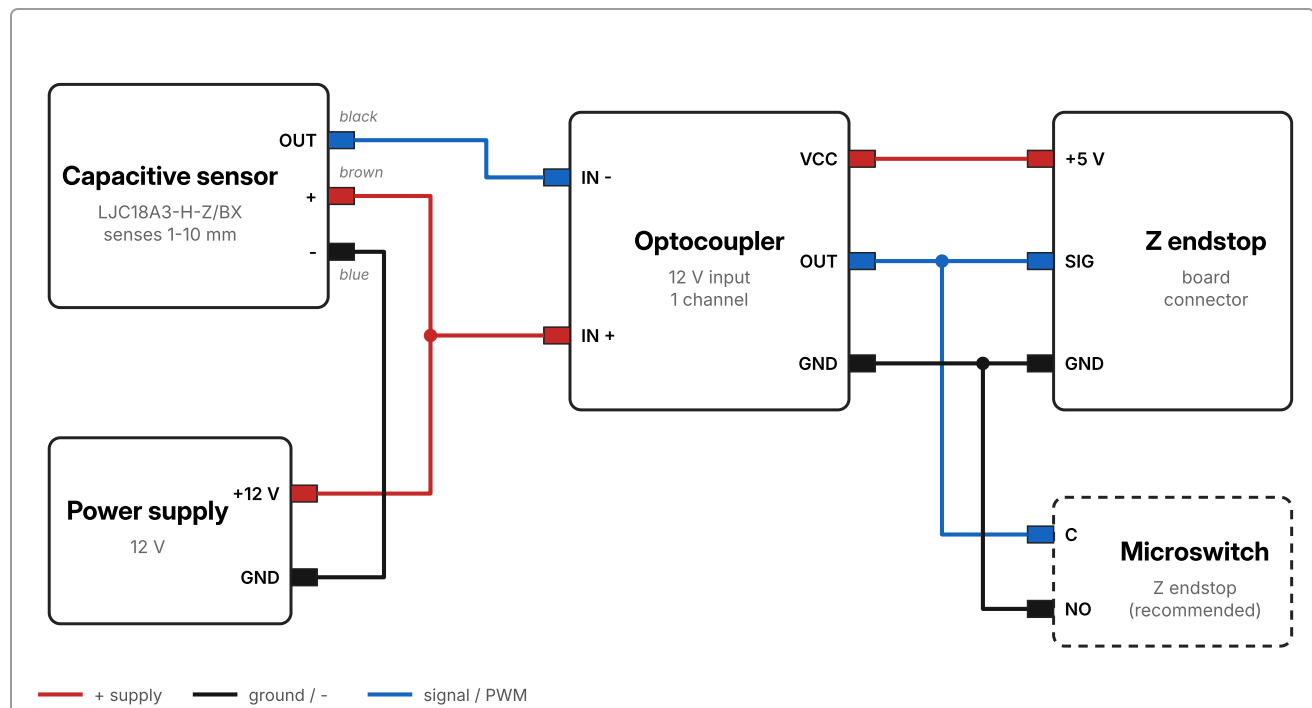
Red: + supply · black: ground / - · blue: signal or PWM

### 4-pin part-cooling fan



Power from an always-on fan header; the PWM line from the FAN0 header (PA8). The tachometer wire is not used.

### Capacitive probe



The 12 V sensor switches an optocoupler, which gives the Z endstop input a clean 5 V signal. A microswitch in parallel is recommended as a backup.

## PART 2

# The Tools

---

The built-in tools, their umbilical contacts and parts, and sheets for the tools you build.

---

The tools

---

BlockOne

---

SwitchFly

---

LightSaber

---

HotJoe

---

DragKnife

---

Tools you add

---

Rhino tools log

---

## The tools

Every tool has a slot number. Slots 1–5 are the built-in tools defined in `variables.cfg`; tools you add in the Tool Manager get slot 6 and up.

| Slot | Tool       | Hardware                                             | Set up a job with                                                                              | Power outputs                                                                |
|------|------------|------------------------------------------------------|------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------|
| 1    | BlockOne   | Single-extruder print head                           | Slicer start G-code ( <code>START_JOB</code> )                                                 | Hotend heater, fans                                                          |
| 2    | SwitchFly  | Two filaments, one stepper, one nozzle               | Slicer start G-code ( <code>START_JOB</code> )                                                 | Hotend heater, fans, path servo                                              |
| 3    | LightSaber | 12 V 80 W 450 nm blue diode laser                    | <code>LASER_JOB_SETUP</code><br><code>FILE=&lt;name&gt;.gcode</code>                           | <code>LASER_INITIALIZE</code><br>rail, <code>SERVO_LASER</code><br>intensity |
| 4    | HotJoe     | 24 V brushless spindle, ER8 collet                   | <code>CNC_JOB_SETUP</code><br><code>FILE=&lt;name&gt;.gcode</code>                             | <code>Spindle_power</code><br>relay,<br><code>SPINDLE_SPEED</code><br>ESC    |
| 5    | DragKnife  | Drag knife for Roland No. 9 and No. 10 blade holders | <code>DRAGKNIFE_JOB_SETUP</code><br><code>FILE=&lt;name&gt;.gcode</code>                       | None                                                                         |
| 6+   | Your tools | Powered, passive or print-head clone                 | <code>TOOL_JOB_SETUP</code><br><code>FILE=&lt;name&gt;.gcode</code><br>(clones use the slicer) | Own pin or a shared output                                                   |

Built-in tools are changed by editing `variables.cfg`. Tools you add are changed in the Tool Manager or with console commands, never by hand-editing `custom_tools.cfg`.

## Notes

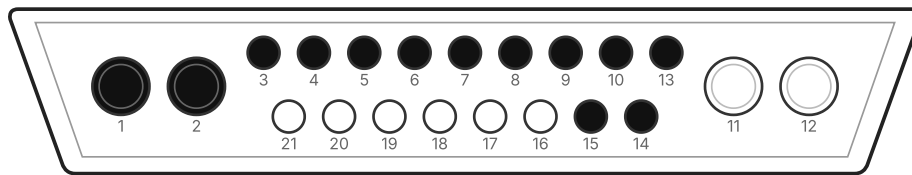
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## Rhino Tools: BlockOne

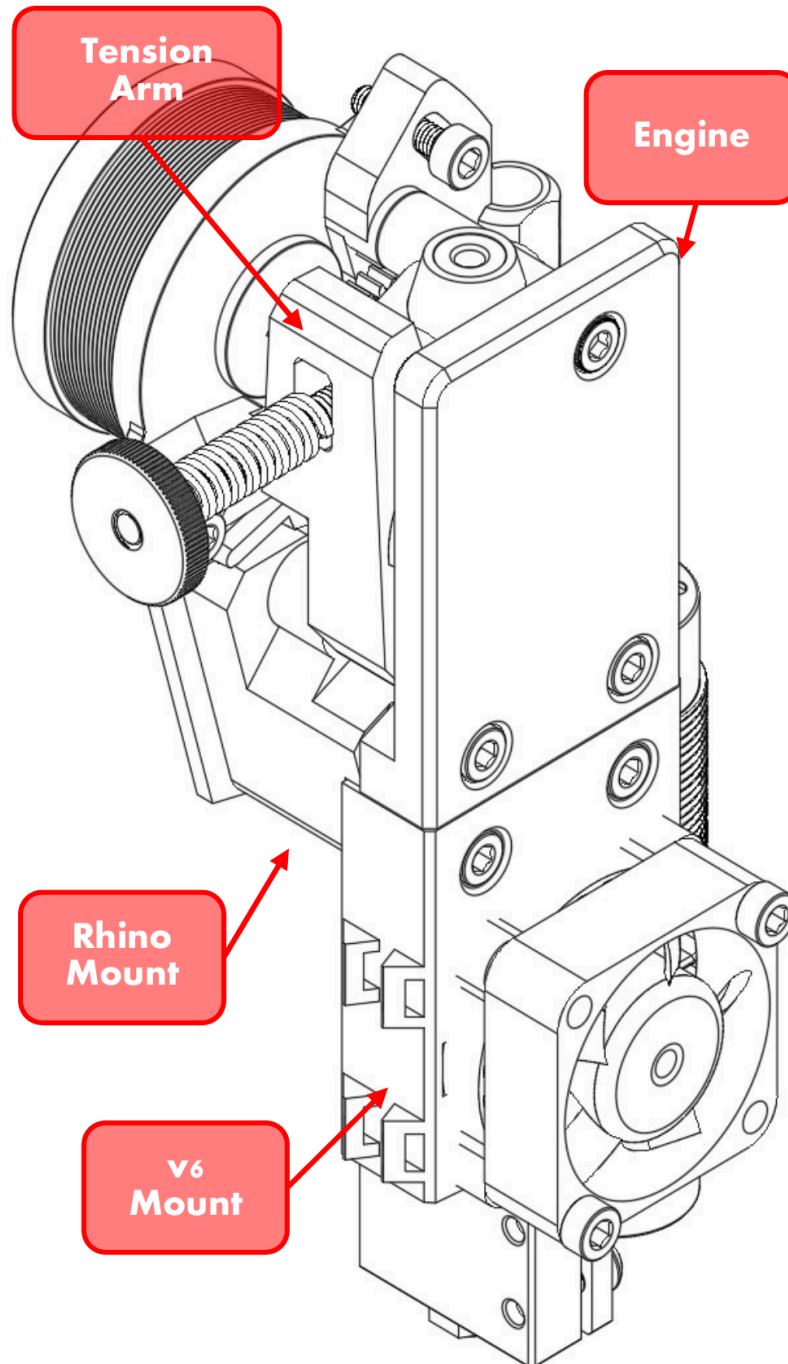
Direct-drive 3D printing, with profiles for 0.4 mm and 0.8 mm nozzles in OrcaSlicer. E3D v6 Volcano hotend at 24 V, BMG drive gear and 1.75 mm filament.

|                           |                                             |
|---------------------------|---------------------------------------------|
| <b>Function</b>           | 3D printing (0.4 and 0.8 mm nozzles)        |
| <b>Slot</b>               | 1                                           |
| <b>Set up a job with</b>  | Slicer start G-code ( START_JOB )           |
| <b>Outputs in Klipper</b> | Hotend heater, hotend fan, part-cooling fan |

### Umbilical connector



| Pin no | Description     | Board pin |
|--------|-----------------|-----------|
| 1      | 24 V hotend     | PA2       |
| 2      | 24 V hotend     |           |
| 3      | Thermistor      | PF4       |
| 4      | Thermistor      |           |
| 5      | 24 V hotend fan | PE5       |
| 6      | 24 V hotend fan |           |
| 7      | Stepper         | A1        |
| 8      | Stepper         | A2        |
| 9      | Stepper         | B1        |
| 10     | Stepper         | B2        |
| 11     |                 |           |
| 12     |                 |           |
| 13     | Probe           |           |
| 14     | Probe           |           |
| 15     | Probe           | PG10      |
| 16     |                 |           |
| 17     |                 |           |
| 18     |                 |           |
| 19     |                 |           |
| 20     |                 |           |
| 21     |                 |           |



\*Part cooling provided by the universal blow-air system.

BlockOne supersedes the Jack Rabbit print head in the repository; the Jack Rabbit files there are outdated. Its materials cover 0.4 and 0.8 mm nozzles.

## Rhino Tools: BlockOne

Parts list · 3D printing (0.4 and 0.8 mm nozzles)

| Qty | Description                            |
|-----|----------------------------------------|
|     | <b>3D-printed pieces:</b>              |
| 1   | BlockOne Rhino Mount                   |
| 1   | BlockOne Engine                        |
| 1   | BlockOne Tension Arm                   |
| 1   | V6 Mount                               |
|     | <b>Off-the-shelf hardware:</b>         |
| 1   | BMG big gear and shaft                 |
| 1   | 625zz sealed bearing                   |
| 2   | MR185 sealed bearing                   |
| 3   | 3 × 5 × 7 brass sleeve bearing         |
| 1   | 24 V E3D v6 Volcano hotend assembly    |
| 1   | J4-HV inductive proximity sensor       |
| 1   | 25 mm NEMA 14 stepper motor            |
| 1   | 30 × 3 mm toy shaft                    |
| 1   | 4 × 8 × 25 mm medium-duty (bed) spring |
| 1   | 21W4 male socket connector             |
|     | <b>Fasteners:</b>                      |
| 6   | M3 × 20 mm socket head cap bolt        |
| 1   | M3 × 30 mm socket head cap bolt        |
| 6   | 2 × 6 × 6 mm M3 square nut             |
| 1   | M5 × 12 mm socket head cap bolt        |
| 1   | M4 × 50 mm socket head cap bolt        |
| 1   | M4 × 2 mm thumb nut                    |
| 1   | M4 washer                              |
| 7   | M3 washers                             |
| 5   | Small zip ties                         |
| 1   | Medium zip tie                         |
|     |                                        |
|     |                                        |
|     |                                        |

Notes · settings that worked, changes, repairs

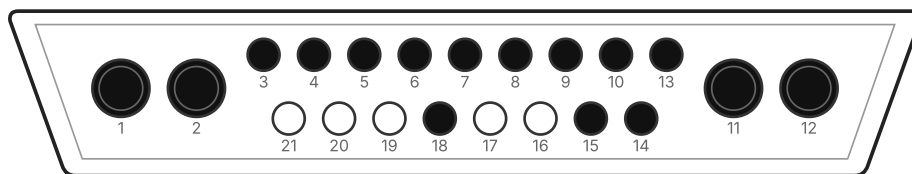
[illegible]

## Rhino Tools: SwitchFly

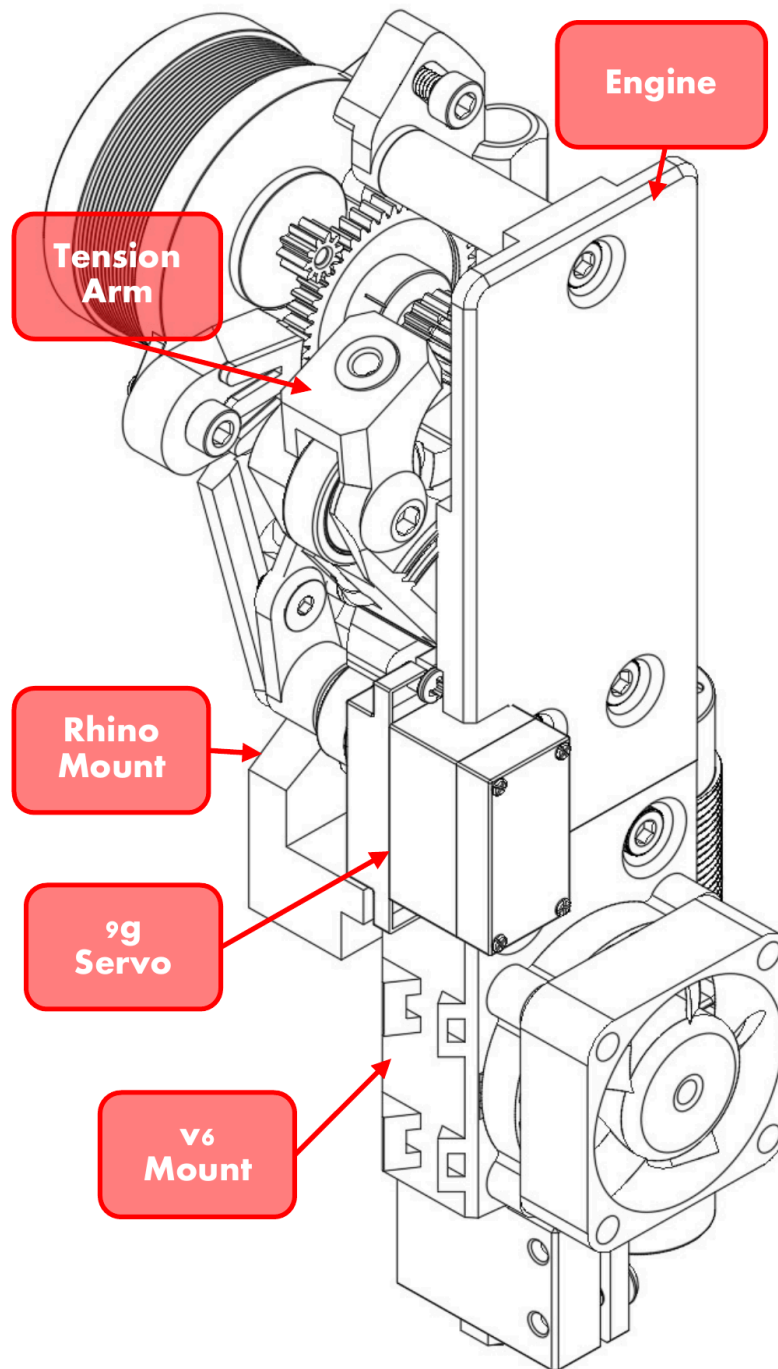
Dual extrusion at its finest: the first true direct-drive, dual-extrusion, single-nozzle extruder for 3D printing. Optimized for the E3D v6 Volcano hotend and BMG drive gear, and rated at 24 V for 1.75 mm filament.

|                           |                                                          |
|---------------------------|----------------------------------------------------------|
| <b>Function</b>           | Dual-extrusion 3D printing (0.4 and 0.8 mm nozzles)      |
| <b>Slot</b>               | 2                                                        |
| <b>Set up a job with</b>  | Slicer start G-code ( START_JOB )                        |
| <b>Outputs in Klipper</b> | Hotend heater, fans, filament-path servo ( SERVO_LASER ) |

### Umbilical connector



| Pin no | Description           | Board pin |
|--------|-----------------------|-----------|
| 1      | 24 V + hotend         | PA2       |
| 2      | 24 V hotend           |           |
| 3      | Thermistor            | PF4       |
| 4      | Thermistor            |           |
| 5      | 24 V hotend fan       | PE5       |
| 6      | 24 V hotend fan       |           |
| 7      | Stepper               | A1        |
| 8      | Stepper               | A2        |
| 9      | Stepper               | B1        |
| 10     | Stepper               | B2        |
| 11     | 24 V + power activate | PA1       |
| 12     | 24 V – power activate |           |
| 13     | Probe                 |           |
| 14     | Probe                 |           |
| 15     | Probe                 | PG10      |
| 16     |                       |           |
| 17     |                       |           |
| 18     | PWM – signal          | PB11      |
| 19     |                       |           |
| 20     |                       |           |
| 21     |                       |           |



\*Part cooling provided by the universal blow-air system.

SwitchFly feeds two filaments through one stepper and one hotend, so there is no second nozzle to align. `SWITCH_T0` and `SWITCH_T1` select the filament with a servo on the shared `SERV0_LASER` output.

## Rhino Tools: SwitchFly

Parts list · Dual-extrusion 3D printing (0.4 and 0.8 mm nozzles)

| Qty | Description                         |
|-----|-------------------------------------|
|     | <b>3D-printed pieces:</b>           |
| 1   | SwitchFly Rhino Mount               |
| 1   | Cam                                 |
| 1   | Linkage Arm                         |
| 1   | Left Tension Arm                    |
| 1   | Right Tension Arm                   |
| 1   | SwitchFly Engine                    |
| 1   | V6 Mount                            |
| 1   | 30 mm Fan Spacer                    |
|     | <b>Off-the-shelf hardware:</b>      |
| 1   | BMG big gear and shaft              |
| 2   | 625zz sealed bearing                |
| 2   | MR185 sealed bearing                |
| 8   | 3 × 5 × 7 brass sleeve bearing      |
| 1   | 24 V E3D v6 Volcano hotend assembly |
| 1   | J4-HV inductive proximity sensor    |
| 5   | Small zip ties                      |
| 1   | Medium zip tie                      |
| 1   | 25 mm NEMA 14 stepper motor         |
| 1   | Ecopower 827 12 g micro servo       |
| 1   | 12 mm control horn                  |
| 1   | 35 × 3 mm toy shaft                 |
| 2   | 30 × 3 mm toy shaft                 |
| 1   | 3 mm × 25 mm expansion spring       |
| 1   | 21W4 male socket connector          |
| 1   | 3-pin servo extension cable         |
| 1   | 24 V 30 mm fan                      |
|     | <b>Fasteners:</b>                   |
| 3   | M3 × 30 mm socket head cap bolt     |
| 4   | M3 × 25 mm socket head cap bolt     |
| 6   | 2 × 6 × 6 mm M3 square nut          |
| 2   | M3 × 6 mm flat head bolt            |
| 2   | M5 × 12 mm button head bolt         |
| 2   | M5 square nut                       |
| 4   | M3 × 12 mm socket head cap bolt     |
| 2   | M2 × 6 mm Phillips head screw       |
| 7   | M3 washers                          |

\*Shaft rotation is negated by the custom G0 and G1 Klipper commands.

\*Ensure the servo default angle is set to 90 degrees in Klipper.

Notes · settings that worked, changes, repairs

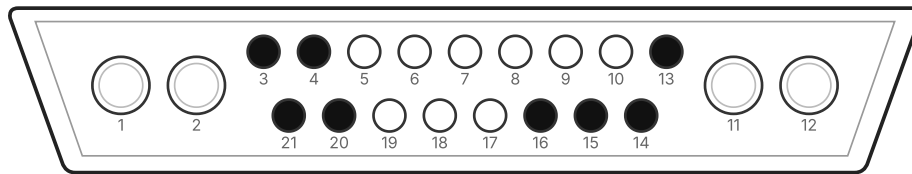
[illegible]

## Rhino Tools: LightSaber

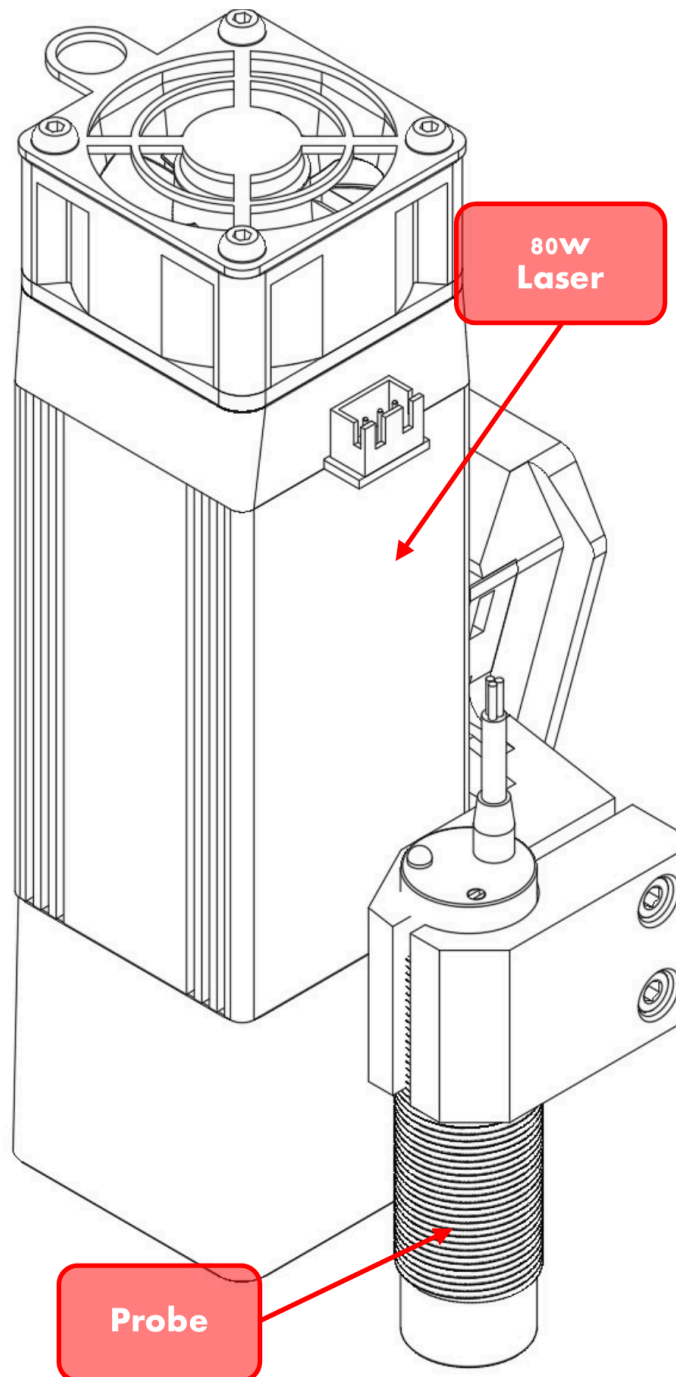
12 V, 80 W, 450 nm blue diode laser cutter and engraver.

|                    |                                                                             |
|--------------------|-----------------------------------------------------------------------------|
| Function           | Laser cutting and engraving                                                 |
| Slot               | 3                                                                           |
| Set up a job with  | <code>LASER_JOB_SETUP FILE=&lt;name&gt;.gcode</code>                        |
| Outputs in Klipper | <code>LASER_INITIALIZE</code> 12 V rail, <code>SERVO_LASER</code> intensity |

### Umbilical connector



| Pin no | Description           | Board pin |
|--------|-----------------------|-----------|
| 1      |                       |           |
| 2      |                       |           |
| 3      | Thermistor            | PF4       |
| 4      | Thermistor            |           |
| 5      |                       |           |
| 6      |                       |           |
| 7      |                       |           |
| 8      |                       |           |
| 9      |                       |           |
| 10     |                       |           |
| 11     |                       |           |
| 12     |                       |           |
| 13     | Probe                 |           |
| 14     | Probe                 |           |
| 15     | Probe                 | PG10      |
| 16     | Laser PWM (5 V)       | PB0       |
| 17     |                       |           |
| 18     |                       |           |
| 19     |                       |           |
| 20     | + 12 V power activate | PD12      |
| 21     | – 12 V power activate |           |



\*Blow-air provided by the universal blow-air system.

A relay switches the laser's 12 V supply and Klipper drives its intensity through the PWM contact. The printed height gauge in the build files sets the focus distance.

**Safety:** wear glasses rated for the laser's wavelength, run extraction, never cut PVC or vinyl, and stay with the machine for the whole cut.

## Parts list · Laser cutting and engraving

[illegible]

Notes · settings that worked, changes, repairs

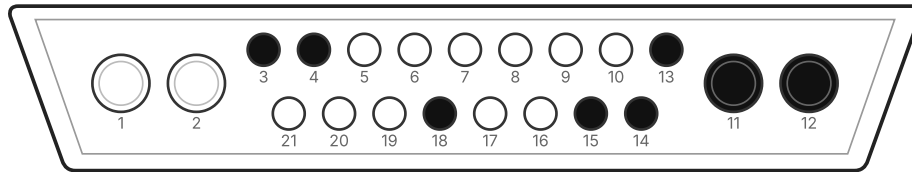
35

## Rhino Tools: HotJoe

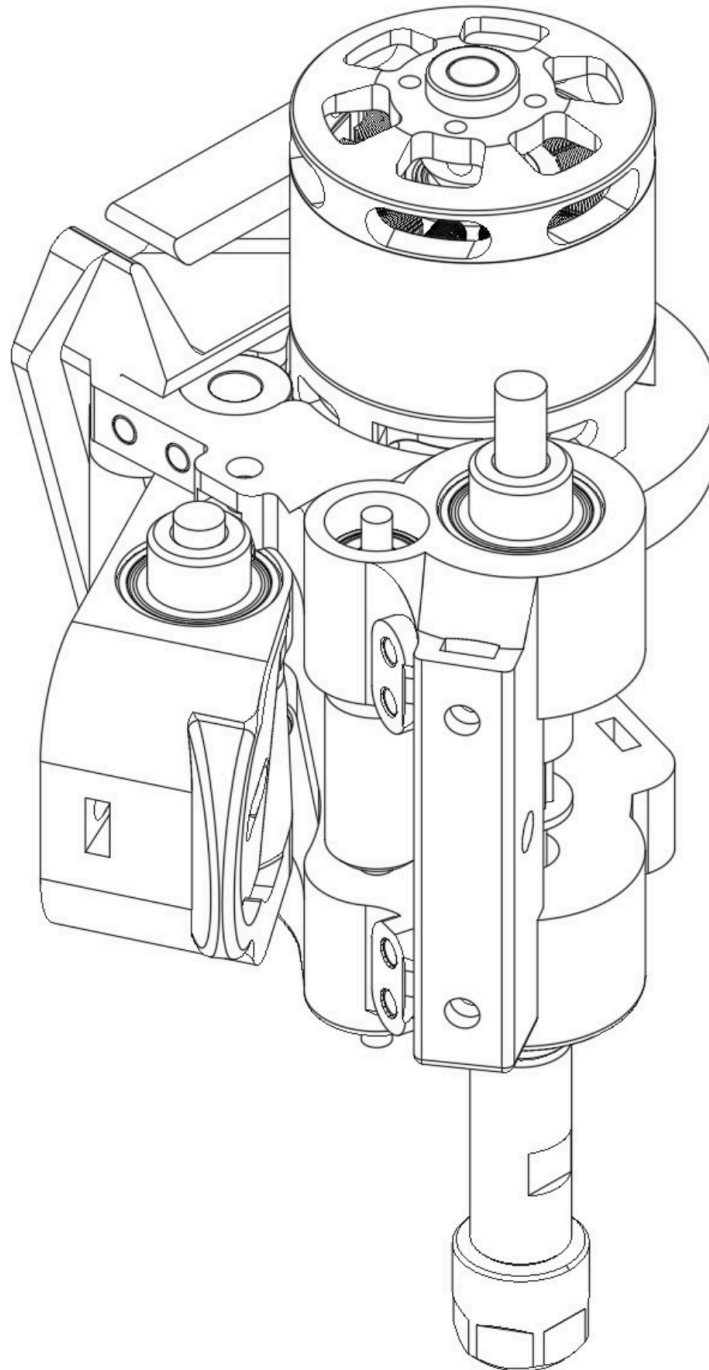
Spindle attachment with an ER8 collet and a 2814 1000 kV brushless motor, driven by an 80 A HobbyWing ESC on 24 V power.

|                           |                                                                         |
|---------------------------|-------------------------------------------------------------------------|
| <b>Function</b>           | CNC spindle for light engraving and milling                             |
| <b>Slot</b>               | 4                                                                       |
| <b>Set up a job with</b>  | <code>CNC_JOB_SETUP FILE=&lt;name&gt;.gcode</code>                      |
| <b>Outputs in Klipper</b> | <code>Spindle_power</code> relay, <code>SPINDLE_SPEED</code> ESC signal |

### Umbilical connector



| Pin no | Description      | Board pin   |
|--------|------------------|-------------|
| 1      |                  |             |
| 2      |                  |             |
| 3      | Thermistor       | <b>PF4</b>  |
| 4      | Thermistor       |             |
| 5      |                  |             |
| 6      |                  |             |
| 7      |                  |             |
| 8      |                  |             |
| 9      |                  |             |
| 10     |                  |             |
| 11     | 24 V relay power | <b>PA1</b>  |
| 12     | 24 V relay power |             |
| 13     | Probe            |             |
| 14     | Probe            |             |
| 15     | Probe            | <b>PG10</b> |
| 16     |                  |             |
| 17     |                  |             |
| 18     | PWM              | <b>PB11</b> |
| 19     |                  |             |
| 20     |                  |             |
| 21     |                  |             |



\*Shown without safety covers.

\*Blow-air provided by the universal blow-air system.

Meant for light-duty milling. A 40 A relay powers the ESC only while the spindle is in use; the speed signal runs at 400 Hz, where 0.3 is stopped (neutral) and about 0.99 is full throttle. Check real speeds at the collet with a tachometer before relying on them. Calibrate the ESC once with `CALIBRATE_ESC` (see *Maintenance*). The build files also include a 70 mm brush holder for dust collection, a blow-air clip and motor and gearbox shields.

## Parts list · CNC spindle for light engraving and milling

[illegible]

Notes · settings that worked, changes, repairs

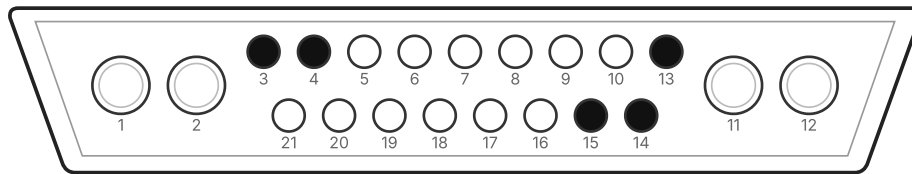
[illegible]

## Rhino Tools: DragKnife

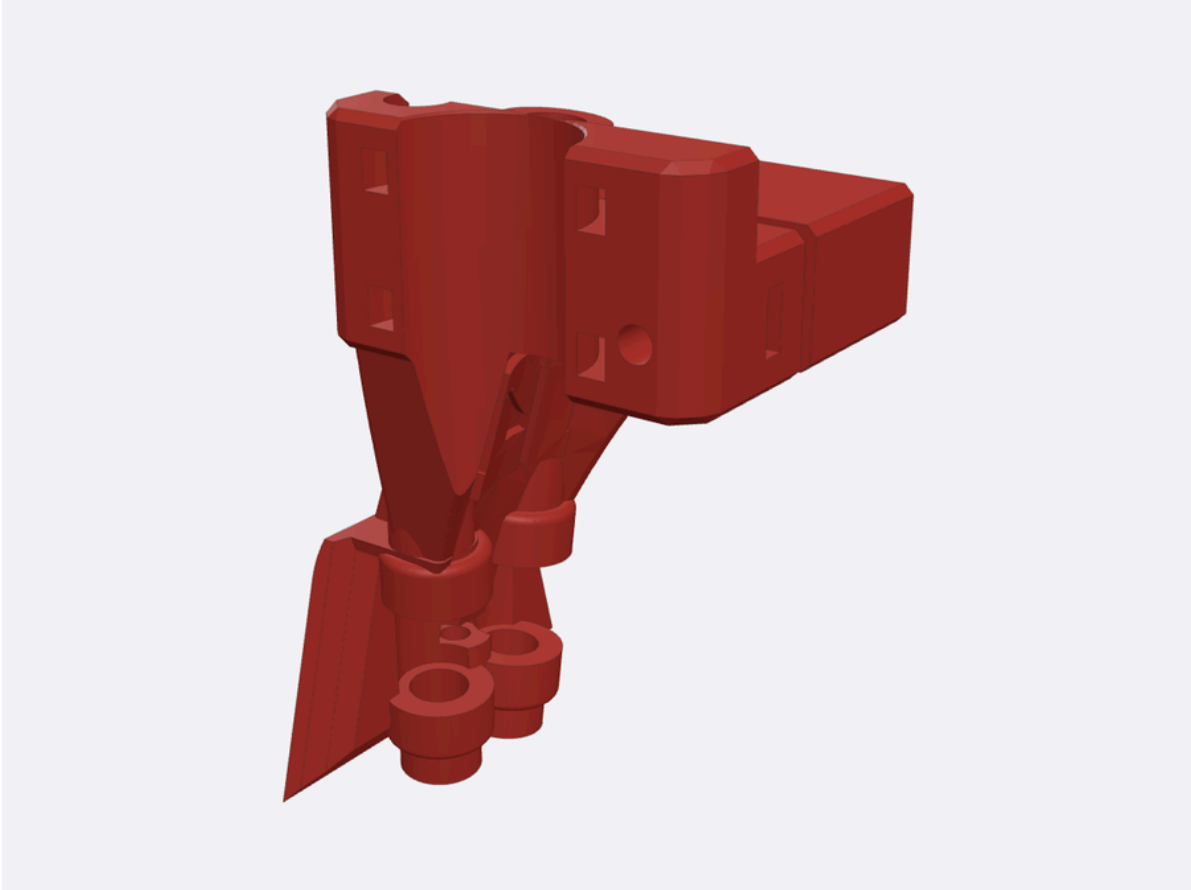
A drag-knife tool designed for Roland No. 9 and No. 10 blade holders.

|                           |                                                          |
|---------------------------|----------------------------------------------------------|
| <b>Function</b>           | Drag-knife cutting (vinyl, cardstock, thin foam)         |
| <b>Slot</b>               | 5                                                        |
| <b>Set up a job with</b>  | <code>DRAGKNIFE_JOB_SETUP FILE=&lt;name&gt;.gcode</code> |
| <b>Outputs in Klipper</b> | None (passive tool)                                      |

### Umbilical connector



| Pin no | Description | Board pin   |
|--------|-------------|-------------|
| 1      |             |             |
| 2      |             |             |
| 3      | Thermistor  | <b>PF4</b>  |
| 4      | Thermistor  |             |
| 5      |             |             |
| 6      |             |             |
| 7      |             |             |
| 8      |             |             |
| 9      |             |             |
| 10     |             |             |
| 11     |             |             |
| 12     |             |             |
| 13     | Probe       |             |
| 14     | Probe       |             |
| 15     | Probe       | <b>PG10</b> |
| 16     |             |             |
| 17     |             |             |
| 18     |             |             |
| 19     |             |             |
| 20     |             |             |
| 21     |             |             |



\*Rendered from the build files.

The drag knife has no power output; cut depth comes from the blade holder's dial and the Z zero set in the job wizard. Its thermistor and probe contacts let the swap wizard confirm the umbilical is seated.

Parts list · Drag-knife cutting (vinyl, cardstock, thin foam)

[illegible]

Notes · settings that worked, changes, repairs

[illegible]

## Tools you add

---

Hot-wire cutters, needle cutters, dispensers, pens and other holders, and extra print-head nozzle sizes are added from the Tool Manager (see *The Tool Manager portal*). A powered tool takes its main power from an umbilical contact it can share, such as the spindle ESC signal on contact 18, or from a new output on a spare pin, and can switch more outputs on with itself.

The next pages are blank tool sheets, two per tool, laid out like the built-in tools' pages. Fill one in for each tool you design. They face each other when the manual lies open: the left page records what the tool is and which umbilical contacts it uses, the right page the settings you enter in the portal (controls, macros, materials) and its parts. Keep the sheet with the tool.

Before wiring a new tool, check **Pins & outputs** in the portal: it shows every contact of the umbilical and what drives it. Only free pins can be given to a new output, and the portal refuses a pin that anything in the config already uses.

## Notes

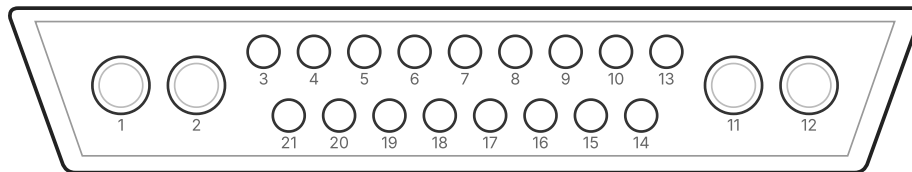
This image shows a single sheet of white paper with horizontal ruling lines. The lines are evenly spaced and run across the width of the page. There is no text or other markings on the paper.

# Rhino Tools: \_\_\_\_\_

New tool sheet 1 · identity and umbilical

|                                |                                                                                                                     |
|--------------------------------|---------------------------------------------------------------------------------------------------------------------|
| <b>Tool name</b>               |                                                                                                                     |
| <b>Slot</b>                    |                                                                                                                     |
| <b>Category and kind</b>       | <input type="checkbox"/> Additive <input type="checkbox"/> Subtractive <input type="checkbox"/> Passive kind: _____ |
| <b>Function</b>                |                                                                                                                     |
| <b>Main power (PWM)</b>        | contact ____ or new pin ____ output _____                                                                           |
| <b>Switch outputs</b>          | contacts / pins _____                                                                                               |
| <b>Power cap / max on-time</b> | _____ % / _____ s                                                                                                   |

**Umbilical connector (shade the contacts the tool uses)**



| Pin no | Description | Board pin | Output / control |
|--------|-------------|-----------|------------------|
| 1      |             |           |                  |
| 2      |             |           |                  |
| 3      |             |           |                  |
| 4      |             |           |                  |
| 5      |             |           |                  |
| 6      |             |           |                  |
| 7      |             |           |                  |
| 8      |             |           |                  |
| 9      |             |           |                  |
| 10     |             |           |                  |
| 11     |             |           |                  |
| 12     |             |           |                  |
| 13     |             |           |                  |
| 14     |             |           |                  |
| 15     |             |           |                  |
| 16     |             |           |                  |
| 17     |             |           |                  |
| 18     |             |           |                  |
| 19     |             |           |                  |
| 20     |             |           |                  |
| 21     |             |           |                  |

# Rhino Tools: \_\_\_\_\_

New tool sheet 1 · portal settings and parts

## Controls

| Control name | What it does | Pin | Off on safe-off          | Macros |
|--------------|--------------|-----|--------------------------|--------|
|              |              |     | <input type="checkbox"/> |        |
|              |              |     | <input type="checkbox"/> |        |
|              |              |     | <input type="checkbox"/> |        |
|              |              |     | <input type="checkbox"/> |        |
|              |              |     | <input type="checkbox"/> |        |

## Macros

| Macro name | Purpose |
|------------|---------|
|            |         |
|            |         |
|            |         |
|            |         |

## Materials

| Material | Power % | Feed mm/min | Z offset mm |
|----------|---------|-------------|-------------|
|          |         |             |             |
|          |         |             |             |
|          |         |             |             |
|          |         |             |             |

## Parts list

| Qty | Part |
|-----|------|
|     |      |
|     |      |
|     |      |
|     |      |
|     |      |
|     |      |
|     |      |

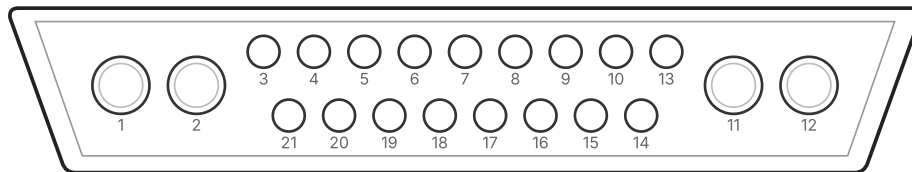
## Notes

## Rhino Tools: \_\_\_\_\_

New tool sheet 2 · identity and umbilical

|                                |                                                                                                                     |
|--------------------------------|---------------------------------------------------------------------------------------------------------------------|
| <b>Tool name</b>               |                                                                                                                     |
| <b>Slot</b>                    |                                                                                                                     |
| <b>Category and kind</b>       | <input type="checkbox"/> Additive <input type="checkbox"/> Subtractive <input type="checkbox"/> Passive kind: _____ |
| <b>Function</b>                |                                                                                                                     |
| <b>Main power (PWM)</b>        | contact ____ or new pin ____ output _____                                                                           |
| <b>Switch outputs</b>          | contacts / pins _____                                                                                               |
| <b>Power cap / max on-time</b> | _____ % / _____ s                                                                                                   |

**Umbilical connector (shade the contacts the tool uses)**



| Pin no | Description | Board pin | Output / control |
|--------|-------------|-----------|------------------|
| 1      |             |           |                  |
| 2      |             |           |                  |
| 3      |             |           |                  |
| 4      |             |           |                  |
| 5      |             |           |                  |
| 6      |             |           |                  |
| 7      |             |           |                  |
| 8      |             |           |                  |
| 9      |             |           |                  |
| 10     |             |           |                  |
| 11     |             |           |                  |
| 12     |             |           |                  |
| 13     |             |           |                  |
| 14     |             |           |                  |
| 15     |             |           |                  |
| 16     |             |           |                  |
| 17     |             |           |                  |
| 18     |             |           |                  |
| 19     |             |           |                  |
| 20     |             |           |                  |
| 21     |             |           |                  |

# Rhino Tools: \_\_\_\_\_

New tool sheet 2 · portal settings and parts

## Controls

| Control name | What it does | Pin | Off on safe-off          | Macros |
|--------------|--------------|-----|--------------------------|--------|
|              |              |     | <input type="checkbox"/> |        |
|              |              |     | <input type="checkbox"/> |        |
|              |              |     | <input type="checkbox"/> |        |
|              |              |     | <input type="checkbox"/> |        |
|              |              |     | <input type="checkbox"/> |        |

## Macros

| Macro name | Purpose |
|------------|---------|
|            |         |
|            |         |
|            |         |
|            |         |

## Materials

| Material | Power % | Feed mm/min | Z offset mm |
|----------|---------|-------------|-------------|
|          |         |             |             |
|          |         |             |             |
|          |         |             |             |
|          |         |             |             |

## Parts list

| Qty | Part |
|-----|------|
|     |      |
|     |      |
|     |      |
|     |      |
|     |      |
|     |      |
|     |      |

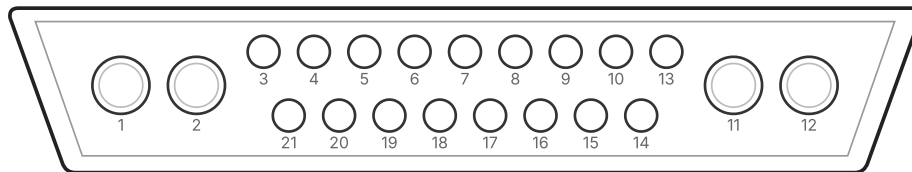
## Notes

## Rhino Tools: \_\_\_\_\_

New tool sheet 3 · identity and umbilical

|                                |                                                                                                                     |
|--------------------------------|---------------------------------------------------------------------------------------------------------------------|
| <b>Tool name</b>               |                                                                                                                     |
| <b>Slot</b>                    |                                                                                                                     |
| <b>Category and kind</b>       | <input type="checkbox"/> Additive <input type="checkbox"/> Subtractive <input type="checkbox"/> Passive kind: _____ |
| <b>Function</b>                |                                                                                                                     |
| <b>Main power (PWM)</b>        | contact ____ or new pin ____ output _____                                                                           |
| <b>Switch outputs</b>          | contacts / pins _____                                                                                               |
| <b>Power cap / max on-time</b> | _____ % / _____ s                                                                                                   |

**Umbilical connector (shade the contacts the tool uses)**



| Pin no | Description | Board pin | Output / control |
|--------|-------------|-----------|------------------|
| 1      |             |           |                  |
| 2      |             |           |                  |
| 3      |             |           |                  |
| 4      |             |           |                  |
| 5      |             |           |                  |
| 6      |             |           |                  |
| 7      |             |           |                  |
| 8      |             |           |                  |
| 9      |             |           |                  |
| 10     |             |           |                  |
| 11     |             |           |                  |
| 12     |             |           |                  |
| 13     |             |           |                  |
| 14     |             |           |                  |
| 15     |             |           |                  |
| 16     |             |           |                  |
| 17     |             |           |                  |
| 18     |             |           |                  |
| 19     |             |           |                  |
| 20     |             |           |                  |
| 21     |             |           |                  |

## Rhino Tools: \_\_\_\_\_

New tool sheet 3 · portal settings and parts

### Controls

| Control name | What it does | Pin | Off on safe-off          | Macros |
|--------------|--------------|-----|--------------------------|--------|
|              |              |     | <input type="checkbox"/> |        |
|              |              |     | <input type="checkbox"/> |        |
|              |              |     | <input type="checkbox"/> |        |
|              |              |     | <input type="checkbox"/> |        |
|              |              |     | <input type="checkbox"/> |        |

### Macros

| Macro name | Purpose |
|------------|---------|
|            |         |
|            |         |
|            |         |
|            |         |

### Materials

| Material | Power % | Feed mm/min | Z offset mm |
|----------|---------|-------------|-------------|
|          |         |             |             |
|          |         |             |             |
|          |         |             |             |
|          |         |             |             |

### Parts list

| Qty | Part |
|-----|------|
|     |      |
|     |      |
|     |      |
|     |      |
|     |      |
|     |      |
|     |      |

### Notes

# Rhino Tools Log

Print heads and nozzles

| Extruder 1                               |                           |
|------------------------------------------|---------------------------|
| Nozzle size                              | Nozzle type               |
| 0.2 / 0.25 / 0.3 / 0.4 / 0.6 / 0.8 / 1.0 | Brass / Steel / Specialty |

| Extruder 2                               |                           |
|------------------------------------------|---------------------------|
| Nozzle size                              | Nozzle type               |
| 0.2 / 0.25 / 0.3 / 0.4 / 0.6 / 0.8 / 1.0 | Brass / Steel / Specialty |

| Extruder 3                               |                           |
|------------------------------------------|---------------------------|
| Nozzle size                              | Nozzle type               |
| 0.2 / 0.25 / 0.3 / 0.4 / 0.6 / 0.8 / 1.0 | Brass / Steel / Specialty |

| Extruder 4                               |                           |
|------------------------------------------|---------------------------|
| Nozzle size                              | Nozzle type               |
| 0.2 / 0.25 / 0.3 / 0.4 / 0.6 / 0.8 / 1.0 | Brass / Steel / Specialty |

| Extruder 5                               |                           |
|------------------------------------------|---------------------------|
| Nozzle size                              | Nozzle type               |
| 0.2 / 0.25 / 0.3 / 0.4 / 0.6 / 0.8 / 1.0 | Brass / Steel / Specialty |

| Extruder 6                               |                           |
|------------------------------------------|---------------------------|
| Nozzle size                              | Nozzle type               |
| 0.2 / 0.25 / 0.3 / 0.4 / 0.6 / 0.8 / 1.0 | Brass / Steel / Specialty |

| Extruder 7                               |                           |
|------------------------------------------|---------------------------|
| Nozzle size                              | Nozzle type               |
| 0.2 / 0.25 / 0.3 / 0.4 / 0.6 / 0.8 / 1.0 | Brass / Steel / Specialty |

| Extruder 8                               |                           |
|------------------------------------------|---------------------------|
| Nozzle size                              | Nozzle type               |
| 0.2 / 0.25 / 0.3 / 0.4 / 0.6 / 0.8 / 1.0 | Brass / Steel / Specialty |

| Extruder 9                               |                           |
|------------------------------------------|---------------------------|
| Nozzle size                              | Nozzle type               |
| 0.2 / 0.25 / 0.3 / 0.4 / 0.6 / 0.8 / 1.0 | Brass / Steel / Specialty |

| SwitchFly 1                              |                           |
|------------------------------------------|---------------------------|
| Nozzle size                              | Nozzle type               |
| 0.2 / 0.25 / 0.3 / 0.4 / 0.6 / 0.8 / 1.0 | Brass / Steel / Specialty |

| SwitchFly 2                              |                           |
|------------------------------------------|---------------------------|
| Nozzle size                              | Nozzle type               |
| 0.2 / 0.25 / 0.3 / 0.4 / 0.6 / 0.8 / 1.0 | Brass / Steel / Specialty |

| SwitchFly 3                              |                           |
|------------------------------------------|---------------------------|
| Nozzle size                              | Nozzle type               |
| 0.2 / 0.25 / 0.3 / 0.4 / 0.6 / 0.8 / 1.0 | Brass / Steel / Specialty |

| SwitchFly 4                              |                           |
|------------------------------------------|---------------------------|
| Nozzle size                              | Nozzle type               |
| 0.2 / 0.25 / 0.3 / 0.4 / 0.6 / 0.8 / 1.0 | Brass / Steel / Specialty |

| SwitchFly 5                              |                           |
|------------------------------------------|---------------------------|
| Nozzle size                              | Nozzle type               |
| 0.2 / 0.25 / 0.3 / 0.4 / 0.6 / 0.8 / 1.0 | Brass / Steel / Specialty |

| SwitchFly 6                              |                           |
|------------------------------------------|---------------------------|
| Nozzle size                              | Nozzle type               |
| 0.2 / 0.25 / 0.3 / 0.4 / 0.6 / 0.8 / 1.0 | Brass / Steel / Specialty |

| SwitchFly 7                              |                           |
|------------------------------------------|---------------------------|
| Nozzle size                              | Nozzle type               |
| 0.2 / 0.25 / 0.3 / 0.4 / 0.6 / 0.8 / 1.0 | Brass / Steel / Specialty |

# Rhino Tools Log

Spindles, lasers, knives and your own tools

## Hot Joe

| Voltage                    | ESC rating                 | Motor size | kV rating | Max RPM |
|----------------------------|----------------------------|------------|-----------|---------|
| <b>12 / 24 / 36 / 48 V</b> | <b>20 / 30 / 50 / 80 A</b> |            |           |         |
|                            |                            |            |           |         |

## Hot Joe (spare)

| Voltage                    | ESC rating                 | Motor size | kV rating | Max RPM |
|----------------------------|----------------------------|------------|-----------|---------|
| <b>12 / 24 / 36 / 48 V</b> | <b>20 / 30 / 50 / 80 A</b> |            |           |         |
|                            |                            |            |           |         |

## Light Saber

| Voltage                    | Wattage | Diode type | Wavelength | Focal distance |
|----------------------------|---------|------------|------------|----------------|
| <b>12 / 24 / 36 / 48 V</b> |         |            |            |                |
|                            |         |            |            |                |

## Light Saber (spare)

| Voltage                    | Wattage | Diode type | Wavelength | Focal distance |
|----------------------------|---------|------------|------------|----------------|
| <b>12 / 24 / 36 / 48 V</b> |         |            |            |                |
|                            |         |            |            |                |

## Drag Knife

| Blade holder | Blade angle | Blade offset | Cut depth | Notes |
|--------------|-------------|--------------|-----------|-------|
|              |             |              |           |       |
|              |             |              |           |       |

Name:

|  |  |  |  |  |
|--|--|--|--|--|
|  |  |  |  |  |
|--|--|--|--|--|

Name:

|  |  |  |  |  |
|--|--|--|--|--|
|  |  |  |  |  |
|--|--|--|--|--|

Name:

|  |  |  |  |  |
|--|--|--|--|--|
|  |  |  |  |  |
|--|--|--|--|--|

## Notes

[illegible]

## PART 3

# Software and Operation

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Installing the software, changing tools, running jobs and the Tool Manager portal.

---

How the software fits together

---

Install and first start

---

Swapping tools

---

Running a job

---

Pause, resume and cancel

---

The Tool Manager portal

---

Pins, outputs and system hardware

---

Admin lists

---

Console commands

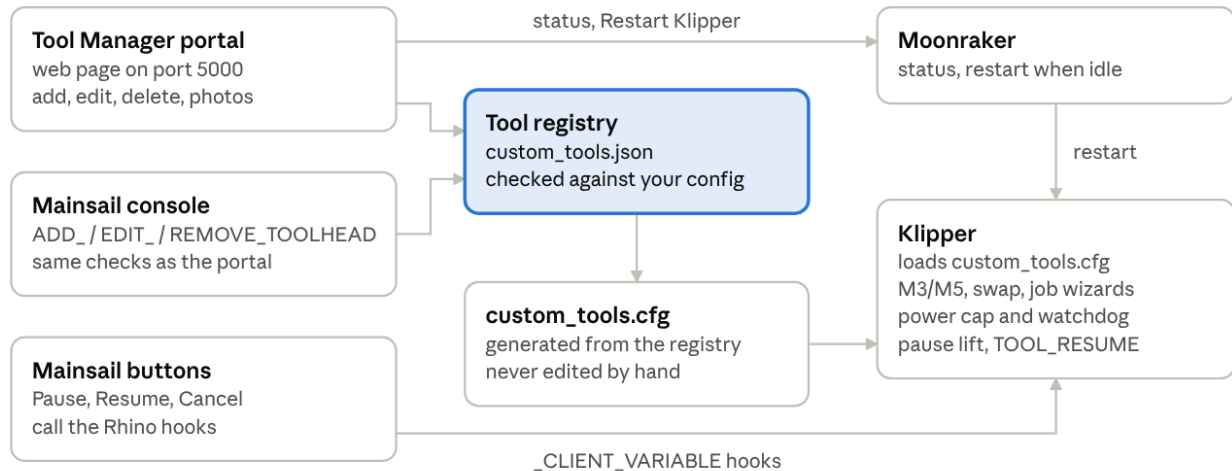
---

Safety systems

---

# How the software fits together

Every tool change goes through the registry and goes live when Klipper restarts



*How the Rhino tool manager connects to Mainsail and Klipper*

You change tools in the portal or the console; both write the same registry, which generates `custom_tools.cfg`. Klipper picks up the change at its next restart, which the portal requests through Moonraker only when no job is running.

# Install and first start

The Rhino software installs from a menu on the Pi, in the style of KIAUH: you upload the zip in Mainsail, type three short commands, then pick numbered options. Every option says what it is about to do and asks before changing anything, and every install backs up your config folder first.

**Before you start:** the printer idle, and a **print head (BlockOne or SwitchFly) mounted**. The first start homes the machine, and only a print head can home Z safely. You need the Pi's address (the one you open Mainsail with) and its login and password; on most Klipper installs the login is `pi`.

1. **Upload the zip.** In Mainsail open **Machine**, press **Upload** and choose `rhino-config-<version>.zip` (for example `rhino-config-1.4.0.zip`). Check the file list shows exactly that name; if the browser renamed it (`... (1).zip`), fix the name and upload again.
2. **Open a terminal on the Pi.** On Windows open **PowerShell**, on a Mac open **Terminal**, and log in with your login and the Pi's address. Answer `yes` the first time, then type the password (it stays invisible).

```
ssh pi@192.168.1.50
```

3. **Open the Rhino menu.** Paste one line at a time:

```
cd ~/printer_data/config
```

```
unzip -o rhino-config-1.4.0.zip -d ~/rhino-1.4.0
```

```
bash ~/rhino-1.4.0/rhino.sh
```

If `unzip` is missing, run `sudo apt install -y unzip` first. The menu looks like this:

```
/=====\  
|          ~~~~ [ Rhino Tool Manager ] ~~~~          |  
|-----|  
| Installed version : not installed                    |  
| This folder has   : 1.4.0    (/home/pi/rhino-1.4.0)  |  
| Portal           : not installed                    |  
|-----|  
| 1) Preview the install      (changes nothing)        |  
| 2) Install / update from this folder                 |  
| 3) Install a new zip (unpack, preview, install)      |  
| 4) Undo an install          (pick a backup)          |  
| 5) Portal status and recent log                      |  
| 6) Restart the portal                                         |  
| |                                                         |  
| Q) Quit                                                         |  
|-----|  
\=====/  
Perform action:
```

4. **Preview: option 1.** It checks your login and config folder, whether a job is running, whether something else uses port 5000 and whether an earlier portal is installed, then lists every step without changing anything. It ends with *Dry run finished*, or with **STOPPED:** and what to do.

5. **Install: option 2**, then `y`. It asks for the password once and works through eight stages, each ending in **OK**:

- backs up the config folder to `~/rhino-backup-<date>`;
- copies the new files, never touching `variables.cfg`, `variables.save`, your positions (`myrhino/positions.cfg`) or the tool files the portal generates;
- keeps the `SAVE_CONFIG` section at the bottom of `printer.cfg`, and never doubles it;
- fixes `/home/pi` paths if your login is different;
- installs Flask from the Pi's own packages, plus Waitress when it is available;
- replaces any earlier portal service (its settings are saved in the backup), starts the portal at every boot, and checks it answers.

It finishes with the portal's address, `http://<printer address>:5000`. Afterwards the menu is always at `bash ~/rhino-manager/rhino.sh`.

6. **First start, in Mainsail**. With the print head still mounted:

- Run `FIRMWARE_RESTART` and wait for *Ready*. A red box names the file and line Klipper did not accept.
- Answer the pop-up asking which tool is mounted (or run `RESTORE_TOOL`).
- Run `CHECK_TOOLHEADS`. When the console line starting `rhino:` says OK, press **Restart now**.

7. **Check Z before the first job**. After any update, check Z (the Rhino uses `gear_ratio: 2:1` on all three Z steppers). Home with the print head, move Z 10 mm and measure it, then run `Z_TILT_ADJUST` and re-check your Z offsets.

**A good place to stop** is after the preview (nothing has changed) or after the first start. Avoid stopping between the install and the first start.

**Updating:** upload the new zip, open the menu, choose **3** and pick it. It unpacks the zip, shows the preview and asks whether to install; answer `y`, then do the first start again. Your tools, photos and maintenance history are kept. **Undoing:** option **4** lists your backups (the oldest is your setup from before the first install), previews the restore and asks first; anything added since is moved aside to `~/rhino-undone-<date>`, not deleted.

**Never home Z with the laser, spindle or drag knife mounted.** Homing Z brings the bed up to the tool. Their macros refuse to, and jobs end with the bed lowered to the safe park height so the machine can always home X and Y after a restart.

The portal has no login. Anyone who can reach port 5000 can add or remove tools, but it cannot run G-code and refuses to restart Klipper during a job. Keep it on your home network, or start it with `--host 127.0.0.1` and reach it through an SSH tunnel.

## Swapping tools

---

Always change tools through the swap wizard. Klipper's safety checks (laser lockout, spindle lockout, job/tool match) all trust the slot the wizard records.

1. Run `START_TOOL_SWAP` (or press its macro button). A pop-up lists every tool; pick the one you are installing.
2. The machine switches off all tool power, the heaters and the part fan, then waits until the hotend **and the bed** are below 40 °C. After a hot print this can take several minutes.
3. It parks the head at the swap position ( `swap_x` / `swap_y` in `myrhino/positions.cfg` ), disables the print head's extruder motor and shows **Swap the toolhead**.
4. Unplug the old tool's umbilical at the top rear of the frame. Undo the knurled nut, remove the M3×50 bolt and lift the tool off the carriage.
5. Slide the new tool down into the carriage's V-groove until it seats, then refit the bolt and nut snug.
6. Plug the new tool's umbilical fully home and route the cable clear of the gantry.
7. Confirm in the pop-up. The wizard checks the thermistor reading on pins 3–4; if it is open it shows **Toolhead not detected** – reseal the umbilical and press **Check again**.
8. The new slot is saved to `variables.save` , so it survives a restart.

Tool swaps are refused while a job is running or paused. All tool power is switched off before the swap starts.

**After any Klipper restart**, a pop-up appears about 8 seconds later: *Klipper restarted – which toolhead is mounted?* Press **Yes** if the tool shown is the one on the machine, or **No – run the swap wizard** if it is not. Until you answer, the tool-mounted safety gates stay locked. You can bring the question back any time with `RESTORE_TOOL` .

## Running a job

Print jobs start from the sliced file. Every other tool starts from a guided setup that walks you through safety checks and setting the work zero, then starts the file for you.

**Pick the material first** (non-print tools): `SET_TOOLHEAD TOOLHEAD=LightSaber MATERIAL=PLYWOOD`. This applies the material's feed, power, acceleration and Z offset, and the setup wizards read it.

### 3D printing (BlockOne, SwitchFly, print-head clones)

1. Slice with the Rhino profile. Its start G-code calls `START_JOB TOOLHEAD="SwitchFly" MATERIAL="PLA" NOZZLE_SIZE=0.4 EXTRUDER=0`.
2. Upload and print from Mainsail as usual.
3. `START_JOB` refuses to run if the file's tool does not match the mounted tool, then homes and shows a confirmation pop-up. It applies the nozzle height saved for that head by the paper test (`NOZZLE_HEIGHT_CALIBRATE`, run once when a head is mounted - never by the job), and keeps it when the file is released.
4. `END_PRINT` switches off whatever tool is active and runs the staged `BED_COOLDOWN`.

Filament change mid-print: `FILAMENT_CHANGE` pauses, walks you through unload, load and purge, then resumes. `FILAMENT_CHANGE`, `PURGE` and `PREHEAT` only run with a print head mounted (they heat the nozzle), and filament change needs the printer homed.

### Setting Z zero (laser, spindle, drag knife, tools you added)

Every non-print job sets its Z zero by eye in the same **Set Z zero** window (the `SET_Z_ZERO` macro, also a Mainsail button). The job setups open it for you.

- **Bed up / Bed down** 10, 1 and 0.1 mm move the bed; *up* brings the material toward the tool. The window shows the bed height and stops at the travel limits (never lower than Z 380, `zero_max_z` in `myrhino/positions.cfg`).
- Spindle, knife and added tools: bring the material up until the tool tip just touches its top surface. LightSaber: bring it to the focus point - the smallest, sharpest dot - using **Focus dot ON (1%)** with laser glasses on.
- **SET – Z zero here** makes that bed height G-code Z0 (it is saved, and nothing moves). The job then runs with Z0 there; the laser runs at Z0 with no extra focus offset.

**After a restart** Klipper does not know where the bed is. If the bed was left parked (every job and swap ends with it lowered), the window first asks **Bed position – Is the bed still down where it was parked?** **Yes** takes the saved park height as Z and homes X/Y; choose **No** if the bed was moved by hand with the power off, then home Z with a print head. Choosing a material, or a job's own `START_JOB` line, never changes the Z zero you set.

**Homing.** Z is only homed with a print head mounted and confirmed - the probe is on the print heads. Any other way of homing Z (Mainsail's Home buttons, `G28`, the old `HOME` macro) is refused with a laser, spindle, knife or added tool mounted, or before the *which toolhead is mounted?* question is answered after a restart. `G28 X Y` always works. When X/Y are homed while Z isn't, Klipper first lowers the bed 10 mm; if the bed's park height was saved, it is updated to match.

## Laser (LightSaber)

Run `LASER_JOB_SETUP FILE=<name>.gcode`, then follow the pop-ups:

1. **Laser setup** – confirm ventilation, glasses, no PVC/vinyl, material flat, extinguisher nearby.
2. **Set Z zero** – move the bed to the focus point and press **SET**; a low-power test mark follows.
3. **Check the test mark** – **Yes** to continue, **No – set focus again** to go back.
4. **Ready to cut** – add or remove passes, then **Start the cut**.

## CNC (HotJoe)

Run `CNC_JOB_SETUP FILE=<name>.gcode`:

1. **CNC setup** – safety confirmations.
2. **Set X/Y zero** – jog to the corner, then **Set X/Y zero here**.
3. **Set Z zero** – bring the material up to the bit tip and press **SET**.
4. The spindle relay switches on and the ESC ramps to 40 %, then to your starting throttle.
5. **Ready to cut** – nudge the throttle  $\pm 5$  %, then **Start the cut**.

## Drag knife (DragKnife)

Run `DRAGKNIFE_JOB_SETUP FILE=<name>.gcode`: verify the material and blade dial, set Z zero on the surface in the **Set Z zero** window, cut a test mark, then **Cut looks good – continue** (or **Cut is wrong – help me adjust** for blade-dial help) and **Start the cut**.

## Tools you added (slot 6+)

Run `T00L_JOB_SETUP FILE=<name>.gcode`:

1. **<tool> setup** – the checklist you wrote in the portal; **All checked – set work zero**.
2. **Set Work Zero** – **Set X/Y zero here**, then **Set Z zero** (the Set Z zero window; the material's Z offset is added on top); **Continue**.
3. **Ready to start** – **Test pulse** fires briefly at low power; then **Start – tool power ON** (power on for the whole file) or **Start – file controls power** (the file's own M3 / M5 switch it). Passive tools just show **Start**.

M3 *S<value>* / M5 in any file control whichever tool is mounted: the laser, the spindle or your powered tool. *S* is scaled against that tool's *S value for full power* and never exceeds its power cap.

## Slicer and CAM settings

**OrcaSlicer (printing)**. Make one printer profile per print head (BlockOne, SwitchFly, and any print head you add). In each profile, **Printer settings > Machine G-code > Machine start G-code** calls `START_JOB`. Type the tool's name once; OrcaSlicer fills in the rest from the filament and nozzle chosen when you slice:

```
START_JOB TOOLHEAD="SwitchFly" MATERIAL="{filament_type[initial_extruder]}" NOZZLE_SIZE=
{nozzle_diameter[initial_extruder]} EXTRUDER={initial_extruder}
```

| START_JOB parameter | OrcaSlicer placeholder              | Becomes                                      |
|---------------------|-------------------------------------|----------------------------------------------|
| TOOLHEAD            | typed in each printer profile       | SwitchFly – must match the mounted tool      |
| MATERIAL            | {filament_type[initial_extruder]}   | PLA, PETG ... – the filament profile's type  |
| NOZZLE_SIZE         | {nozzle_diameter[initial_extruder]} | 0.4 – the printer profile's nozzle           |
| EXTRUDER            | {initial_extruder}                  | 0 or 1 – the first filament used (SwitchFly) |

Together they pick the tool's material profile, such as `PLA_0_4`. **Filament type** (in the filament profile's settings) must be one of the tool's materials – PLA, PETG, ABS or ASA for BlockOne and SwitchFly; a type such as *PLA-CF* stops the job at the start with *Material PLA-CF\_0\_4 not defined for toolhead BlockOne*. End the file (**Machine end G-code**) with `END_PRINT`.

**LaserWeb 4 (laser)**. Use these machine settings. They replace the older ones in the repository's *Laserweb Setup* note, which drove pins (`LASERPOWER_pin`, `VACPUMP_pin`) that no longer exist; going through `M3` / `M5` keeps the laser lockout, the pause handling and the power scaling.

| LaserWeb field  | Value                                                                                                        |
|-----------------|--------------------------------------------------------------------------------------------------------------|
| Gcode Start     | <code>G21</code> and <code>G90</code> – homing, focus and work zero are done by <code>LASER_JOB_SETUP</code> |
| Gcode End       | <code>M5</code>                                                                                              |
| Gcode Homing    | leave blank                                                                                                  |
| Tool On         | leave blank                                                                                                  |
| Tool Off        | <code>M5</code>                                                                                              |
| Laser Intensity | <code>M3 S</code>                                                                                            |
| PWM MIN S VALUE | <code>0</code>                                                                                               |
| PWM MAX S VALUE | <code>1000</code> (matches <code>laser_s_max</code> in <code>variables.cfg</code> )                          |

**Drag-knife files** must not contain `M3`: it is refused while the DragKnife is mounted. Use a separate LaserWeb profile with *Laser Intensity* and *Tool Off* left blank.

**CNC files**. Spindle speed is set in `CNC_JOB_SETUP` (the throttle buttons), not by `M3 S`. An `M3` in the file leaves a running spindle alone; `M5` at the end stops it.

## Pause, resume and cancel

Pause and cancel work from Mainsail's own buttons for every tool. Resume works from Mainsail's button only for print jobs; laser, CNC, drag-knife and added-tool jobs resume with the **Resume job** button in the Rhino pop-up (or `TOOL_RESUME`).

Mainsail's Resume refuses whenever the extruder is cold ("not hot enough"), and with a non-print tool mounted it always is. `TOOL_RESUME` does the same job without that check, and adds the steps a cutting tool needs.

|                  | Print head                     | Laser                                                              | CNC spindle                                                                | Drag knife / passive            | Powered tool you added                                   |
|------------------|--------------------------------|--------------------------------------------------------------------|----------------------------------------------------------------------------|---------------------------------|----------------------------------------------------------|
| On <b>Pause</b>  | Mainsail parks as usual        | Lift 10 mm, laser off, park                                        | Lift 10 mm while spinning, spindle off, park                               | Lift 10 mm, park                | Lift 10 mm, power off, park                              |
| Pop-up           | None                           | <b>&lt;tool&gt; job paused</b> with <b>Resume job / Cancel job</b> | same                                                                       | same                            | same                                                     |
| Resume with      | Mainsail <b>Resume</b>         | <b>Resume job</b> or <code>TOOL_RESUME</code>                      | same                                                                       | same                            | same                                                     |
| On resume        | Mainsail as usual              | Back over the spot, down slowly, laser back to its level           | Back over the spot, spindle ramps 40 % → its speed (4 s), then down slowly | Back over the spot, down slowly | Back over the spot, down slowly, power back to its level |
| On <b>Cancel</b> | Tool off, lift 10 mm, home X/Y | same                                                               | same                                                                       | same                            | same                                                     |

**How the resume moves.** The head travels to the paused X/Y at its lifted height, drops to 2 mm above the paused height, then makes the last 2 mm at 5 mm/s. Power comes back only once the head is on the paused spot — never at the park corner.

**If Mainsail's Resume is pressed by mistake** on a non-print job, nothing moves: the Rhino pop-up comes back and the console says to use `TOOL_RESUME`.

**Refusals.** `TOOL_RESUME` refuses if nothing is paused, or if the recorded tool differs from the one the job paused with. On a print job it simply hands over to Mainsail's Resume.

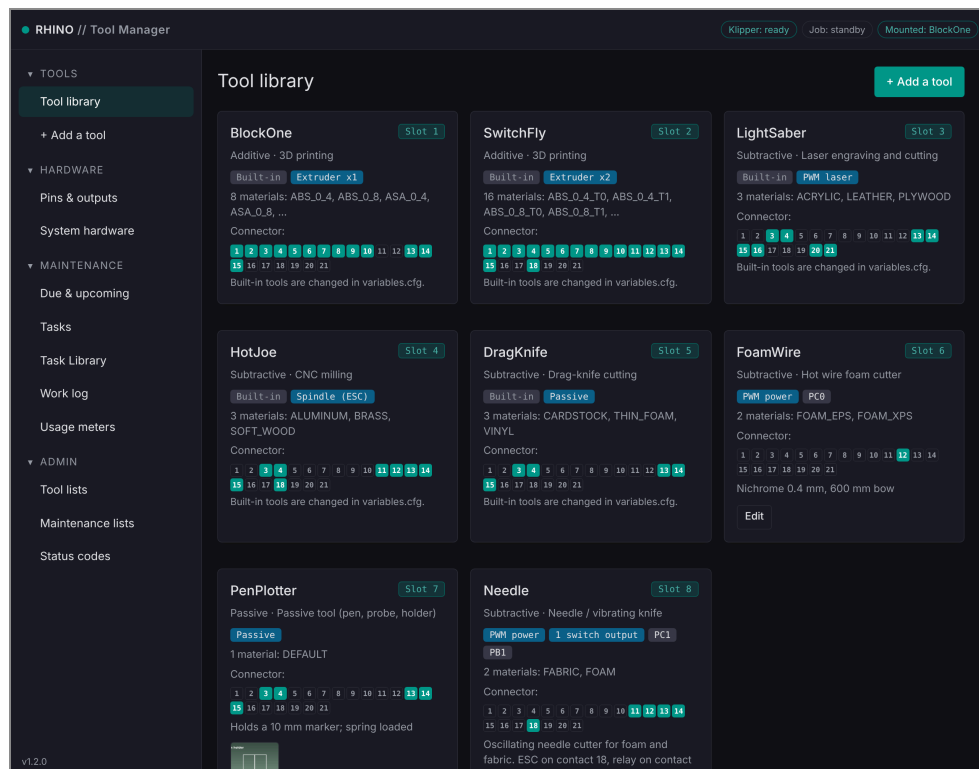
The 10 mm lift is set by `variable_lift` in `_RHINO_PAUSE_HOOK` (`tools/tool_power.cfg`); it is reduced automatically near the top of Z travel.

# The Tool Manager portal

Open `http://<printer address>:5000`. The portal adds, edits, renames and deletes your own tools, records which umbilical contacts each one uses, sets up system hardware, tracks maintenance, and restarts Klipper when changes need loading. The sidebar groups it into **Tools**, **Hardware**, **Maintenance** and **Admin**; click a group's name to fold it away. The screenshots come from a test run with three example tools (FoamWire, PenPlotter, Needle), three pieces of system hardware and placeholder photos. Built-in tools are read-only here; your tools have an **Edit** button.

## Tool library

Every tool is a card: slot, category and kind, outputs, materials, notes and photos. The row of numbered boxes is the tool's pin map: the umbilical contacts it uses are filled in, as on the tool pages in Part 2. The chips at the top right show Klipper's state, the job state and the mounted tool. Built-in tools are read-only here; your tools have an **Edit** button.



Tool library with the built-in and added tools

## Adding a tool

Press **+ Add a tool**. The wizard's steps depend on the kind of tool: powered tools get all seven, passive tools get a Connector step instead of Outputs and Limits, print-head clones get Hardware. Every kind gets Controls and Macros.

**Step 1 · Tool** – say whether the tool is **Additive** (adds material: print heads, paste or glue dispensers), **Subtractive** (removes or cuts material: lasers, spindles, hot wires, knives) or **Passive** (pens, probes, cameras, holders). Then pick the kind of tool within it, type a name and optional notes. The kind fills in safe starting values; the list of kinds is yours to edit under **Admin > Tool lists**. The name is what G-code and slicers use: 2–24 letters, digits or `_`, starting with a letter.

**Add a tool** [X]

1. Tool 2. Outputs 3. Limits 4. Materials 5. Controls 6. Macros 7. Review

WHAT KIND OF TOOL IS IT?

**Additive**  
Adds material: print heads, paste or glue dispensers.

**Subtractive**  
Removes or cuts material: lasers, spindles, hot wires, knives.

**Passive**  
Does not make or cut anything: pens, probes, cameras, holders.

WHICH SUBTRACTIVE TOOL?

Needle / vibrating knife

Oscillating needle or knife driven by a PWM output.

NAME

Needle

2-24 characters: letters, digits and `_`, starting with a letter. This is the name used in G-code and slicers.

NOTES (OPTIONAL)

Oscillating needle cutter for foam and fabric. ESC on contact 18, relay on contact 11.

Wiring, supplier, settings that worked... shown on the tool card. Not sent to Klipper.

Cancel Next

Step 1: category, kind, name and notes

Mistakes are caught before you move on, with the rule spelled out:

**Add a tool** [X]

1. Tool 2. Outputs 3. Limits 4. Materials 5. Controls 6. Macros 7. Review

WHAT KIND OF TOOL IS IT?

**Additive**  
Adds material: print heads, paste or glue dispensers.

**Subtractive**  
Removes or cuts material: lasers, spindles, hot wires, knives.

**Passive**  
Does not make or cut anything: pens, probes, cameras, holders.

WHICH SUBTRACTIVE TOOL?

Needle / vibrating knife

Oscillating needle or knife driven by a PWM output.

NAME

2 needle

2-24 characters: letters, digits and `_`, starting with a letter. This is the name used in G-code and slicers.

NOTES (OPTIONAL)

Wiring, supplier, settings that worked... shown on the tool card. Not sent to Klipper.

Name must start with a letter, use only letters, digits and `_` and be 2-24 characters.

Cancel Next

A name that breaks the rules is refused at step 1

**Step 2 · Outputs** – one row per output the tool needs. Most tools have a main power feed and a signal or two, so press + **Add an output** for each. Every row has a job and a source:

- **Main power level (PWM)** – the output M3 S sets, scaled and capped. Exactly one per tool.
- **Switch on with the tool** – an on/off output that turns on when the tool's power does and off with it: a relay, a driver enable line, a vacuum.

The source is a **contact of the umbilical** that a Klipper output already drives (contact 18, the spindle ESC signal, for example), a **new output on a spare pin**, a piece of **system hardware**, or another output in the config. Below the outputs, tick the other contacts the tool uses – returns, supplies, the probe – so its pin map is complete. Contacts its outputs drive are ticked for you.

**Add a tool** [X]

1. Tool 2. **Outputs** 3. Limits 4. Materials 5. Controls 6. Macros 7. Review

Which umbilical contacts power and switch this tool. Most tools have a main power feed and a signal or two: add one row per output. The main power level is what M3 S sets; switch outputs turn on with the tool and off with it (an enable line, a relay, a vacuum).

**OUTPUTS**

|                                |                                |        |
|--------------------------------|--------------------------------|--------|
| Main power level (PWM) ▾       | Contact 18 · PB11 · SPINDL ▾   | Remove |
| Switch on with the tool (on) ▾ | Contact 11 · PA1 · Spindle_r ▾ | Remove |

+ Add an output

**CONTACTS THIS TOOL USES**

|                                                              |                                                                           |
|--------------------------------------------------------------|---------------------------------------------------------------------------|
| <input type="checkbox"/> 1 Hotend heater +                   | <input type="checkbox"/> 2 Hotend heater -                                |
| <input type="checkbox"/> 3 Hotend thermistor                 | <input type="checkbox"/> 4 Hotend thermistor                              |
| <input type="checkbox"/> 5 Hotend fan +                      | <input type="checkbox"/> 6 Hotend fan -                                   |
| <input type="checkbox"/> 7 Stepper A1                        | <input type="checkbox"/> 8 Stepper A2                                     |
| <input type="checkbox"/> 9 Stepper B1                        | <input type="checkbox"/> 10 Stepper B2                                    |
| <input checked="" type="checkbox"/> 11 24 V power activate + | <input checked="" type="checkbox"/> 12 24 V power activate -              |
| <input checked="" type="checkbox"/> 13 Probe -               | <input checked="" type="checkbox"/> 14 Probe +                            |
| <input type="checkbox"/> 15 Probe signal                     | <input type="checkbox"/> 16 Laser PWM (5 V)                               |
| <input type="checkbox"/> 17 Spare                            | <input checked="" type="checkbox"/> 18 PWM signal 24 V - (ESC signal t... |
| <input type="checkbox"/> 19 Spare                            | <input type="checkbox"/> 20 12 V power activate +                         |
| <input type="checkbox"/> 21 12 V power activate -            |                                                                           |

Contacts its outputs and controls drive are ticked for you. Tick the returns, supplies and signal lines it also uses.

Back Cancel **Next**

Step 2: a PWM signal on contact 18, a relay on contact 11, and the contacts the tool uses

**Step 3 · Limits** – the power cap (a hard ceiling whatever the G-code asks), minimum and idle levels, the watchdog's maximum on-time, and the S value that means full power. Sharing the ESC channel sets the idle level to its neutral 30%.

**Add a tool** [Close]

1. Tool 2. Outputs **3. Limits** 4. Materials 5. Controls 6. Macros 7. Review

These limits protect the tool. Percentages are of the output's full range. Start low and raise them while watching the tool.

|             |                 |              |
|-------------|-----------------|--------------|
| POWER CAP % | MINIMUM POWER % | IDLE LEVEL % |
| 100         | 0               | 30           |

Hard ceiling, whatever the G-code asks for.

Output at the lowest non-zero request.

Output when the tool is OFF. Leave 0 unless the driver needs a neutral signal.

|                 |                        |
|-----------------|------------------------|
| MAX TIME ON (S) | S VALUE FOR FULL POWER |
| 900             | 1000                   |

Watchdog: output is cut after this long without a new power command.

G-code M3 S<value> is scaled against this (1000 = S1000 is 100%).

[Back] [Cancel] [Next]

Step 3: power limits and watchdog

**Step 4 · Materials** – the presets you pick when starting a job (power, feed, Z offset), and the pre-job checklist shown before every job with this tool. The material name box suggests the names listed under **Admin > Tool lists**.

**Add a tool** [Close]

1. Tool 2. Outputs 3. Limits **4. Materials** 5. Controls 6. Macros 7. Review

Materials are the presets you pick when starting a job: what to cut or draw, and how fast.

| MATERIAL | POWER % | FEED MM/MIN | Z OFFSET MM |        |
|----------|---------|-------------|-------------|--------|
| FOAM     | 60      | 600         | 0           | Remove |
| FABRIC   | 40      | 500         | 0           | Remove |

+ Add material

PRE-JOB CHECKLIST (ONE ITEM PER LINE)

Needle or blade is firmly seated  
Material is clamped flat  
Hands and cables clear of the cutting path

Shown as a confirmation list before every job with this tool.

[Back] [Cancel] [Next]

Step 4: materials and checklist

**Step 5 · Controls** – extra outputs and inputs the tool brings that are not simply switched with it: a fan with its own speed, a servo, a lid switch. Press **+ Add a control** and answer the guide's questions: you describe what the control does, and the portal works out which Klipper section that needs and shows the exact text it will write.

**Add a tool** ×

1. Tool 2. Outputs 3. Limits 4. Materials **5. Controls** 6. Macros 7. Review

Extra outputs and inputs this tool brings: a vacuum relay, a cooling fan, a servo, a lid switch... Describe what each one does and the portal works out the Klipper section for it. The tool's main power output is separate (the Hardware step).

**needle\_vacuum**  
Vacuum pump relay - on/off - PC1 (inverted) - off when made safe -  
macros: NEEDLE\_VACUUM\_ON, NEEDLE\_VACUUM\_OFF Edit Delete

**needle\_lid**  
Lid switch - input - PB1 (inverted) - on trigger: pause the job Edit Delete

+ Add a control

Back Cancel Next

*Step 5: the tool's extra controls*

The control guide asks, in order:

1. **Name and purpose.** Lower-case letters, digits and `_`, such as `needle_vacuum`. G-code uses the name (`SET_PIN PIN=needle_vacuum`), so it must not match anything else in your config.
2. **Which way does the signal go?** Rhino controls something on the tool, or the tool sends a signal to Rhino.
3. **What should Rhino do with it?** Pick the closest match:

| You pick                     | Klipper section                                  | Typical use                                   |
|------------------------------|--------------------------------------------------|-----------------------------------------------|
| Switch it on and off         | <code>output_pin</code>                          | Relay, solenoid, vacuum pump, light, valve    |
| Set a level                  | <code>output_pin</code> , <code>pwm: True</code> | Driver board, LED dimmer, small heater        |
| Run it like a fan            | <code>fan_generic</code>                         | Fan or blower set by speed, with a kick-start |
| Run it while a heater is hot | <code>heater_fan</code>                          | Hotend or heat-break cooling                  |
| Move it to an angle          | <code>servo</code>                               | Gate, pen lift, deflector                     |
| The tool sends a signal      | <code>gcode_button</code>                        | Lid switch, limit switch, foot pedal, sensor  |

4. **Which pin?** Only free tool-connector pins are listed. If `myrhino/umbilical.json` says what each pin is (`"type": "logic"` or `"type": "mosfet"`), pins that cannot do the job are greyed out.
5. **How does it behave?** Active-low wiring and power-up state for a switch; maximum level and PWM cycle time for a level or a fan; heater and turn-on temperature for a heater fan; pulse widths for a servo.
6. **Safety and shortcuts.** Switch it off whenever the machine is made safe (end of job, cancel, tool swap, `EMERGENCY_STOP`; ticked by default); switch it off or leave it on if Klipper shuts down; and make `NAME_ON` / `NAME_OFF` (or `NAME_SET`, `NAME_ANGLE`) macros, optionally only while this tool is mounted.

The top of the guide: name, direction, what it does and the pin.

New control - Needle

CONTROL NAME

WHAT IS IT?

needle\_vacuum

Vacuum pump relay

Lower-case letters, digits and \_ Used in G-code, e.g. SET\_PIN  
PIN=needle\_vacuum

1. WHICH WAY DOES THE SIGNAL GO?

☒ Rhino controls something on the tool  
A relay, fan, light, servo, driver...

☐ The tool sends a signal to Rhino  
A button, door switch, limit switch, sensor...

2. WHAT SHOULD RHINO DO WITH IT?

☒ Switch it on and off  
Relay, solenoid, vacuum pump, light, valve - anything that is simply on or off.

☐ Set a level (power, brightness, speed)  
A driver board, LED dimmer or heater that takes a PWM level from 0 to 100%.

☐ Run it like a fan  
A fan or blower you set by speed %, with a kick-start to get it spinning.

☐ Run it automatically while a heater is hot  
Hotend or heat-break cooling: Kripper turns it on above a temperature by itself.

☐ Move it to an angle (hobby servo)  
A servo that swings a gate, lifts a pen or flips a deflector.

3. WHICH PIN ON THE TOOL CONNECTOR?

PIN

PC1

Not described in umbilical.json

4. HOW DOES IT BEHAVE?

☒ On means the pin is pulled LOW (active-low)  
Common on relay boards that switch when their input goes to ground.

AT POWER-UP

Off

Cancel

Add control

*A new control: name, direction and what it does*

The bottom of the guide: safety options, the macros it makes, and the exact text written to `custom_tools.cfg`. **Add control** saves only a valid entry; a name already in use, a taken pin or two controls on one pin are refused, with the reason in the preview box.

**New control - Needle**

A servo that swings a gate, lifts a pen or flips a deflector.

3. WHICH PIN ON THE TOOL CONNECTOR?

PIN

PC1

Not described in umbilical.json

4. HOW DOES IT BEHAVE?

☒ On means the pin is pulled LOW (active-low)  
Common on relay boards that switch when their input goes to ground.

AT POWER-UP

Off

5. SAFETY AND SHORTCUTS

☒ Switch it off whenever the machine is made safe  
End of job, cancel, tool swap and emergency stop (TOOL\_SAFE\_OFF).

IF KLIPPER SHUTS DOWN OR LOSES THE BOARD

☒ Switch it off (recommended)

☐ Leave it on  
Only for things that are safe left running, such as cooling.

☒ Make macros for it: NEEDLE\_VACUUM\_ON, NEEDLE\_VACUUM\_OFF  
Shown on the Macros tab; use them in job files or the console.

☒ Those macros only switch it on while this tool is mounted

☒ Switching it off always works.

WRITTEN TO CUSTOM\_TOOLS.CFG

```
[output_pin needle_vacuum]
# control of toolhead Needle: Vacuum pump relay
pin: PC1
value: 0
shutdown_value: 0

[gcode_macro NEEDLE_VACUUM_ON]
# made by control needle_vacuum of Needle
description: Needle: needle_vacuum on
gcode:
{% if printer["gcode_macro SWAP_TOOL"].current_tool|int != 8 %}
{ action_raise_error("NEEDLE_VACUUM_ON: Needle is not mounted (slot 8). Swap tools first.") }
{% endif %}
```

Cancel Add control

*Safety options and the text written to `custom_tools.cfg`*

For a signal from the tool, the guide asks how it is wired. A switch to ground turns on the pin's pull-up; a sensor that drives the line needs none. Then it asks what happens when the signal triggers and when it is released: show a message, pause the job (only while one is running), emergency stop, or run a command or macro.

**New control - Needle**

☐ Rhino controls something on the tool  
A relay, fan, light, servo, driver...

☒ The tool sends a signal to Rhino  
A button, door switch, limit switch, sensor...

2. WHICH PIN ON THE TOOL CONNECTOR?

PIN  
PB1

Not described in umbilical.json

3. HOW DOES IT BEHAVE?

WHAT IS CONNECTED?

☒ A switch or button that connects the pin to ground  
The pin's pull-up is turned on, so it reads 'off' until pressed.

☐ A sensor or board that drives the line itself  
No pull-up.

☐ Swap pressed and released  
Tick if it reports the opposite of what you expect (normally-closed switches).

WHEN IT TRIGGERS  
Pause the job

WHEN IT IS RELEASED  
Show a message

MESSAGE  
lid closed

WRITTEN TO custom\_tools.cfg

```
[gcode_button needle_lid]
# control of toolhead Needle: Lid switch
pin: ^!PB1
press gcode:
(% if printer.print_stats.state == "printing" %)
RESPOND TYPE:error MSG="needle_lid triggered - pausing the job"
PAUSE
(% endif %)
release gcode:
RESPOND MSG="needle_lid: lid closed"
```

needle\_lid: make sure PB1 is a plain logic pin, not a MOSFET power output (describe it in myrhino/umbilical.json and the portal checks this for you).

Cancel Add control

*A lid switch that pauses the job when opened*

**Step 6 · Macros** – G-code macros that belong to this tool: a test cut, a purge, a park move. They are written to `custom_tools.cfg` with the tool and removed with it. Macros your controls made are listed underneath.

**Add a tool**

1. Tool 2. Outputs 3. Limits 4. Materials 5. Controls 6. Macros 7. Review

G-code macros that belong to this tool: a purge, a test cut, a park move... They are written to custom\_tools.cfg with the tool and removed with it.

**NEEDLE\_TEST\_CUT**  
Run the needle for 2 s with the vacuum on - only while mounted

Edit Delete

**Made by your controls**  
NEEDLE\_VACUUM\_ON NEEDLE\_VACUUM\_OFF  
Change these on the Controls tab.

+ Add a macro

Back Cancel Next

*Step 6: the tool's macros*

Press + **Add a macro**. **Macro name**: letters, digits and `_`, saved in capitals because Klipper ignores letter case. **Purpose**: one line, shown in Mainsail's macro list. **Only run when this tool is mounted**: with another tool on the machine the macro refuses with a message. **Macro**: Klipper G-code and Jinja exactly as in a `[gcode_macro]`. The box underneath shows exactly what will be written.

**Refused**: a name used by any macro or Klipper command (`SET_PIN`, `home`, `PAUSE`), G-code words such as `M3`, and template errors such as a missing `{% endif %}`. **Warned**: `SAVE_CONFIG`, `FIRMWARE_RESTART` or `M112` in the body, an unrecognised command (usually a typo), and text after `#` or `;`, which Klipper drops as a comment.

**New macro - Needle**

**MACRO NAME**  
needle\_test\_cut

**PURPOSE**  
Run the needle for 2 s with the vacuum on

Letters, digits and `_`. Klipper ignores case, so it is saved in capitals. Must not clash with any other macro or command. One line. Shown in Mainsail's macro list and console autocomplete.

☒ Only run when this tool is mounted (refused with a message otherwise)

**MACRO**

```
NEEDLE_VACUUM_ON
M3 S{params.S|default(300)|int}
G4 P2000
M5
NEEDLE_VACUUM_OFF
```

Klipper G-code and Jinja templates (`{% if %}`, `{params.X}`) exactly as in a `[gcode_macro]`. Checked before it is saved.

**WRITTEN TO CUSTOM\_TOOLS.CFG**

```
[gcode_macro NEEDLE_TEST_CUT]
# macro of toolhead Needle
description: Run the needle for 2 s with the vacuum on
gcode:
{% if printer["gcode_macro SWAP_TOOL"].current_tool|int != 8 %}
{ action_raise_error("NEEDLE_TEST_CUT: Needle is not mounted (slot 8). Swap tools first.") }
{% endif %}
NEEDLE_VACUUM_ON
M3 S{params.S|default(300)|int}
G4 P2000
M5
NEEDLE_VACUUM_OFF
```

Cancel Add macro

*A macro with its live preview*

**Step 7 · Review** – the portal checks everything against your live Klipper config, warns about shared outputs, and tells you the slot the tool will get. Add up to six photos here, then **Create tool**.

**Add a tool** [X]

1. Tool 2. Outputs 3. Limits 4. Materials 5. Controls 6. Macros 7. Review

Name: Needle  
Kind: Subtractive - Needle / vibrating knife  
Main power: contact 18 (PB11, SPINDLE\_SPEED)  
Switch outputs: contact 11 (PA1, Spindle\_power)  
Limits: cap 100%, min 0%, idle 30%, max 900s on, S1000 = full  
Materials: FOAM, FABRIC  
Controls: needle\_vacuum (on/off, PC1), needle\_lid (input, PB1)  
Macros: NEEDLE\_TEST\_CUT, NEEDLE\_VACUUM\_ON, NEEDLE\_VACUUM\_OFF

SPINDLE\_SPEED is also used by: HotJoe spindle ESC (idles at 0.3 = neutral). Only one of those tools is mounted at a time, but wire the tool to match.

Spindle\_power is also used by: HotJoe spindle relay.

needle\_lid: make sure PB1 is a plain logic pin, not a MOSFET power output (describe it in myrhino/umbilical.json and the portal checks this for you).

Ready: this becomes slot 8.

PHOTOS (OPTIONAL, UP TO 6)  
Choose Files 2 files

Back Cancel Create tool

*Step 7: review, with the outputs, controls and macros listed*

Press **Add maintenance tasks** on the confirmation and the portal offers the Task Book that fits the new tool (see *Task Library and Task Books* in Part 4).

**Assign the book "Needle cutter"** [X]

ASSIGN IT TO

Tool - Needle

Adds 2 task(s), linked to the library: editing a library task updates them. "Whole machine" puts each task in its own area.

Only toolheads are listed: these tasks count a tool's own use.

Cancel Assign book

*The Task Book offered for a new needle cutter*

## Restarting Klipper to load changes

A new or changed tool is saved immediately but is not live until Klipper restarts. A yellow banner lists what is waiting, from the portal or the console; press **Restart Klipper** when the machine is idle. Changes Klipper never sees – notes, photos, the pin map – are saved without asking for a restart.

### Tool changes are saved but not live yet.

Klipper has to restart to load them. The machine is idle, so it is safe now.

- edited Needle: notes

Restart Klipper

*Changes waiting; the machine is idle, so the restart is allowed*

While a job is printing or paused, **Restart Klipper** is greyed out and the banner says why, with the file name, a progress bar, time elapsed and time left, and the clock time when a restart becomes possible. Changes made during a job are saved and queued; only the restart waits, so it can never kill a job.

### Tool changes are saved but not live yet.

Klipper loads them at its next restart, which waits until the job finishes - restarting now would stop it.

- edited Needle: notes

Running: bracket\_v3.gcode

38% · 47 m 01 s elapsed · about 1 h 14 m left · restart possible at about 9:22 AM

Restart Klipper

☐ Restart automatically when this job finishes

*A print running: progress, time left and when a restart is possible*

Paused, the countdown stops and the banner shows how long the job has been paused.

### Tool changes are saved but not live yet.

Klipper loads them at its next restart, which waits until the job is finished (it is paused now) - restarting now would stop it.

- edited Needle: notes

Paused: bracket\_v3.gcode

38% · 47 m 00 s elapsed · about 1 h 15 m left · paused for 6 s - the timer is stopped

Restart Klipper

☒ Restart automatically when this job finishes

*The same job paused*

Time left comes from the slicer's estimate for 3D prints. Laser, CNC and cutter files usually carry none, so the portal works it out from file progress, the way Mainsail does, and says so.

### Tool changes are saved but not live yet.

Klipper loads them at its next restart, which waits until the job finishes - restarting now would stop it.

- edited Needle: notes

Running: sign\_v2.nc

25% · 15 m 01 s elapsed · about 44 m 41 s left · restart possible at about 8:52 AM · (estimated from file progress)

Restart Klipper

☒ Restart automatically when this job finishes

*A laser job: time left estimated from file progress*

**Restart automatically when this job finishes** is off by default. Tick it and the portal restarts Klipper 30 seconds after the job completes or is cancelled, which leaves time for `END_PRINT` or `CANCEL_PRINT` to finish. After an error it does not restart, so you can see what happened. A restart started anywhere else (Mainsail, `RESTART_FOR_TOOLS`) also clears the banner. The console commands (`ADD_TOOLHEAD` and the rest) stay refused during a job: they run a shell command that Klipper waits for, which would stall the job's G-code.

## Editing a tool

Press **Edit** on the tool's card. On the **Settings** tab you can rename it and change its power limits and materials; the legend above the limits lists its outputs and the contacts they use.

**Edit Needle**

**Settings** Controls (2) Macros (3)

NAME: Needle KIND: NEEDLE (powered)

Renaming changes the name used in G-code and slicers.

POWER LIMITS - OUTPUTS: POWER: SPINDLE\_SPEED (CONTACT 18) - SWITCH: SPINDLE\_POWER (CONTACT 11)

POWER CAP %: 100 MINIMUM %: 0 IDLE %: 30

MAX TIME ON (S): 900 S FOR FULL POWER: 1000

**MATERIALS**

| MATERIAL | POWER % | FEED MM/MIN | Z OFFSET MM |        |
|----------|---------|-------------|-------------|--------|
| FABRIC   | 40      | 500         | 0           | Remove |
| FOAM     | 60      | 600         | 0           | Remove |

+ Add material

**PRE-JOB CHECKLIST (ONE ITEM PER LINE)**

Needle or blade is firmly seated  
Material is clamped flat  
Hands and cables clear of the cutting path

**CONTACTS THIS TOOL USES**

|                                                  |                                                           |                                                            |                                                            |
|--------------------------------------------------|-----------------------------------------------------------|------------------------------------------------------------|------------------------------------------------------------|
| <input type="checkbox"/> 1 Hotend heater +       | <input type="checkbox"/> 2 Hotend heater -                | <input type="checkbox"/> 3 Hotend thermistor               | <input type="checkbox"/> 4 Hotend thermistor               |
| <input type="checkbox"/> 5 Hotend fan +          | <input type="checkbox"/> 6 Hotend fan -                   | <input type="checkbox"/> 7 Stepper A1                      | <input type="checkbox"/> 8 Stepper A2                      |
| <input type="checkbox"/> 9 Stepper B1            | <input type="checkbox"/> 10 Stepper B2                    | <input checked="" type="checkbox"/> 11 24 V power activ... | <input checked="" type="checkbox"/> 12 24 V power activ... |
| <input checked="" type="checkbox"/> 13 Probe -   | <input checked="" type="checkbox"/> 14 Probe +            | <input type="checkbox"/> 15 Probe signal                   | <input type="checkbox"/> 16 Laser PWM (5 V)                |
| <input type="checkbox"/> 17 Spare                | <input checked="" type="checkbox"/> 18 PWM signal 24 V... | <input type="checkbox"/> 19 Spare                          | <input type="checkbox"/> 20 12 V power activa...           |
| <input type="checkbox"/> 21 12 V power activa... |                                                           |                                                            |                                                            |

Contacts its outputs and controls drive are ticked for you. Tick the returns, supplies and signal lines it also uses.

**NOTES**

Delete tool Cancel Save changes

*Edit window: name, outputs, limits and materials*

Further down are the checklist, the contacts the tool uses (tick more as you wire them), notes and photos. **x** on a photo deletes it; **Choose Files** adds more. Photos are kept in `~/printer_data/rhino_data/images/`, outside the config folder, so config backups stay small.

The 'Edit Needle' window contains the following sections:

- MAX TIME ON (S)**: 900
- S FOR FULL POWER**: 1000
- MATERIALS**:
 

| MATERIAL | POWER % | FEED MM/MIN | Z OFFSET MM |        |
|----------|---------|-------------|-------------|--------|
| FABRIC   | 40      | 500         | 0           | Remove |
| FOAM     | 60      | 600         | 0           | Remove |
- + Add material**
- PRE-JOB CHECKLIST (ONE ITEM PER LINE)**:
  - Needle or blade is firmly seated
  - Material is clamped flat
  - Hands and cables clear of the cutting path
- CONTACTS THIS TOOL USES**:
 

|                                                  |                                                           |                                                            |                                                            |
|--------------------------------------------------|-----------------------------------------------------------|------------------------------------------------------------|------------------------------------------------------------|
| <input type="checkbox"/> 1 Hotend heater +       | <input type="checkbox"/> 2 Hotend heater -                | <input type="checkbox"/> 3 Hotend thermistor               | <input type="checkbox"/> 4 Hotend thermistor               |
| <input type="checkbox"/> 5 Hotend fan +          | <input type="checkbox"/> 6 Hotend fan -                   | <input type="checkbox"/> 7 Stepper A1                      | <input type="checkbox"/> 8 Stepper A2                      |
| <input type="checkbox"/> 9 Stepper B1            | <input type="checkbox"/> 10 Stepper B2                    | <input checked="" type="checkbox"/> 11 24 V power activ... | <input checked="" type="checkbox"/> 12 24 V power activ... |
| <input checked="" type="checkbox"/> 13 Probe -   | <input checked="" type="checkbox"/> 14 Probe +            | <input type="checkbox"/> 15 Probe signal                   | <input type="checkbox"/> 16 Laser PWM (5 V)                |
| <input type="checkbox"/> 17 Spare                | <input checked="" type="checkbox"/> 18 PWM signal 24 V... | <input type="checkbox"/> 19 Spare                          | <input type="checkbox"/> 20 12 V power activa...           |
| <input type="checkbox"/> 21 12 V power activa... |                                                           |                                                            |                                                            |
- NOTES**: Oscillating needle cutter for foam and fabric. ESC on contact 18, relay on contact 11.
- PHOTOS**: Two photo thumbnails with 'x' delete buttons. Below them is a 'Choose Files' button and 'No file chosen' text.
- Buttons**: Delete tool, Cancel, Save changes.

Edit window: checklist, contacts, notes and photos

The **Controls** and **Macros** tabs work exactly like steps 5 and 6 of the wizard; their changes are saved with **Save changes** and are live after the restart.

The 'Edit Needle' window, Controls tab, shows:

- Settings** | **Controls (2)** | **Macros (3)**
- Extra outputs and inputs this tool brings: a vacuum relay, a cooling fan, a servo, a lid switch... Describe what each one does and the portal works out the Klipper section for it. The tool's main power output is separate (the Hardware step).
- needle\_vacuum**: Vacuum pump relay - on/off · PC1 (inverted) · off when made safe · macros: NEEDLE\_VACUUM\_ON, NEEDLE\_VACUUM\_OFF. **Edit** **Delete**
- needle\_lid**: Lid switch - input · PB1 (inverted) · on trigger: pause the job. **Edit** **Delete**
- + Add a control**
- Buttons**: Delete tool, Cancel, Save changes.

Edit window, Controls tab

## Deleting a tool

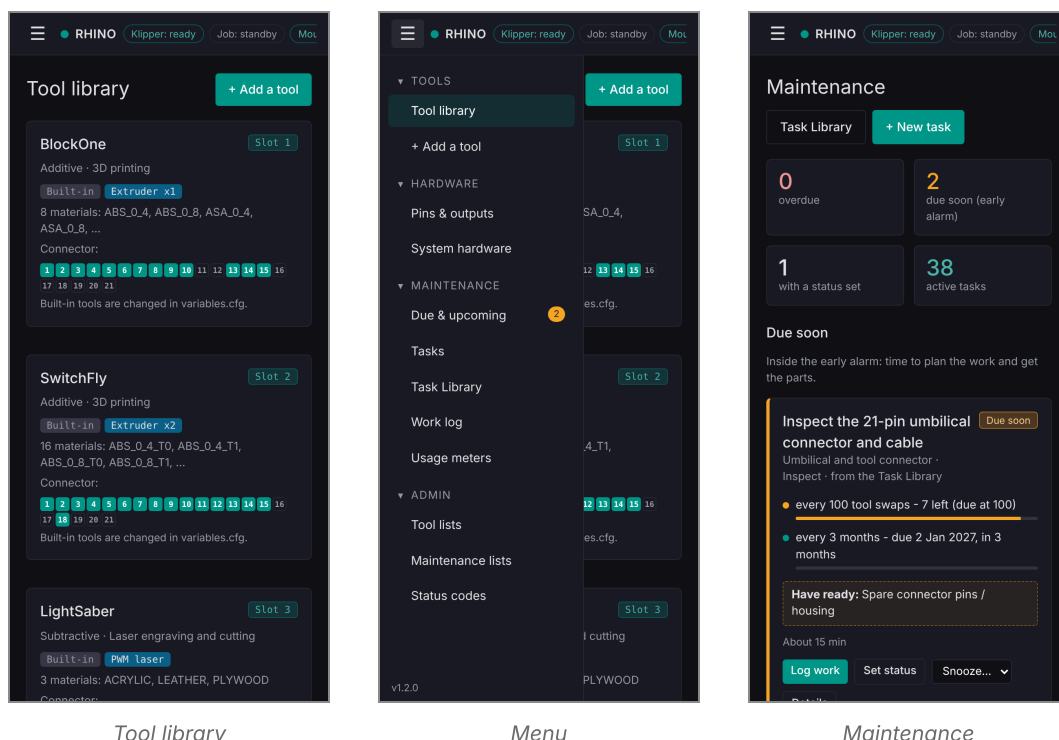
In the edit window press **Delete tool**. The button changes to **Click again to delete <name>**; press it within 4 seconds to confirm. The tool's pins are freed, its photos removed and its maintenance tasks archived. Restart Klipper afterwards.



*Two-click delete confirmation*

## On a phone

The portal works on a phone: the menu folds behind , and the status chips scroll sideways in the top bar.



*Tool library*

*Menu*

*Maintenance*

# Pins, outputs and system hardware

**Hardware > Pins & outputs** starts with the tool connector: all 21 contacts of the umbilical, the board pin behind each, what drives it in the Klipper config and which tools use it. A contact whose board pin nothing in the config uses is highlighted – it means the pin map and `printer.cfg` disagree (contact 20 here; see *Still to check on the machine*). The contacts come from the Rhino pin map; describe different wiring with "contacts" in `myrhino/umbilical.json`.

| Every contact of the tool connector, the board pin behind it, what drives it in the Klipper config and which tools use it. |           |                                                    |                                                                                     |                                                                        |
|----------------------------------------------------------------------------------------------------------------------------|-----------|----------------------------------------------------|-------------------------------------------------------------------------------------|------------------------------------------------------------------------|
| CONTACT                                                                                                                    | BOARD PIN | FUNCTION                                           | DRIVEN BY                                                                           | TOOLS USING IT                                                         |
| 1                                                                                                                          | PA2       | Hotend heater +<br>Switched output                 | tools/3dp.cfg [extruder]                                                            | BlockOne, SwitchFly                                                    |
| 2                                                                                                                          | -         | Hotend heater -<br>Return / -                      | -                                                                                   | BlockOne, SwitchFly                                                    |
| 3                                                                                                                          | PF4       | Hotend thermistor<br>Signal                        | tools/3dp.cfg [extruder]                                                            | BlockOne, SwitchFly, LightSaber, HotJoe, DragKnife, PenPlotter         |
| 4                                                                                                                          | -         | Hotend thermistor<br>Return / -                    | -                                                                                   | BlockOne, SwitchFly, LightSaber, HotJoe, DragKnife, PenPlotter         |
| 5                                                                                                                          | PE5       | Hotend fan +<br>Switched output                    | tools/3dp.cfg [heater_fan fan1]                                                     | BlockOne, SwitchFly                                                    |
| 6                                                                                                                          | -         | Hotend fan -<br>Return / -                         | -                                                                                   | BlockOne, SwitchFly                                                    |
| 7                                                                                                                          | -         | Stepper A1<br>Stepper                              | -                                                                                   | BlockOne, SwitchFly                                                    |
| 8                                                                                                                          | -         | Stepper A2<br>Stepper                              | -                                                                                   | BlockOne, SwitchFly                                                    |
| 9                                                                                                                          | -         | Stepper B1<br>Stepper                              | -                                                                                   | BlockOne, SwitchFly                                                    |
| 10                                                                                                                         | -         | Stepper B2<br>Stepper                              | -                                                                                   | BlockOne, SwitchFly                                                    |
| 11                                                                                                                         | PA1       | 24 V power activate +<br>Switched output           | Spindle_power                                                                       | SwitchFly, HotJoe, Needle                                              |
| 12                                                                                                                         | -         | 24 V power activate -<br>Return / -                | -                                                                                   | SwitchFly, HotJoe, FoamWire, Needle                                    |
| 13                                                                                                                         | -         | Probe -<br>Return / -                              | -                                                                                   | BlockOne, SwitchFly, LightSaber, HotJoe, DragKnife, Needle, PenPlotter |
| 14                                                                                                                         | -         | Probe +<br>Supply +                                | -                                                                                   | BlockOne, SwitchFly, LightSaber, HotJoe, DragKnife, Needle, PenPlotter |
| 15                                                                                                                         | PG10      | Probe signal<br>Signal                             | printer.cfg [probe]                                                                 | BlockOne, SwitchFly, LightSaber, HotJoe, DragKnife, PenPlotter         |
| 16                                                                                                                         | PB0       | Laser PWM (5 V)<br>Signal                          | SERVO_LASER                                                                         | LightSaber                                                             |
| 17                                                                                                                         | -         | Spare<br>Spare                                     | -                                                                                   | -                                                                      |
| 18                                                                                                                         | PB11      | PWM signal 24 V - (ESC<br>signal to GND)<br>Signal | SPINDLE_SPEED                                                                       | SwitchFly, HotJoe, Needle                                              |
| 19                                                                                                                         | -         | Spare<br>Spare                                     | -                                                                                   | -                                                                      |
| 20                                                                                                                         | PD12      | 12 V power activate +<br>Switched output           | -<br>Nothing in the Klipper config uses PD12 - check the wiring against printer.cfg | LightSaber                                                             |
| 21                                                                                                                         | -         | 12 V power activate -<br>Return / -                | -                                                                                   | LightSaber                                                             |

*The 21 contacts, what drives them and which tools use them*

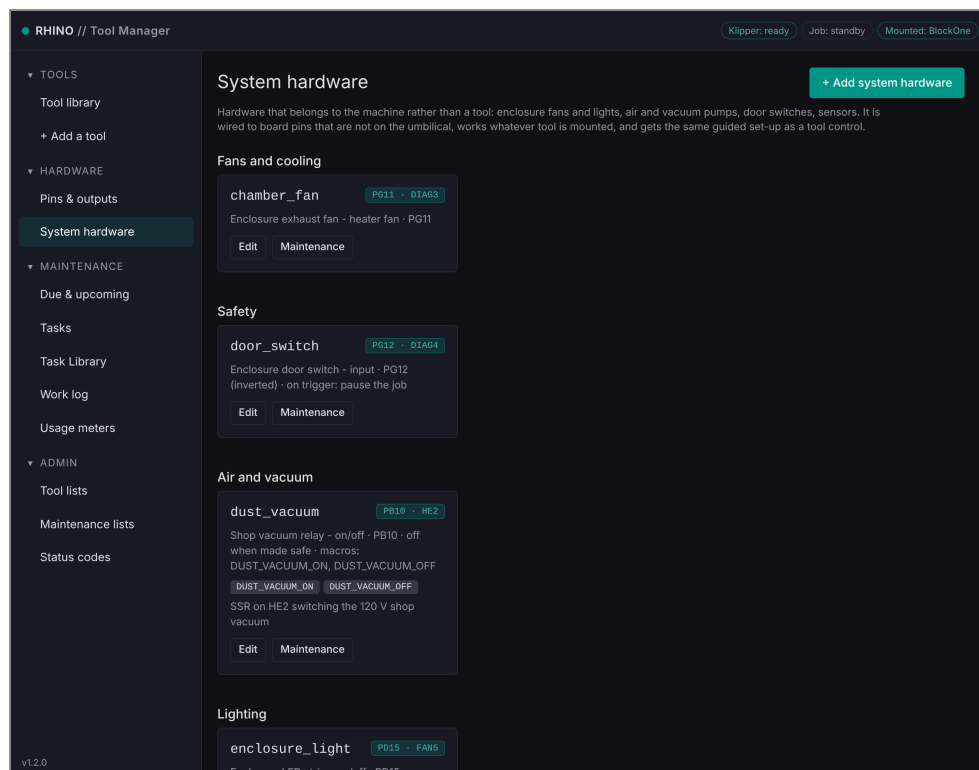
Further down: the spare pins a new tool output may claim (Free, In use with the file and section, or Reserved by a tool), the outputs tools can share with their contact and idle level, and the system hardware.

| Output channels tools can share                                           |        |                 |            |                                                     |
|---------------------------------------------------------------------------|--------|-----------------|------------|-----------------------------------------------------|
| Existing outputs a new powered tool may drive instead of using a new pin. |        |                 |            |                                                     |
| OUTPUT                                                                    | TYPE   | CONTACT         | IDLE LEVEL | ALSO USED BY                                        |
| FoamWire_PWR                                                              | PWM    | -               | 0          | -                                                   |
| LASER_INITIALIZE                                                          | PWM    | -               | 0          | LightSaber laser power rail                         |
| SERVO_LASER                                                               | PWM    | 16              | 0          | LightSaber laser intensity / SwitchFly path servo   |
| SPINDLE_SPEED                                                             | PWM    | 18              | 0.3        | HotJoe spindle ESC (idles at 0.3 = neutral)         |
| Spindle_power                                                             | On/off | 11              | 0          | HotJoe spindle relay                                |
| Umbilical_LED                                                             | PWM    | -               | 0          | umbilical LED strip data line (not a power output!) |
| dust_vacuum                                                               | On/off | system hardware | 0          | -                                                   |
| enclosure_light                                                           | On/off | system hardware | 0          | -                                                   |

*Output channels tools can share*

## System hardware

Some hardware belongs to the machine rather than a tool: enclosure fans and lights, an air or vacuum pump, a door switch, a sensor. It is wired to board pins that are *not* on the umbilical and works whatever tool is mounted. **Hardware > System hardware** lists it by category; **+ Add system hardware** opens the same guide as a tool control.



*System hardware, grouped by category*

The differences from a tool control: the pin list offers free **board headers** (FAN5, HE2, DIAG4...) and refuses any pin on the umbilical; you can type another free pin; you choose a **category** (edited under **Admin > Tool lists**); and its macros have no mounted-tool guard. **Switch it off whenever the machine is made safe** is off by default, since lights and fans usually stay on. A powered tool can switch a piece of system hardware on with itself – a shop vacuum with a cutter – by choosing it as a **Switch on with the tool** output in step 2. Each piece can carry maintenance tasks (**Maintenance** on its card).

**New system hardware**

NAME: **door\_switch** WHAT IS IT?: **Enclosure door switch**

Lower-case letters, digits and \_ Used in G-code, e.g. SET\_PIN  
PIN=enclosure\_light.

CATEGORY: **Safety** NOTES (OPTIONAL): **24 V strip on the FAN5 header, 2 A**

Edit the categories under Admin > Tool lists.

1. WHICH WAY DOES THE SIGNAL GO?

☐ Rhino controls it  
A relay, fan, light, servo, driver...

☒ It sends a signal to Rhino  
A button, door switch, limit switch, sensor...

2. WHICH BOARD PIN IS IT WIRED TO?

PIN: **PG12 - DIAG4** OR TYPE ANOTHER PIN: **PE11**

Free headers on the board that are not on the umbilical. Any free board pin, e.g. on the EXP headers. Checked against the whole config.

3. HOW DOES IT BEHAVE?

WHAT IS CONNECTED?

☒ A switch or button that connects the pin to ground  
The pin's pull-up is turned on, so it reads 'off' until pressed.

☐ A sensor or board that drives the line itself  
No pull-up.

☐ Swap pressed and released  
Tick if it reports the opposite of what you expect (normally-closed switches).

WHEN IT TRIGGERS: **Pause the job**

WHEN IT IS RELEASED: **Do nothing**

WRITTEN TO CUSTOM\_TOOLS.CFG

Cancel Add hardware

*An enclosure door switch that pauses the job*

# Admin lists

The **Admin** group sets what the portal's drop-down menus offer. Every list is checked as you save: a kind of tool, category or kind of work that something still uses cannot be removed (rename it instead).

## Tool lists

**Kinds of tool** are grouped under Additive, Subtractive and Passive – what step 1 of Add a tool offers. Each has a name, the help line shown in the wizard and what it needs: a print head (copies an extruder), a power output, or nothing. What it needs is fixed once the kind exists; built-in kinds can be renamed but not removed.

### Kinds of tool

Step 1 of Add a tool asks Additive, Subtractive or Passive, then offers the kinds listed under it. What a kind needs (a print head, a power output, or nothing) is fixed once it exists; its name, help and starting values can change.

#### ADDITIVE

Adds material: print heads, paste or glue dispensers.

| NAME                                         | IT NEEDS                          | HELP SHOWN IN THE WIZARD                |
|----------------------------------------------|-----------------------------------|-----------------------------------------|
| 3D-print head (clone of an existing one)     | A print head (copies an extruder) | A new nozzle size or head that reuses : |
| Built in                                     |                                   | Remove                                  |
| Paste or glue dispenser                      | A power output                    | A pump or plunger driven by a PWM or    |
| ▶ Starting values for new tools of this kind |                                   |                                         |
| Built in                                     |                                   | Remove                                  |

+ Add an additive kind

#### SUBTRACTIVE

Removes or cuts material: lasers, spindles, hot wires, knives.

| NAME                                         | IT NEEDS            | HELP SHOWN IN THE WIZARD                  |
|----------------------------------------------|---------------------|-------------------------------------------|
| Hot wire foam cutter                         | A power output      | Heated wire for cutting foam. Power is    |
| ▶ Starting values for new tools of this kind |                     |                                           |
| Built in - used by FoamWire                  |                     | Remove                                    |
| Needle / vibrating knife                     | A power output      | Oscillating needle or knife driven by a f |
| ▶ Starting values for new tools of this kind |                     |                                           |
| Built in - used by Needle                    |                     | Remove                                    |
| Other powered cutter                         | A power output      | Any other cutting tool driven by a PWM    |
| ▶ Starting values for new tools of this kind |                     |                                           |
| Built in                                     |                     | Remove                                    |
| Unpowered blade or knife                     | Nothing (unpowered) | Cuts by moving through the material: a    |
| ▶ Starting values for new tools of this kind |                     |                                           |
| Built in                                     |                     | Remove                                    |

+ Add an subtractive kind

#### PASSIVE

Does not make or cut anything: pens, probes, cameras, holders.

| NAME                                         | IT NEEDS            | HELP SHOWN IN THE WIZARD              |
|----------------------------------------------|---------------------|---------------------------------------|
| Passive tool (pen, probe, holder)            | Nothing (unpowered) | No power output: a pen, probe, camera |
| ▶ Starting values for new tools of this kind |                     |                                       |
| Built in - used by PenPlotter                |                     | Remove                                |

+ Add a passive kind

Save kinds of tool

Admin > Tool lists: kinds of tool by category

Open **Starting values for new tools of this kind** to set the power cap, maximum on-time, materials (one per line: name, power %, feed) and pre-job checklist that a new tool of this kind starts with.

The screenshot shows a configuration window for a 'Hot wire foam cutter'. It has three tabs: 'NAME', 'IT NEEDS', and 'HELP SHOWN IN THE WIZARD'. The 'NAME' tab is active, showing the tool name 'Hot wire foam cutter'. Below the tabs, there is a section titled 'Starting values for new tools of this kind'. This section contains four input fields: 'POWER CAP %' with the value '50', 'MAX TIME ON (S)' with the value '600', 'MATERIALS (ONE PER LINE: NAME POWER% FEED)' with two lines of text 'FOAM\_EPS 40% 300' and 'FOAM\_XPS 50% 250', and 'PRE-JOB CHECKLIST (ONE PER LINE)' with two lines of text 'Wire installed, taut and clear of the stock' and 'Foam is clamped flat and cannot shift'. At the bottom left, it says 'Built in · used by FoamWire', and at the bottom right, there is a 'Remove' button.

| NAME                 | IT NEEDS       | HELP SHOWN IN THE WIZARD               |
|----------------------|----------------|----------------------------------------|
| Hot wire foam cutter | A power output | Heated wire for cutting foam. Power is |

▼ Starting values for new tools of this kind

| POWER CAP % | MAX TIME ON (S) |
|-------------|-----------------|
| 50          | 600             |

| MATERIALS (ONE PER LINE: NAME POWER% FEED) | PRE-JOB CHECKLIST (ONE PER LINE)            |
|--------------------------------------------|---------------------------------------------|
| FOAM_EPS 40% 300                           | Wire installed, taut and clear of the stock |
| FOAM_XPS 50% 250                           | Foam is clamped flat and cannot shift       |

Built in · used by FoamWire Remove

Starting values of the hot-wire kind

The same page holds the material names suggested in the material tables, by category; the nozzle sizes offered when you add a print head; and the system hardware categories.

The screenshot shows a 'Material names' configuration window. It has a title bar 'Material names' and a subtitle 'Suggested in the material tables of Add a tool and Edit, by category. Any other name can still be typed.' Below the subtitle, there are three columns of material names: 'ADDITIVE' (PLA, PETG, ABS, ASA, TPU, NYLON, PC, PASTE, GLUE), 'SUBTRACTIVE' (PLYWOOD, MDF, HARDWOOD, ACRYLIC, LEATHER, CARDSTOCK, VINYL, FABRIC, FOAM\_EPS, FOAM\_XPS, ALUMINIUM, PCB), and 'PASSIVE' (PAPER, CARD, DEFAULT). At the bottom, there is a note 'Separate names with commas. Letters, digits and \_ - spaces become \_' and a 'Save material names' button.

| ADDITIVE                                         | SUBTRACTIVE                                                                                            | PASSIVE              |
|--------------------------------------------------|--------------------------------------------------------------------------------------------------------|----------------------|
| PLA, PETG, ABS, ASA, TPU, NYLON, PC, PASTE, GLUE | PLYWOOD, MDF, HARDWOOD, ACRYLIC, LEATHER, CARDSTOCK, VINYL, FABRIC, FOAM_EPS, FOAM_XPS, ALUMINIUM, PCB | PAPER, CARD, DEFAULT |

Separate names with commas. Letters, digits and \_ - spaces become \_

Save material names

Material name suggestions by category

## Maintenance lists

The machine areas tasks are grouped by, the kinds of work (Inspect, Clean...) with how many tasks use each, and the names of the four priority levels.

### Machine areas

Machine tasks are grouped by area (the 'For' list of a task). An area with tasks cannot be removed.

|                              |        |
|------------------------------|--------|
| Motion X/Y                   | Remove |
| Motion Z                     | Remove |
| Frame and enclosure          | Remove |
| Electronics                  | Remove |
| Umbilical and tool connector | Remove |
| Other                        | Remove |

[+ Add area](#) [Save areas](#)

### Kinds of work

Inspect, clean, lubricate... the 'Kind of work' list of a task.

|           |            |        |
|-----------|------------|--------|
| Adjust    | 1 task(s)  | Remove |
| Calibrate | 2 task(s)  | Remove |
| Clean     | 9 task(s)  | Remove |
| Inspect   | 11 task(s) | Remove |
| Lubricate | 2 task(s)  | Remove |
| Other     | 1 task(s)  | Remove |
| Replace   | 4 task(s)  | Remove |

[+ Add kind of work](#) [Save kinds of work](#)

### Priorities

Four levels, most urgent last. They order the Mainsail reminder; rename them to suit you.

| LOW | NORMAL | HIGH | CRITICAL |
|-----|--------|------|----------|
| Low | Normal | High | Critical |

[Save priorities](#)

*Admin > Maintenance lists*

**Status codes** are described with maintenance tracking in Part 4.

# Console commands

Everything the portal does can also be typed in the Mainsail console. Lines starting `rhino:` are the tool manager's replies; nothing changes if a command is refused.

## Managing tools

| Command                                                                               | What it does                                                                                                     |
|---------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------|
| <code>ADD_TOOLHEAD NAME=FoamWire KIND=HOT_WIRE PIN=PB1</code>                         | Add a powered tool on a new pin                                                                                  |
| <code>ADD_TOOLHEAD NAME=Needle KIND=NEEDLE POWER_PIN=SPINDLE_SPEED</code>             | Add a powered tool on a shared output                                                                            |
| <code>ADD_TOOLHEAD NAME=Wide KIND=PRINT_HEAD BASE=Block0ne NOZZLES=0.6,1.0</code>     | Add a print-head clone with new nozzle sizes                                                                     |
| <code>EDIT_TOOLHEAD NAME=FoamWire CAP=0.4 MAX_ON_S=300</code>                         | Change limits (also <code>MIN=</code> , <code>IDLE=</code> , <code>S_MAX=</code> , <code>DEFAULT_POWER=</code> ) |
| <code>EDIT_TOOLHEAD NAME=FoamWire MATERIAL=FOAM_EPS POWER=0.45 FEED=280 ZOFF=0</code> | Add or change a material                                                                                         |
| <code>EDIT_TOOLHEAD NAME=FoamWire DELMATERIAL=FOAM_XPS</code>                         | Remove a material                                                                                                |
| <code>RENAME_TOOLHEAD NAME=FoamWire NEWNAME=HotWire</code>                            | Rename                                                                                                           |
| <code>REMOVE_TOOLHEAD NAME=FoamWire</code>                                            | Remove (asks you to confirm)                                                                                     |
| <code>LIST_TOOLHEADS</code>                                                           | Every tool with slot, type and materials                                                                         |
| <code>TOOLHEAD_PINS</code>                                                            | Which connector pins are free, used or reserved                                                                  |
| <code>CHECK_TOOLHEADS</code>                                                          | Re-check the registry against the config and regenerate <code>custom_tools.cfg</code>                            |
| <code>RESTART_FOR_TOOLS</code>                                                        | Restart Klipper to load tool changes (refused during a job)                                                      |
| <code>TOOLS_MENU</code>                                                               | Reminder of where to add, edit and remove tools                                                                  |

## Running the machine

| Command                                                                                                     | What it does                                                                                      |
|-------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------|
| <code>START_TOOL_SWAP</code>                                                                                | Guided tool change                                                                                |
| <code>RESTORE_TOOL</code>                                                                                   | Confirm the mounted tool after a restart                                                          |
| <code>SET_TOOLHEAD TOOLHEAD=&lt;tool&gt; MATERIAL=&lt;material&gt;</code>                                   | Select tool and material for the next job                                                         |
| <code>LASER_JOB_SETUP / CNC_JOB_SETUP / DRAGKNIFE_JOB_SETUP / TOOL_JOB_SETUP FILE=&lt;name&gt;.gcode</code> | Guided job setup for that tool                                                                    |
| <code>RHINO_POSITIONS</code>                                                                                | List the positions in <code>myrhino/positions.cfg</code> and check them against the travel limits |

| Command                    | What it does                                                                                             |
|----------------------------|----------------------------------------------------------------------------------------------------------|
| G28 / G28 X Y              | Home. Z only with a print head confirmed; G28 X Y always                                                 |
| SET_Z_ZERO                 | Set Z zero by eye for the laser (at focus), spindle, drag knife or an added tool – Bed up/down, then SET |
| TOOL_RESUME                | Resume a paused non-print job                                                                            |
| M3 S<value> / M5           | Mounted tool on / off                                                                                    |
| SET_TOOL_POWER POWER=<0-1> | Set an added powered tool's output (fraction of its cap)                                                 |
| FILAMENT_CHANGE            | Guided filament change                                                                                   |
| BED_COOLDOWN               | Staged, material-aware bed cooldown                                                                      |
| EMERGENCY_STOP             | All tool power off (see Safety systems)                                                                  |
| BACKUP_CFG MSG=<note>      | Push the config folder to your GitHub backup                                                             |
| SWITCH_T0 / SWITCH_T1      | SwitchFly: select filament path 1 or 2                                                                   |
| Z_TILT_ADJUST              | Tram the bed with the three Z motors (home first)                                                        |
| CALIBRATE_ESC              | Guided spindle ESC throttle calibration (bit removed)                                                    |
| PM_STATUS                  | List maintenance that is overdue or due soon (live)                                                      |
| PM_REMINDER                | Show the start-up maintenance pop-up again                                                               |

# Safety systems

---

Each of these is automatic; none needs switching on.

- **Tool must match.** `START_JOB` refuses a file sliced for a different tool. The laser and spindle commands refuse to fire unless LightSaber or HotJoe is the recorded tool, and switch their outputs off when they refuse.
- **Shared pin guard.** `SERVO_LASER` is the laser's intensity signal on LightSaber and SwitchFly's filament servo otherwise; servo commands are blocked while LightSaber is mounted.
- **Power cap.** An added powered tool never exceeds its cap, whatever `M3 S` or `SET_TOOL_POWER` asks for.
- **Watchdog.** Every 5 seconds while an added powered tool is on: with no job running it is cut after the tool's *Max time on*; during a pause it is held off; if the tool is no longer mounted everything is switched off.
- **Everything off on the way out.** Tool swap, `CANCEL_PRINT`, `END_PRINT` and `EMERGENCY_STOP` all switch off the laser rail and signal, the spindle relay (ESC to 0.3 neutral) and every added tool.
- **Pause makes it safe.** See *Pause, resume and cancel*: lifted clear and powered off before parking.
- **Nothing changes mid-job.** Tool swaps, adding/editing/removing tools and Klipper restarts from the portal are all refused while a job is printing or paused.

**EMERGENCY\_STOP is a power-off, not a halt.** It cuts tool power but does not stop motion. To stop the machine dead, use Mainsail's emergency-stop button (`M112`), which shuts Klipper down; then `FIRMWARE_RESTART`.

**Start low.** Every starting power value (hot wire 40–50 %, caps, maximum on-times) is a placeholder. Test a new tool from a current-limited supply first, then raise its values in the portal while watching it.

**Hardware layer.** Heavy loads (laser, spindle, bed, vacuum pump) are switched through relays, so a controller fault cannot hold them on through the board itself. The spindle ESC's signal falls to neutral (0.3) and its relay opens if Klipper shuts down. Close the enclosure doors while the laser or spindle runs. Smoke detection is planned in the build notes but not yet fitted, so never leave a laser, hot-wire or spindle job unattended.

## Notes

This image shows a single sheet of white paper with horizontal ruling lines. The lines are evenly spaced and run across the width of the page. There is no text or other markings on the paper.

PART 4

# Maintenance

---

Tracking preventive maintenance in the portal, and the procedures themselves.

|                             |
|-----------------------------|
| Maintenance tracking        |
| Task Library and Task Books |
| Status codes                |
| Maintenance procedures      |

# Maintenance tracking

---

The portal keeps track of preventive maintenance for the machine, every tool and the system hardware. It tells you what is overdue, what is coming up and what parts to have ready. It never blocks a job, a swap or a command. Open it from **Maintenance** in the sidebar: **Due & upcoming**, **Tasks**, **Task Library**, **Work log** and **Usage meters**.

## Getting started

Open the **Task Library** and assign a **Task Book** to the whole machine and to each tool: *Rhino motion system and frame* on the machine, *3D print head* on BlockOne and SwitchFly, *Laser module* on LightSaber, and so on. The intervals are starting points; tune them to how your machine wears (see *Task Library and Task Books*). When you add a tool, the confirmation offers the book that fits it.

## Due & upcoming

Tasks are grouped by status. The tiles at the top count them. Each card shows every rule with a progress bar, how far off it is, and an estimate in days at your usual rate of use. Once the early alarm goes off, the card lists the parts to **Have ready**.

**RHINO // Tool Manager**

Klipper: ready | Job: standby | Mounted: BlockOne | Maintenance: 1 overdue

**Maintenance** Task Library + New task

1 overdue | 2 due soon (early alarm) | 1 with a status set | 34 active tasks

**Overdue**  
Past the due point. Nothing is blocked - fit it in when it suits you.

**Clean the nozzle and check it for wear** Overdue  
BlockOne - Clean - from the Task Library  
every 100 working hours - 4.0 h over (due at 100 h)  
**Have ready:** Brass brush  
About 10 min  
Log work Set status  
Snooze... Details

**Due soon**  
Inside the early alarm: time to plan the work and get the parts.

**Inspect the 21-pin umbilical connector and cable** Due soon  
Umbilical and tool connector - Inspect - from the Task Library  
every 100 tool swaps - 7 left (due at 100)  
every 3 months - due 2 Jan 2027, in 3 months  
**Have ready:** Spare connector pins / housing  
About 15 min  
Log work Set status  
Snooze... Details

**Clean the laser lens / protective window** In progress Due soon  
LightSaber - Clean - High priority - from the Task Library  
In progress - Lens soaking in IPA  
Set Oct 2, 2026 - ends by itself Oct 5, 2026  
every 10 working hours - 0.8 h left (due at 10 h)  
**Have ready:** Lens tissue, Isopropyl alcohol  
About 5 min  
Log work Change status  
Details

**Pending - Waiting on parts**  
Parts are on order. Say what in the note. The task keeps ageing so it comes back as soon as you clear it.

**Check and tension the X/Y belts** Pending - Waiting on parts Overdue  
Motion X/Y - Adjust - from the Task

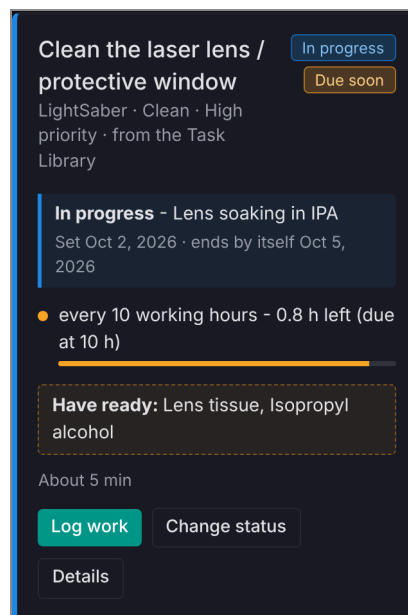
v1.2.0

Due & upcoming

| Group                     | Meaning                                                                                                    |
|---------------------------|------------------------------------------------------------------------------------------------------------|
| Overdue                   | Past the due point                                                                                         |
| Due soon                  | Inside the early alarm: time to plan the work and order parts                                              |
| A status code's own group | Tasks set to a status shown in its own group, such as <i>Pending – Waiting on parts</i>                    |
| Snoozed                   | Put off on purpose                                                                                         |
| Coming up                 | The next eight tasks by how close they are to due, plus tasks set to a status shown with the upcoming ones |

**Log work** records what happened. **Set status** puts a status code on the task. **Snooze** puts an overdue or due-soon task off for 1 day to 2 weeks. **Details** opens the task. The sidebar badge and a chip in the top bar show the count: red for overdue, amber for due soon.

A card with a status code set shows it at the top in its colour, with the note, when it was set and when it ends by itself. Here *In progress* ends after three days; the task is still due soon underneath.



A task set to *In progress*, with its note

## Logging work

**Log work** shows the task's steps and safety notes. **What happened?** is a list: **Done**; **Skipped – not needed this time** if you checked and nothing needed doing; or one of your status codes. Done and Skipped start the schedule again from here, and so do a part's hours, so a new nozzle starts at zero. You can backdate the entry and add notes; the current meter readings are saved with it.

The screenshot shows a dark-themed dialog box titled "Log work - Clean the nozzle and check it for wear". At the top, it says "BlockOne · 4.0 h over (due at 100 h)". Below this is a section "Steps" with a list: "1. Heat the nozzle", "2. Brush the outside clean", and "3. Check the tip for wear or a widened hole". A yellow box contains the text "Safety: The nozzle is hot." Below this are two dropdown menus: "WHAT HAPPENED?" set to "Done" and "WHEN" set to "10/02/2026". A small text line says "The schedule starts again from now (a part replaced starts its hours again too)." Below that is a "NOTES (OPTIONAL)" section with a text box containing "Brushed clean; tip shows no wear." At the bottom, there is a "Cancel" button and a green "Save to the work log" button. A small note at the bottom says "The current meter readings are saved with this entry."

*Logging a nozzle clean as done*

Choosing a status code instead sets that status on the task and logs it. The help line says what the status does; *Pending – Waiting on parts* needs a note saying what is on order.

This screenshot shows the same "Log work" dialog, but with "WHAT HAPPENED?" set to "Pending - Waiting on parts". Below this dropdown, a text line reads: "Parts are on order. Say what in the note. The task keeps ageing so it comes back as soon as you clear it. The task keeps ageing. A note is required." The "NOTES (REQUIRED)" section now has a text box containing "0.4 mm hardened nozzle ordered". At the bottom, the buttons are "Cancel" and a green "Set status and log it" button. A small note at the bottom says "The status shows on the task until it is cleared, logged done, or ends by its own rule (Admin > Status codes)."

*Logging a task as waiting on parts*

**Set status** on a card does the same without the steps, and **Clear the status** takes it off again.

Status - Clean the nozzle and check it for wear

STATUS

Deferred to next shutdown

NOTE (OPTIONAL)

Do it with the nozzle change at the weekend

Shown with the upcoming tasks. Left out of the Mainsail reminder.

Cancel

Set status

*Setting a status from the card*

## Tasks

Every task in one table, sorted overdue first, with its status code and status. Library tasks show *Library* and the Task Book they came from. **Show** filters it: overdue and due soon, with a status set, one machine area, one tool or piece of system hardware, or archived. Choosing an area or a tool also shows its usage meters.

RHINO // Tool Manager

Klipper: ready

Job: standby

Mounted: BlockOne

Maintenance: 2 due soon

TOOLS

Tool library

+ Add a tool

HARDWARE

Pins & outputs

System hardware

MAINTENANCE

Due & upcoming

Tasks

Task Library

Work log

Usage meters

ADMIN

Tool lists

Maintenance lists

Status codes

Tasks

Task Library

+ New task

SHOW

Every task

| TASK                                                                                                       | FOR                          | SCHEDULE                                     | STATUS                                                                           | LAST DONE |          |
|------------------------------------------------------------------------------------------------------------|------------------------------|----------------------------------------------|----------------------------------------------------------------------------------|-----------|----------|
| Check and tension the X/Y belts<br><small>Library - Rhino motion system and frame</small>                  | Motion X/Y                   | every 100 production hours or every month    | Pending - Waiting on parts<br>Overdue (due at 100 h)<br>12 h over (due at 100 h) | never     | Log work |
| Clean the laser lens / protective window<br><small>Library - Laser module</small>                          | LightSaber                   | every 10 working hours                       | In progress<br>Due soon<br>0.8 h left (due at 10 h)                              | never     | Log work |
| Inspect the 21-pin umbilical connector and cable<br><small>Library - Rhino motion system and frame</small> | Umbilical and tool connector | every 100 tool swaps or every 3 months       | Due soon<br>7 left (due at 100)                                                  | never     | Log work |
| Back up the Klipper config (BACKUP.CFG)<br><small>Library - Rhino motion system and frame</small>          | Electronics                  | every month                                  | Scheduled<br>due 2 Nov 2026, in 4 weeks                                          | never     | Log work |
| Check frame bolts and squareness<br><small>Library - Rhino motion system and frame</small>                 | Frame and enclosure          | every 6 months                               | Scheduled<br>due 2 Apr 2027, in 6 months                                         | never     | Log work |
| Check laser focus and alignment<br><small>Library - Laser module</small>                                   | LightSaber                   | every 50 working hours                       | Scheduled<br>41 h left (due at 50 h)                                             | never     | Log work |
| Check spindle runout and bearing noise<br><small>Library - CNC spindle</small>                             | HotJoe                       | every 100 working hours                      | Scheduled<br>100 h left (due at 100 h)                                           | never     | Log work |
| Check the hot-wire tension and replace a kinked wire<br><small>Library - Hot wire cutter</small>           | FoamWire                     | every 10 working hours                       | Scheduled<br>10 h left (due at 10 h)                                             | never     | Log work |
| Check the hotend screws, heater and thermistor wiring<br><small>Library - 3D print head</small>            | BlockOne                     | every 200 working hours or every 6 months    | Scheduled<br>96 h left (due at 200 h)                                            | never     | Log work |
| Check the Z belt drive (3:1 reduction)<br><small>Library - Rhino motion system and frame</small>           | Motion Z                     | every 500 production hours or every 6 months | Scheduled<br>386 h left (due at 500 h)                                           | never     | Log work |
| Clean and lubricate the X/Y linear guides<br><small>Library - Rhino motion system and frame</small>        | Motion X/Y                   | every 200 production hours or every 3 months | Scheduled<br>88 h left (due at 200 h)                                            | never     | Log work |
| Clean and lubricate the Z leadscrews<br><small>Library - Rhino motion system and frame</small>             | Motion Z                     | every 250 production hours or every 3 months | Scheduled<br>198 h left (due at 250 h)                                           | never     | Log work |

*Tasks*

## Creating a task

+ **New task** opens the task editor:

- **Task** – a short title. It is also shown in the Mainsail reminder.
- **For** – a machine area, a tool or a piece of system hardware.
- **Kind of work** and **Priority** – used for sorting and the card's label (the lists are under **Admin › Maintenance lists**).
- **When is it due?** – up to four rules. The task is due when the first of them is reached.
- **Start counting from** (new tasks only) – now, or the date it was last done, with the meter readings at that time.
- **How to do it** – steps, parts and consumables, tools needed, time needed, safety notes and links.
- **Reminders** – include it in the start-up reminder, or pause the task.

**New maintenance task** [X]

**TASK**  
Replace the laser module fan

**FOR**  
Machine - Motion X/Y

Short - it is also shown in the Mainsail reminder.

**KIND OF WORK**  
Inspect

**PRIORITY**  
Normal

**WHEN IS IT DUE? (WHICHEVER RULE COMES FIRST)**

Every so often every 3 months early alarm 14 days before  
[Remove]

After so much use every 100 h Machine: Production hours early alarm  
10 h before [Remove]

+ Add another rule

**TIME RULES COUNT...**

☒ **From when it was last done**  
Done late? The next one moves later too.

☐ **On a fixed calendar**  
Same dates every time, e.g. the 1st of every month.

**START COUNTING FROM**

☒ **Now - treat it as just done**

☐ **The last time it was done**

**HOW TO DO IT**

**STEPS (ONE PER LINE)**

**PARTS AND CONSUMABLES (ONE PER LINE)**

**TOOLS NEEDED**  
2.5 mm hex key, belt tension gauge

[Cancel] [Save]

*A new task with two rules*

| Rule              | Example                                       | Early alarm       |
|-------------------|-----------------------------------------------|-------------------|
| Every so often    | every 3 months                                | 14 days before    |
| After so much use | every 100 working hours of this tool          | 10 h before       |
| After each event  | after each emergency stop or Klipper shutdown | due straight away |
| Once, on a date   | 1 Dec 2026                                    | 7 days before     |

Time rules count from when the task was last done, so doing it late moves the next one later. Choose **On a fixed calendar** to keep the same dates every time, such as the 1st of every month. A usage rule's early alarm is in the meter's own unit; leave it empty for 10% of the interval. Events are a tool swap, this tool being mounted, a job, and an emergency stop or shutdown. **Add this task to the Task Library**, at the bottom of an existing task, turns it into a library task and links it.

## Work log

Every entry, newest first: work done or skipped, status codes set and cleared, and meter changes. Filter it by machine area, tool or system hardware.

| WHEN                  | FOR        | TASK                                     | WHAT                                   | NOTES                                 |
|-----------------------|------------|------------------------------------------|----------------------------------------|---------------------------------------|
| Oct 2, 2026, 12:50 PM | BlockOne   | Clean the nozzle and check it for wear   | Done (was overdue)                     | Brushed clean; tip shows no wear.     |
| Oct 2, 2026, 12:50 PM | LightSaber | Clean the laser lens / protective window | Status set: In progress                | Lens soaking in IPA                   |
| Oct 2, 2026, 12:50 PM | Motion XY  | Check and tension the X/Y belts          | Status set: Pending - Waiting on parts | OT2 6 mm belt ordered, arrives Friday |
| Oct 2, 2026, 12:50 PM | BlockOne   | Meter set: Working hours                 | 0 → 104                                |                                       |
| Oct 2, 2026, 12:50 PM | Machine    | Meter set: Tool swaps                    | 0 → 93                                 |                                       |
| Oct 2, 2026, 12:50 PM | LightSaber | Meter set: Working hours                 | 0 → 9                                  |                                       |
| Oct 2, 2026, 12:50 PM | Machine    | Meter set: Production hours              | 0 → 112                                | Hours before tracking started         |

Work log

## Usage meters

Usage rules count these meters. The portal reads them from Moonraker every few seconds and from Moonraker's job history.

**RHINO // Tool Manager** Klipper: ready Job: standby Mounted: BlockOne Maintenance: 2 due soon

**Usage meters**

**Machine**  
Tracking since Oct 2, 2026. Use Set if the machine had hours before tracking started.

| METER                                                                                | READING | AVERAGE |
|--------------------------------------------------------------------------------------|---------|---------|
| <b>Machine up hours</b><br>Klipper running and ready (counted while the portal runs) | 0.0 h   | - Set   |
| <b>Production hours</b><br>A job running. Pauses are not counted.                    | 112 h   | - Set   |
| <b>Jobs</b><br>Finished jobs from Moonraker's history                                | 0       | - Set   |
| <b>Filament used</b><br>From Moonraker's job history                                 | 0.0 m   | - Set   |
| <b>Tool swaps</b><br>Times a different tool was mounted (connector and dock wear)    | 93      | - Set   |
| <b>Emergency stops and shutdowns</b><br>Klipper entering shutdown                    | 0       | - Set   |

**BlockOne** hours mounted 0.0 h - production hours 0.0 h

**SwitchFly** hours mounted 0.0 h - production hours 0.0 h

**LightSaber** hours mounted 0.0 h - production hours 0.0 h

**HotJoe** hours mounted 0.0 h - production hours 0.0 h

**DragKnife** hours mounted 0.0 h - production hours 0.0 h

**FoamWire** hours mounted 0.0 h - production hours 0.0 h

**PenPlotter** hours mounted 0.0 h - production hours 0.0 h

**Needle** hours mounted 0.0 h - production hours 0.0 h

**Reminders**  
Show overdue and due-soon tasks in Mainsail after Klipper starts  
☒ Once per start, after the "which tool is mounted?" question. It never blocks a job. PM\_STATUS in the console lists them any time. A status code can leave its tasks out (Admin > Status codes).

v12.0

Usage meters

| Meter                         | Machine | Each tool   | Counts                                                                                                                                                    |
|-------------------------------|---------|-------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------|
| Machine up hours              | yes     | –           | Klipper running and ready                                                                                                                                 |
| Production hours              | yes     | yes         | A job running. Pauses are not counted.                                                                                                                    |
| Hours mounted                 | –       | yes         | The tool on the machine while Klipper is up                                                                                                               |
| Working hours                 | –       | yes         | Print heads: nozzle heater on. LightSaber: laser on. HotJoe: spindle running. Added powered tools: output on. Passive tools and DragKnife: running a job. |
| Jobs                          | yes     | yes         | Finished jobs, from the job history                                                                                                                       |
| Filament used                 | yes     | print heads | From the job history                                                                                                                                      |
| Tool swaps / Times mounted    | yes     | yes         | A different tool mounted. A restart is not a swap.                                                                                                        |
| Emergency stops and shutdowns | yes     | –           | Klipper entering shutdown                                                                                                                                 |

The first time the portal runs, it imports the jobs already in Moonraker's history as the machine's starting reading. If the machine had hours before that, press **Set** on a meter and note why; the change goes in the work log. Jobs that ran while the portal was stopped are filled in from the history the next time it starts. System hardware tasks count the machine's meters. The page also has the start-up reminder switch.

## Reminders in Mainsail

After Klipper starts, once you answer *which toolhead is mounted?*, Mainsail shows a **Maintenance due** pop-up with up to eight overdue and due-soon tasks, overdue first. It appears once per start; close it with **OK**. Snoozed tasks, tasks with the reminder unticked and tasks set to a status code that leaves them out of the reminder are not shown.

The portal keeps `myrhino/maintenance_status.cfg` up to date, and Klipper reads it when it starts. At any other time, type `PM_STATUS` in the console for the live list, or `PM_REMINDER` to show the start-up pop-up again. Renaming a tool moves its tasks and meters with it. Removing a tool or a piece of system hardware archives its tasks (filter **Archived**) and keeps their work-log entries.

# Task Library and Task Books

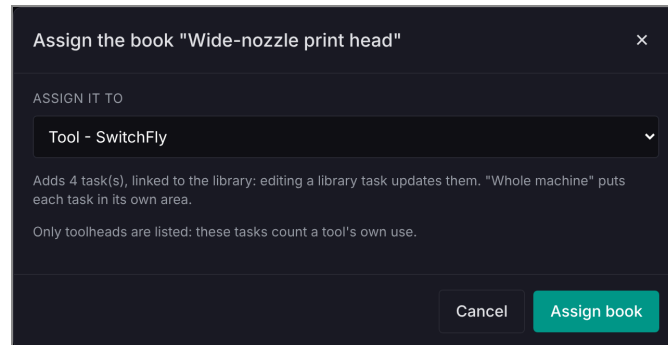
The **Task Library** holds task definitions that are not tied to anything yet – the starter tasks for the Rhino and each kind of tool, and any you add. A **Task Book** is a named group of library tasks. Assign a book to a toolhead, a piece of system hardware or the whole machine, and it carries all of its tasks there.

The **Task Books** tab shows each book, its tasks and where it is assigned. **Assign...** puts it on something more; the **x** on an assignment takes it off, which archives the tasks it added there (their work log is kept). Books whose tasks count a tool's own use, such as working hours, can only go on toolheads.

The screenshot shows the RHINO Tool Manager interface. On the left is a sidebar with navigation options: TOOLS (Tool library, + Add a tool), HARDWARE (Pins & outputs, System hardware), MAINTENANCE (Due & upcoming, Tasks, Task Library, Work log, Usage meters), and ADMIN (Tool lists, Maintenance lists, Status codes). The main area is titled 'Task Library' and contains a list of task books. Each task book card includes a title, a brief description, a list of tasks, an 'ON (TOOLHEADS ONLY)' section with an assignment button (e.g., 'BlockOne'), and 'Assign...' and 'Edit' buttons. The task books shown are: 3D print head, CNC spindle, Drag knife, Hot wire cutter, Laser module, Needle cutter, Passive tool, and Powered tool. The interface also shows status indicators at the top: 'Klipper: ready', 'Job: standby', 'Mounted: BlockOne', and 'Maintenance: 2 due soon'.

Task Books, with where each is assigned

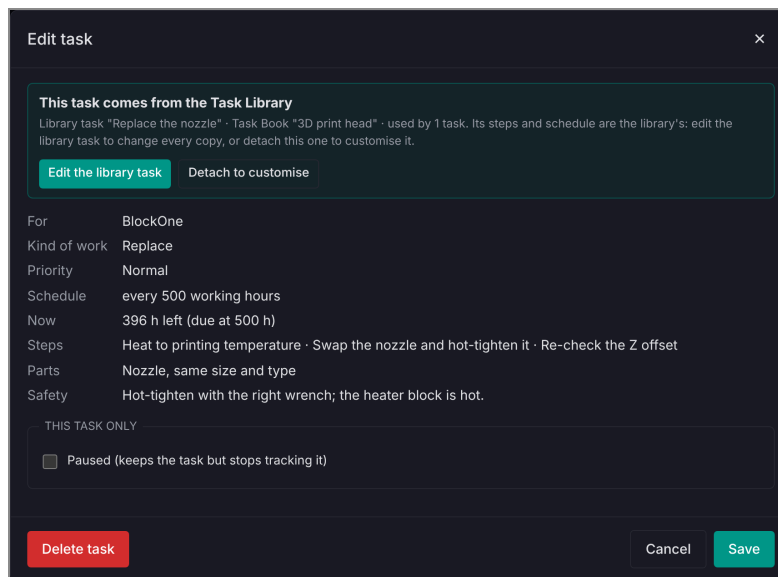
**Assign...** asks where. **Whole machine** puts each task in its own machine area (belts in Motion X/Y, leadscrews in Motion Z...). Assigning a book twice adds nothing twice.



*Assigning a Task Book*

## Linked tasks

Tasks that come from the library stay **linked** to it. Their steps and schedule are the library's, so **Details** on a linked task shows them read-only with two choices: **Edit the library task**, which changes every copy at once, or **Detach to customise**, which turns this one task into an ordinary task you can edit on its own. Pausing a linked task affects only that task. Each copy keeps its own schedule: when it was last done, its readings and its status.



*A linked task*

The **Library tasks** tab lists every library task: its schedule, what it can go on, the books it is in and the tasks made from it. **Edit** opens the editor; saving says how many linked tasks it updates. **Add to...** puts one library task on something without a book. **+ Library task** adds your own; its **Area when the book goes on the whole machine** decides where it lands with **Whole machine**. Deleting a library task keeps the tasks made from it as ordinary tasks.

● RHINO // Tool Manager

Klipper: ready

Job: standby

Mounted: BlockOne

Maintenance: 2 due soon

▼ TOOLS

Tool library

+ Add a tool

▼ HARDWARE

Pins & outputs

System hardware

▼ MAINTENANCE

Due & upcoming 2

Tasks

**Task Library**

Work log

Usage meters

▼ ADMIN

Tool lists

Maintenance lists

Status codes

v12.0

Task Library

+ Library task + Task Book

Library tasks are task definitions that are not tied to anything yet. A Task Book groups library tasks under one name; assign the book to a toolhead (or the whole machine) and it carries all of its tasks there. Those tasks stay linked to the library, so editing a library task updates every copy.

Task Books (9) Library tasks (29)

| LIBRARY TASK                                                                             | SCHEDULE                                      | GOES ON                                          | IN BOOKS                      | USED BY                      |      |           |
|------------------------------------------------------------------------------------------|-----------------------------------------------|--------------------------------------------------|-------------------------------|------------------------------|------|-----------|
| <b>Back up the Klipper config (BACKUP_CFG)</b><br>Other - starter                        | every month                                   | Anything (machine: Electronics)                  | Rhino motion system and frame | Electronics                  | Edit | Add to... |
| <b>Check and tension the X/Y belts</b><br>Adjust - starter                               | every 100 production hours or every month     | Anything (machine: Motion X/Y)                   | Rhino motion system and frame | Motion X/Y                   | Edit | Add to... |
| <b>Check frame bolts and squareness</b><br>Inspect - starter                             | every 6 months                                | Anything (machine: Frame and enclosure)          | Rhino motion system and frame | Frame and enclosure          | Edit | Add to... |
| <b>Check the Z belt drive (3:1 reduction)</b><br>Inspect - starter                       | every 500 production hours or every 6 months  | Anything (machine: Motion Z)                     | Rhino motion system and frame | Motion Z                     | Edit | Add to... |
| <b>Clean and lubricate the X/Y linear guides</b><br>Lubricate - starter                  | every 200 production hours or every 3 months  | Anything (machine: Motion X/Y)                   | Rhino motion system and frame | Motion X/Y                   | Edit | Add to... |
| <b>Clean and lubricate the Z leadscrews</b><br>Lubricate - starter                       | every 250 production hours or every 3 months  | Anything (machine: Motion Z)                     | Rhino motion system and frame | Motion Z                     | Edit | Add to... |
| <b>Clean dust from the electronics bay and fans</b><br>Clean - starter                   | every 1000 machine up hours or every 3 months | Anything (machine: Electronics)                  | Rhino motion system and frame | Electronics                  | Edit | Add to... |
| <b>Clean the enclosure LED strip and diffuser</b><br>Clean                               | every 3 months                                | Anything (machine: Frame and enclosure)          | -                             | enclosure light              | Edit | Add to... |
| <b>Inspect and clean the tool</b><br>Clean - starter                                     | every 3 months                                | Anything                                         | Passive tool                  | PenPlotter                   | Edit | Add to... |
| <b>Inspect the 21-pin umbilical connector and cable</b><br>Inspect - starter             | every 100 tool swaps or every 3 months        | Anything (machine: Umbilical and tool connector) | Rhino motion system and frame | Umbilical and tool connector | Edit | Add to... |
| <b>Inspect the machine after an emergency stop or shutdown</b><br>Inspect - starter      | after an emergency stop or Klipper shutdown   | Anything (machine: Other)                        | Rhino motion system and frame | Other                        | Edit | Add to... |
| <b>Inspect wiring, connectors and cable chains</b><br>Inspect - starter                  | every 6 months                                | Anything (machine: Electronics)                  | Rhino motion system and frame | Electronics                  | Edit | Add to... |
| <b>Level the gantry (Z_TILT_ADJUST) and check the first layer</b><br>Calibrate - starter | every month                                   | Anything (machine: Motion Z)                     | Rhino motion system and frame | Motion Z                     | Edit | Add to... |

Library tasks

Editing a library task: the box at the top lists the tasks it will update.

Edit library task

Used by 1 task

Saving updates every one of them: LightSaber. Their schedules carry on from where they are.

TASK

Clean the laser lens / protective window

Short - it is also shown in the Mainsail reminder.

AREA WHEN THE BOOK GOES ON THE WHOLE MACHINE

Choose when assigned...

Ignored on a toolhead or system hardware.

KIND OF WORK

Clean

PRIORITY

High

WHEN IS IT DUE? (WHICHEVER RULE COMES FIRST)

After so much use

every

10

h

The tool's own: Working hours

early alarm

2

h before

+ Add another rule

A rule that counts the tool's own use makes this a toolhead-only task.

HOW TO DO IT

STEPS (ONE PER LINE)

Laser and power rail off

Wipe the lens with lens tissue and isopropyl alcohol

Check for burn marks or pitting

PARTS AND CONSUMABLES (ONE PER LINE)

Lens tissue

Isopropyl alcohol

Listed as 'Have ready' once the early alarm goes off.

TOOLS NEEDED

2.5 mm hex key, belt tension gauge

TIME NEEDED (MINUTES)

5

SAFETY NOTES

Laser off and LASER\_INITIALIZE off before touching the module. Wear laser goggles for any test fire.

LINKS (MANUALS, VIDEOS - ONE PER LINE)

REMINDEES

Delete library task

Cancel

Save and update 1 task

Editing the laser-lens library task

+ **Task Book** makes a book: a name, an optional description and the library tasks in it, found with the search box. Tasks you add to a book that is already assigned are added there too; tasks you take out are archived there.

New Task Book

BOOK NAME: Wide-nozzle print head

DESCRIPTION (OPTIONAL): Print heads with 0.8 mm and larger nozzles

TASKS IN THE BOOK

Find a library task... 4 selected

- ☐ Clean the drag knife bearing every 50 production hours
- ☒ Clean the extruder drive gears every 1000 m of filament
- ☐ Clean the laser lens / protective window every 10 working hours
- ☐ Clean the laser module fan and air assist every 25 working hours or every month
- ☒ Clean the nozzle and check it for wear every 100 working hours
- ☐ Clear chips from the spindle mount and Z axis every 10 jobs
- ☐ Inspect or replace the drag-knife blade every 20 production hours
- ☐ Inspect the tool and its power cable every 50 working hours or every 3 months
- ☒ Inspect the tool connector pins and latch every 50 mounts
- ☐ Replace the needle or blade every 20 working hours
- ☒ Replace the nozzle every 500 working hours

Need a task that is not here? Add it with + Library task first.

Cancel Save book

*A new Task Book for wide-nozzle print heads*

## Status codes

**Admin > Status codes** lists two kinds of status. The **built-in statuses** – Overdue, Due soon, Scheduled, Done, Snoozed and Paused – are worked out from each task's schedule, the snooze and the pause switch. Rename them or change their colour; they cannot be removed.

| Built-in statuses                                                                                                                  |            |                                                                                          |                                                       |
|------------------------------------------------------------------------------------------------------------------------------------|------------|------------------------------------------------------------------------------------------|-------------------------------------------------------|
| Worked out from each task's schedule, the snooze and the pause switch. Rename them or change their colour; they cannot be removed. |            |                                                                                          |                                                       |
| STATUS                                                                                                                             | NAME SHOWN | COLOUR                                                                                   | WHEN                                                  |
| overdue                                                                                                                            | Overdue    |  Red    | Past the due point.                                   |
| due_soon                                                                                                                           | Due soon   |  Amber  | Inside the early alarm: plan the work, get the parts. |
| ok                                                                                                                                 | Scheduled  |  Teal   | Not due yet.                                          |
| done                                                                                                                               | Done       |  Grey   | A one-off task that has been done.                    |
| snoozed                                                                                                                            | Snoozed    |  Indigo | Due, but put off for a few days.                      |
| paused                                                                                                                             | Paused     |  Grey   | Kept, but not tracked.                                |

*Built-in statuses: name and colour*

**Your status codes** are set by hand, from a task card or the Log work list: *Pending – Waiting on parts, In progress, Deferred to next shutdown*, or any you add with **+ Add a status code**. Each answers four questions, and the portal applies the answers as rules.

| NAME                                                                                                                                                                                                                                                                                                                                                                                                     | COLOUR                                                                                     | DESCRIPTION (SHOWN WHEN IT IS CHOSEN)    |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------|------------------------------------------|
| Pending - Waiting on parts                                                                                                                                                                                                                                                                                                                                                                               |  Purple | Parts are on order. Say what in the note |
| <div>1. WHERE DOES THE TASK SHOW?</div> <div> <input type="radio"/> With the due tasks           <input type="radio"/> With the upcoming tasks           <input checked="" type="radio"/> In its own group         </div>                                                                                                                                                                                |                                                                                            |                                          |
| <div>2. WHAT HAPPENS TO ITS SCHEDULE?</div> <div> <input checked="" type="radio"/> Keeps running - the task still ages toward overdue           <input type="radio"/> Pauses - time and use while it is set do not count           <input type="radio"/> Counts as done - the interval starts again         </div>                                                                                       |                                                                                            |                                          |
| <div>3. WHEN DOES IT END BY ITSELF?</div> <div> <input checked="" type="radio"/> Never - until someone changes it           <input type="radio"/> After a number of days           <input type="radio"/> When the machine or tool is used again         </div> <p>When it ends, the task goes back to its built-in status. Logging the task done also ends it.</p>                                       |                                                                                            |                                          |
| <div>4. ALSO</div> <div> <input type="checkbox"/> Keep it in the Mainsail reminder<br/>The task still shows after a Klipper start when it is due.           <input checked="" type="checkbox"/> A note is required<br/>Whoever sets it has to say why (which parts, who, when).           <input checked="" type="checkbox"/> Offer it in the Log work list<br/>Next to Done and Skipped.         </div> |                                                                                            |                                          |
| Remove                                                                                                                                                                                                                                                                                                                                                                                                   |                                                                                            |                                          |

*The rules of Pending – Waiting on parts*

| Question                         | Choices                                                                               | What it does                                                                                                                                                                                                                                                 |
|----------------------------------|---------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1. Where does the task show?     | With the due tasks · in its own group · with the upcoming tasks                       | Where Due & upcoming lists it. Only <i>with the due tasks</i> keeps it in the overdue and due-soon counts.                                                                                                                                                   |
| 2. What happens to its schedule? | Keeps running · pauses · counts as done                                               | <i>Keeps running</i> : the task still ages toward overdue. <i>Pauses</i> : time and use while the status is set do not count; the due point moves later by that much when it ends. <i>Counts as done</i> : the interval starts again when the status is set. |
| 3. When does it end by itself?   | Never · after a number of days · when the machine or tool is used again               | When it ends, the task goes back to its built-in status and the end is logged. Logging the task done also ends it.                                                                                                                                           |
| 4. Also                          | Keep it in the Mainsail reminder · a note is required · offer it in the Log work list | Whether the start-up pop-up still shows the task when due; whether whoever sets it must say why; whether it appears next to Done and Skipped.                                                                                                                |

A status code that tasks are set to cannot be removed until they are cleared. The three codes the portal starts with:

| Status code                | Shows                   | Schedule      | Ends                           |
|----------------------------|-------------------------|---------------|--------------------------------|
| Pending – Waiting on parts | own group               | keeps running | never (note required)          |
| In progress                | with the due tasks      | keeps running | after 3 days                   |
| Deferred to next shutdown  | with the upcoming tasks | pauses        | when the machine is used again |

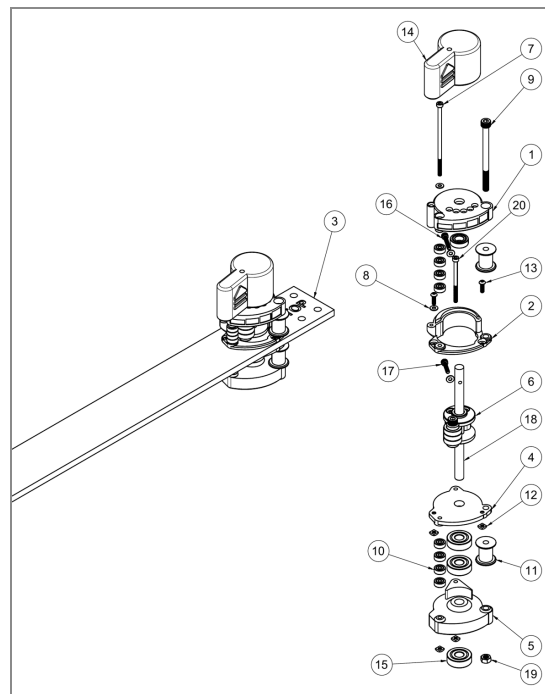
# Maintenance procedures

Four items are scheduled in the build notes; no intervals are set yet, so record what you find and when.

| Item               | What to do                                                                                                         |
|--------------------|--------------------------------------------------------------------------------------------------------------------|
| XY belts           | Check for stretch and uneven tension; re-tension as below. Replacements: 15 mm GT2, glass-fibre core, 3 m per belt |
| Bearings and rails | Check the X rail and the bearings in the idlers and tensioners for roughness or play                               |
| Z leadscrews       | Clean and re-lubricate all three leadscrews                                                                        |
| Under the bed      | Clear dust, chips and filament from the underside of the machine                                                   |

## Tensioning the XY belts

Each belt has a cam tensioner on the rear frame bar with an adjustment knob that indexes into one of five settings.



*Belt tensioner assembly, exploded (item 14 is the adjustment knob)*

1. Loosen both belt clamps on the X-carriage.
2. Turn both tensioners to their lowest setting.
3. Pull each belt through its carriage clamp until it is snug with no slack – not tight – while keeping the gantry square to both Y rails. The holes in the linear rails are a quick reference.
4. Tighten the belt clamps.

5. Turn both tensioner knobs to the same setting.

## Tramming the bed

With a print head mounted, home with `G28`, then run `Z_TILT_ADJUST`. Klipper probes three points and levels the bed with its three Z motors, retrying up to five times until the points agree within 0.05 mm. Run it after any work on the leadscrews or the bed.

## Calibrating the spindle ESC

Do this once when an ESC is new or replaced, or if spindle speeds look wrong. **Remove the bit from the collet first.**

1. Mount HotJoe and run `CALIBRATE_ESC`.
2. Press **Step 2: Send full throttle (1.0)**. This switches the spindle relay on with a full-throttle signal, so the ESC powers up and plays its start-up tones.
3. As soon as the tones finish, press **Step 4: Send low endpoint (0.3)**.
4. The ESC confirms with a tone. Its throttle range now matches Klipper's 0.3–1.0.

## Notes

This image shows a single sheet of white paper with horizontal ruling lines. The lines are evenly spaced and run across the width of the page. There is no text or other markings on the paper.

## PART 5

# Reference

---

When something goes wrong, where things live, and what is left to check.

---

Troubleshooting

---

Files, backups and tests

---

Build reference

---

Still to check on the machine

---

# Troubleshooting

| Symptom                                                          | Likely cause                                               | Fix                                                                                          |
|------------------------------------------------------------------|------------------------------------------------------------|----------------------------------------------------------------------------------------------|
| Mainsail's Resume says <i>extruder not hot enough</i>            | A non-print tool's job is paused                           | Use <b>Resume job</b> in the Rhino pop-up, or <code>TOOL_RESUME</code>                       |
| Console: <i>resume it with TOOL_RESUME</i> after pressing Resume | Same – the extruder happened to be warm                    | <code>TOOL_RESUME</code>                                                                     |
| Job refuses to start: <i>HARDWARE MISMATCH</i>                   | File sliced for a different tool than the one recorded     | <code>START_TOOL_SWAP</code> to the right tool, or answer the restart pop-up correctly       |
| Laser or spindle command: <i>CRITICAL SAFETY BLOCK</i>           | Recorded tool is not LightSaber / HotJoe                   | Check the restart pop-up answer; run <code>RESTORE_TOOL</code>                               |
| New tool not listed in Mainsail / <i>Unknown command</i>         | Klipper not restarted since adding it                      | <b>Restart Klipper</b> in the portal, or <code>RESTART_FOR_TOOLS</code>                      |
| Portal shows <b>Moonraker offline</b>                            | Moonraker stopped, or portal pointed at the wrong address  | Restart Moonraker; the portal still edits tools but cannot restart Klipper                   |
| Portal: <i>Restarting now would kill it</i>                      | A job is printing or paused                                | Wait for the job to finish or cancel it                                                      |
| <code>REMOVE_TOOLHEAD</code> : <i>is the mounted tool</i>        | You cannot remove the tool on the machine                  | Swap to another tool first                                                                   |
| Tool commands: <i>blocked while a job is printing</i>            | Registry changes are locked during jobs                    | Finish or cancel the job                                                                     |
| Add wizard offers no free pins                                   | All three free connector pins (PB1, PC0, PC1) are taken    | Share an existing output, or list more umbilical pins in <code>myrhino/umbilical.json</code> |
| Swap wizard: <i>Toolhead not detected</i>                        | Umbilical not seated                                       | Re-seat the 21-pin umbilical and press verify                                                |
| Added tool switched itself off                                   | Watchdog: on longer than <i>Max time</i> on with no job    | Raise <i>Max time</i> on in the portal, or run it from a job file                            |
| ABS/ASA bed cooldown stops early                                 | Klipper's 600 s idle timeout                               | Add <code>[idle_timeout]</code> with <code>timeout: 1800</code>                              |
| <code>rhino_toolgen</code> or backup commands fail               | Login is not <code>pi</code>                               | Fix the <code>/home/pi</code> paths (see <i>Install</i> )                                    |
| Swap wizard stays on Cooling Down & Parking                      | It waits until the hotend and the bed are both below 40 °C | Wait; after a hot print the bed is usually the slow one                                      |
| Drag-knife file stops with M3 rejected                           | The CAM profile writes laser power commands                | Use a drag-knife LaserWeb profile with Laser Intensity and Tool Off blank                    |
| Spindle speed range wrong, or ESC will not arm                   | ESC endpoints not calibrated to Klipper's 0.3–1.0 signal   | Run <code>CALIBRATE_ESC</code> with the bit removed                                          |

| Symptom                                         | Likely cause                                                                                              | Fix                                                                                                         |
|-------------------------------------------------|-----------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------|
| Control or macro refused: <i>already exists</i> | The name is used by another macro, a Klipper command or a config object (Klipper ignores case for macros) | Choose another name; the message says who has it                                                            |
| A pin is greyed out in the controls guide       | <code>umbilical.json</code> marks it as a MOSFET output, which cannot read a signal or drive a servo      | Use a logic pin, or correct the pin's type                                                                  |
| Banner says <i>time left unknown</i>            | The file has no slicer estimate and is under 2% done                                                      | Wait a moment; it estimates from progress after that                                                        |
| Meters do not move                              | The portal service is not running (meters are counted by it)                                              | <code>sudo systemctl status rhino-portal</code> ; missed jobs are filled in from the history when it starts |
| No maintenance pop-up after a restart           | Nothing was due, the reminder is switched off, or the tool-restore question was not answered yet          | Answer the question, or run <code>PM_STATUS</code>                                                          |

# Files, backups and tests

To change one thing, open the one file that owns it.

| To change                                                             | Edit                                                                                                                                                                        |
|-----------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Built-in tools, their materials and slots                             | <code>variables.cfg</code>                                                                                                                                                  |
| Positions: park, tool swap, safe height, Set Z zero limit, prime line | <code>myrhino/positions.cfg</code> - the one place; every macro reads it                                                                                                    |
| Your added tools                                                      | The portal or console commands (stored in <code>myrhino/custom_tools.json</code> ; <code>custom_tools.cfg</code> is generated – never edit it)                              |
| Starting values, caps, checklists, allowed characters                 | <code>rhino/presets.py</code>                                                                                                                                               |
| Which pins reach the tool connector                                   | <code>myrhino/umbilical.json</code> (copy <code>umbilical.example.json</code> )                                                                                             |
| M3 / M5 , power cap, watchdog, safe-off, pause/resume                 | <code>tools/tool_power.cfg</code>                                                                                                                                           |
| What Pause / Resume / Cancel add to Mainsail's buttons                | <code>_CLIENT_VARIABLE</code> and <code>_RHINO_CANCEL</code> in <code>client_macros.cfg</code>                                                                              |
| Tool swap and restart prompt                                          | <code>toolchanger.cfg</code> , <code>guided/tool_swap_wizard.cfg</code>                                                                                                     |
| Job setup wizards                                                     | <code>guided/*_workflow.cfg</code>                                                                                                                                          |
| Portal screens                                                        | <code>rhino/portal/static/portal.js</code> , <code>templates/index.html</code> , <code>portal.css</code>                                                                    |
| A tool's controls and macros                                          | The portal (Controls and Macros tabs). They are written into <code>custom_tools.cfg</code> ; the rules are in <code>rhino/extras.py</code>                                  |
| Maintenance tasks, work log, meters                                   | The portal. Stored in <code>myrhino/maintenance/</code> ( <code>tasks.json</code> , <code>history.json</code> , <code>meters.json</code> ), so the GitHub backup keeps them |
| Starter task intervals                                                | <code>rhino/maint/library.py</code> (only affects tasks added afterwards)                                                                                                   |
| The start-up maintenance pop-up                                       | <code>guided/maintenance.cfg</code> ; <code>myrhino/maintenance_status.cfg</code> is generated - never edit it                                                              |

**Positions.** Where the head parks, where it goes for a tool swap, how far the bed lowers before a swap or park, the lowest the Set Z zero window goes, and where the prime line starts and how long it is are all in `myrhino/positions.cfg`. Change a number, save, then `FIRMWARE_RESTART`; every macro (and Mainsail's own pause park) uses the new number. Type `RHINO_POSITIONS` in the console to list them - it also checks each one against the travel limits in `printer.cfg`, and the same check runs at every start with any problem shown in red. Updates never overwrite this file. The travel limits themselves, where Z homes and the probe offset stay in `printer.cfg` because Klipper needs them; the paper test goes to the probe's spot worked out from those.

**Backups.** `BACKUP_CFG MSG=<note>` pushes the config folder to your GitHub backup through `klipper-backup`. The tool registry keeps its last 10 versions in `myrhino/registry_backups/`. Photos live outside the config folder in `~/printer_data/rhino_data/images/`, so they are not in the GitHub backup.

**Tests.** `sh dev/run_all.sh` runs a static check of the whole config and 583 automated checks: the macros in a simulated Klipper (against the real Mainsail macros); motion tests that follow where the bed and head go and run every macro and button from seven machine states with each toolhead against five safety rules; the portal, tool controls and macros, maintenance and the Admin lists; every portal screen opened in a real browser, with an accessibility scan and keyboard checks; and the installer run on a fake Pi. It needs `pip install jinja2 flask` (and Playwright for the browser test), and runs on every change pushed to GitHub. The simulator knows the travel limits, not the shape of your tools or stock, so every release also has on-machine checks - the plan is in `docs/testing.md`, and the ones for the current version are in *Still to check on the machine*.

**Trying the portal without the printer.** `python3 dev/demo_server.py` runs the portal against a copy of the config and a pretend Moonraker. Every screen in this manual can be tried there without touching the machine.

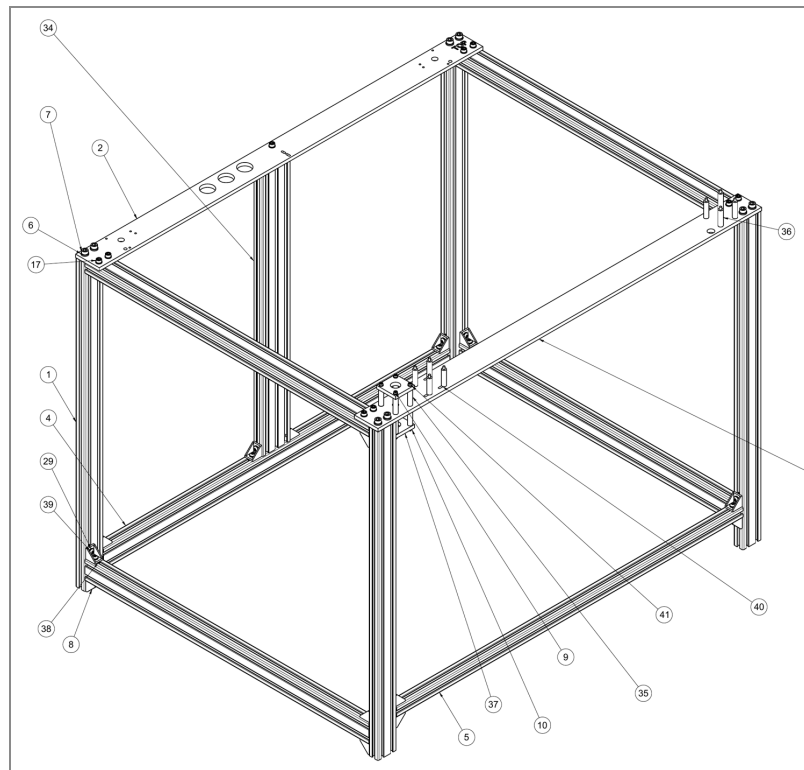
## Build reference

The printable parts, CAD (STEP) files and assembly drawings are in the `Rhino-3d-Printer` repository. Print structural parts in ABS or ASA.

| Repository folder         | What is there                                                                                                                                   |
|---------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>frame</code>        | Frame assembly drawing and STEP model;<br><code>frame/panels</code> – 3 mm ACP enclosure panels, doors and hinges, front spacer assembly        |
| <code>motion/xy</code>    | XY joiners, X-carriage and Back Bone, stepper mounts, rear centre idler, belt tensioners, gantry router-template guide                          |
| <code>motion/z</code>     | Leadscrew carriages and Z stepper mounts                                                                                                        |
| <code>bed</code>          | Vacuum bed heat spreader                                                                                                                        |
| <code>toolheads</code>    | BlockOne (build files to come), SwitchFly, LightSaber laser mount and height gauge, HotJoe spindle parts, DragKnife; Jack Rabbit kept as legacy |
| <code>electrical</code>   | Controller, relays, umbilical and wiring diagrams                                                                                               |
| <code>filament-box</code> | "Cheeky Monkey" roll stand and runout switch (unfinished)                                                                                       |
| <code>docs</code>         | Software, safety (planned smoke detection) and maintenance notes                                                                                |
| <code>archive</code>      | 2023 Klipper configs for the old SKR GTR board, ESC, fan and heated-bed notes – superseded by this config package                               |

## Frame

The frame is 2020/2040/2060 aluminium extrusion with 2" × ¼" aluminium flat bar across the top; some components mount straight to the flat bar, using printed drilling templates for hole placement.



Frame assembly drawing

| Item | Qty | Part                                              |
|------|-----|---------------------------------------------------|
| 1    | 8   | 2040 extrusion, 24 in                             |
| 2    | 1   | Rear top frame flat bar, 2 in × 0.25 in aluminium |
| 3    | 1   | Front top frame flat bar, aluminium               |
| 4    | 1   | 2040 extrusion, 744 mm                            |
| 5    | 1   | 2020 extrusion, 744 mm                            |
| 6    | 8   | M6 washer                                         |
| 7    | 8   | M6 × 20 mm hex socket head screw                  |
| 8    | 22  | Cast corner bracket                               |
| 9    | 4   | M3 × 85 mm socket head screw                      |
| 10   | 4   | M3 hex nut                                        |
| 17   | 9   | M5 × 12 mm screw                                  |
| 29   | 53  | M5 washer                                         |

| Item | Qty | Part                                  |
|------|-----|---------------------------------------|
| 34   | 1   | 2060 extrusion, 550 mm                |
| 35   | 9   | M3 × 30 mm spacer                     |
| 36   | 8   | M3 × 30 mm standoff                   |
| 37   | 2   | Motion system bracket, aluminium      |
| 38   | 44  | M5 hammer nut                         |
| 39   | 44  | M5 × 10 mm button head hex drive bolt |
| 40   | 8   | M3 × 12 mm socket head screw          |
| 41   | 16  | M3 washer                             |

The frame notes recommend socket-head rather than button-head bolts wherever a bolt is torqued hard, because button heads strip more easily.

## Still to check on the machine

The automated tests can't move the machine, so each release is checked on it. Work through these in order, with a hand on the emergency stop; stop at any test that doesn't pass and note its number. The full plan, with what counts as a pass, is in `docs/testing.md`.

### Install and start

- ☐ **A1** Install 1.4.0 with menu option 3 – the preview shows no STOPPED, and the portal answers on port 5000.
- ☐ **A2** `FIRMWARE_RESTART` with a print head mounted – Klipper ready, no config errors.
- ☐ **A3** The *which toolhead is mounted?* question appears; `CHECK_TOOLHEADS` lists every tool.

### Motion and homing

- ☐ **B1** Jog X and Y 100 mm and measure – 100 mm  $\pm 0.5$ . About 50 mm means a step setting is wrong: stop and report it.
- ☐ **B2** Jog Z 10 mm and measure – 10 mm  $\pm 0.1$  (gear\_ratio 2:1). Then `Z_TILT_ADJUST`.
- ☐ **B5** Cancel a print – the bed lowers to Z150 or more, X/Y home, the head parks at X551 Y381 without touching the frame.

### Homing guard (new in 1.4.0)

- ☐ **A6** LightSaber mounted: Mainsail's Home All is refused with a message and nothing moves; Home X and Home Y work. BlockOne mounted: Home All homes as before.
- ☐ **C4** Restart with the bed parked, then swap: the console says homing X/Y lowered the bed 10 mm and the park height was updated; `SET_Z_ZERO` then shows the new height.
- ☐ **A7** `RHINO_POSITIONS` lists the positions and says they are inside the travel limits. Change `park_x` in `myrhino/positions.cfg` to 540, restart, cancel a print: the head parks at X540. Put it back.

### Tool swaps

- ☐ **C1** Swap BlockOne to LightSaber – waits for 40 °C, lowers the bed, never homes Z.
- ☐ **C2** Confirm a swap with the umbilical unplugged – *Toolhead not detected*; plug in, **Check again**.
- ☐ **C3** Start a BlockOne file with the laser mounted – refused, nothing moves.

### Printing

- ☐ **D1-D3** Paper test, then a test cube with the Orca profiles: the PLA question closes when answered, 210/65 °C, a 120 mm prime line, a slow first layer, the bed parked at the end. The first layer is laid at the paper-test height (1.4.0 keeps it when the file is released).

## Set Z zero

- ☐ **E1-E2** `SET_Z_ZERO` with LightSaber or HotJoe – the buttons move the bed the amount shown, Bed down stops at Z380, SET moves nothing, and `G1 Z5 F300` then lifts the tool 5 mm above the surface.
- ☐ **E3** Restart with the bed parked, run `SET_Z_ZERO` – the *Bed position* question; **Yes** moves nothing until a Bed button, then X/Y home. Needs `[force_move] enable_force_move: True` (in `toolchanger.cfg`).
- ☐ **E4** Restart after moving the bed below Z150 – no question; home Z with a print head.
- ☐ **E6** After a CNC or drag-knife setup, swap to BlockOne and print – no X/Y shift.

## Laser, CNC and drag-knife jobs

- ☐ **F1-F3** Laser at low power on scrap; a CNC air cut 10 mm above the stock first; a drag-knife test line. Run the first file from the LaserWeb settings in *Slicer and CAM settings* at low power.
- ☐ **F4** Pause and resume each tool type – the 10 mm lift clears the work, and the spindle is back to speed before it re-enters the cut.
- ☐ Power values – tune every placeholder power and on-time on the real tools.

## Emergency stop and backups

- ☐ **H1-H2** `EMERGENCY_STOP` with the focus dot on – laser off at once; with HotJoe – relay open, ESC quiet.
- ☐ **I1** Klipper-Backup shows your config files in GitHub again after removing the stray `.git` folder.

## Settings to know

- ☐ Idle timeout – add `[idle_timeout] timeout: 1800` if you use the full ABS/ASA cooldown.
- ☐ Connector pins – only PB1, PC0 and PC1 are free; list any other spare umbilical pins in `myrhino/umbilical.json`.
- ☐ `LEDFLASH` takes about 1.35 s, which also delays the macros that call it (cancel, emergency stop).

## Wiring to confirm

- ☐ **Spindle and servo PWM contact.** The umbilical pin map puts the ESC signal (PB11) on contact 18, but the original HotJoe and SwitchFly sheets listed the PWM signal on contact 19. This manual and the portal use 18, which matches the pin map and `printer.cfg`; check the connector and correct the pages if it is 19.
- ☐ **Laser 12 V power pin.** The pin map lists PD12 (Fan2) for the 12 V power activate on contacts 20–21; `printer.cfg` drives `LASER_INITIALIZE` from PD13 (Fan3 on the Octopus v1.1).
- ☐ **X and Y endstops.** The wiring sheets give X-limit PG6 and Y-limit PG9; `printer.cfg` has `stepper_x` on PG9 and `stepper_y` on PG6.
- ☐ **SwitchFly servo power.** The SwitchFly page lists 24 V power activate on contacts 11–12, while its servo wiring diagram feeds the regulator from 12 V activate.
- ☐ **Fan labels.** On the Octopus v1.1 PA8 is FAN0 and PE5 is FAN1; the pin sheets label them Fan5 and Fan0. The pins are right; only the labels differ.